PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong

In partial fulfillment for the requirements in Capstone 1 and Research Subject

Ella Marie Tabobo

Feljan Katipunan

Evelyn Tamonan

Uki Mae Sotelo

Alexa Elaine Roldan

Business Informatics

October 2026

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

ABSTRACT

BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong

This study developed BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong, a Business Intelligence-based system designed to address challenges in sales transaction management, inventory monitoring, business reporting, and revenue analysis. Kape Tubong previously relied on traditional methods for recording sales and preparing reports, which required considerable time and effort and increased the possibility of inaccurate records, duplicate entries, and reporting delays. To address these challenges, the study developed a centralized web-based system that integrated sales, product, inventory, and business records into a unified platform. The developed system incorporated user authentication, sales transaction processing, product management, inventory management, reporting, and Business Intelligence dashboards. It provided graphical representations of revenue trends, sales performance, product performance, and inventory information. The system was developed using the Agile software development methodology, which involved planning, requirements analysis, system design, development, testing, deployment, and user evaluation.

The findings showed that BrewInsight BI achieved its intended objectives by providing an organized platform for recording sales transactions, managing products, monitoring inventory, and generating business reports. The sales transaction module simplified purchase recording through automatic computation and database storage, while the inventory module automatically updated ingredient quantities based on completed sales. The reporting and analytics module generated sales, inventory, transaction history, and product performance reports, reducing the effort required for manual report preparation. The Business Intelligence dashboard transformed sales data into visual information, including revenue trends, best-selling and slow-moving products, sales summaries, and inventory information. The evaluation results indicated that the system functioned according to its intended purpose and was found by the business owner to be easy to use, reliable, and appropriate for daily business operations.

The study concluded that BrewInsight BI addressed the identified challenges in sales management, inventory monitoring, business reporting, and revenue analysis. By integrating centralized data management, automated processes, and Business Intelligence analytics, the system provided organized and accessible business information that supported more informed decision-making and improved operational efficiency.

TABLE OF CONTENTS

CHAPTER I 5
INTRODUCTION 5
1.1 Background of the Study 5
1.2 Statement of the Problem 7
1.3 Objectives of the Study 9
1.4 Scope and Delimitation 11
1.5 Significance of the Study 14
1.6 Definition of Terms 18
CHAPTER II 22
REVIEW OF RELATED LITERATURE AND STUDIES 22
2.1 Foreign Studies 22
2.2 Local Studies 31
2.3 Related Literature 38
2.4 Synthesis of the Reviewed Literature 45
CHAPTER III 48
METHODOLOGY 48
3.1 Research Design 48
3.2 System Development Methodology 49
3.3 System Architecture 61
3.4 Tools and Technologies Used 62
3.5 Data Gathering Techniques 62
3.6 Testing Procedures 63
3.7 Evaluation Criteria 66
CHAPTER IV 78
SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION 78
4.2 Overview of the Developed System 78
4.3 System Development Methodology 79
4.4 System Architecture 90
4.5 Hardware and Software Requirements 92

4.6 Database Design....95
CHAPTER V 102
CONCLUSION AND RECOMMENDATIONS 102
5.1 Introduction....102
5.2 Summary of Findings....102
5.3 Conclusion 103
5.4 Recommendations....104
5.5 Impact of the Study....105
5.6 Summary....105

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CHAPTER I

INTRODUCTION

1.1 Background of the Study

The Food and Beverage (F&B) industry experienced significant growth over the past decade because of changing consumer lifestyles, rapid urbanization, and the increasing demand for convenient dining and beverage services. Among the businesses in this sector, coffee shops became one of the fastest-growing industries because they served not only as places where customers purchased beverages but also as venues for studying, meetings, remote work, and social interaction. As customer expectations continued to evolve, coffee shop businesses were expected to maintain high standards of service quality, operational efficiency, and customer satisfaction to remain competitive. The growing presence of both local and international coffee brands also intensified market competition, encouraging businesses to adopt more efficient operational strategies and modern management practices. As consumer preferences continued to shift toward convenience, speed, and personalized service, coffee shop owners increasingly relied on technology to improve their daily operations and overall business performance.

Modern coffee shops are no longer focused only on selling beverages. They also provided customers with comfortable spaces for studying, working remotely, conducting business meetings, and spending time with family and friends. Because of these changing customer needs, businesses were required to manage different operational activities efficiently while maintaining consistent product quality and customer service. The increasing use of digital technologies and online customer engagement raised customer expectations for faster transactions and better service experiences. As competition continued to increase, businesses recognized the importance of using technology to improve operational efficiency, strengthen customer satisfaction, and support long-term business growth.

As coffee shop businesses expanded into multiple branches, managing daily operations became more challenging. Opening additional kiosks and outlets increased the complexity of handling customer transactions, monitoring inventory, scheduling employees, recording sales, and evaluating branch performance. Tasks that were manageable in a single-location business became more difficult when operations were distributed across several branches. Businesses therefore required a centralized system that could provide consistent monitoring and standardized operational procedures across all locations. Without proper coordination, differences in transaction recording, inventory monitoring, and reporting practices affected service quality and business performance. Centralized operational management became essential in maintaining consistency and ensuring that all branches operated according to the same business standards.

Managing several branches also introduced additional administrative responsibilities. Branch managers sometimes followed different reporting procedures when standardized monitoring systems were unavailable. Communication gaps between management and branch personnel delayed important operational decisions, while fragmented business records made it difficult to evaluate the overall performance of the organization. As the number of branches increased,

comparing branch performance and identifying operational issues became more complicated. These challenges demonstrated the importance of implementing integrated information systems that provided centralized monitoring, improved communication, and supported informed decision-making across multiple business locations.

Kape Tubong experienced continuous growth by expanding from a single coffee outlet into several kiosks together with a main branch. This expansion reflected the increasing popularity of the business and the growing demand for its products. As business operations expanded, larger volumes of sales transactions and operational data were generated from different locations. Without an integrated information system, managing and monitoring these records became increasingly difficult. Sales information from different branches remained separated, making report consolidation more time-consuming and reducing management's ability to monitor overall business performance efficiently. Maintaining the consistency, accuracy, and accessibility of business information also became more challenging as the organization continued to grow.

The continuous expansion of Kape Tubong created new operational requirements that demanded better information management practices. Higher transaction volumes increased the complexity of monitoring revenue performance, evaluating branch operations, and consolidating business reports. Manual coordination between branches and management became less efficient as the organization expanded, while fragmented data limited management's ability to obtain a complete view of business performance. These operational challenges highlighted the need for a centralized digital platform capable of supporting business growth through accurate reporting, efficient data management, and improved operational monitoring.

Like many small and medium-sized enterprises (SMEs), Kape Tubong initially relied on traditional and semi-manual methods to manage its daily operations. Sales transactions were recorded through spreadsheets and standalone Point-of-Sale (POS) systems used independently by each branch. While these methods were sufficient during the early stages of the business, they became less effective as the company expanded. Since every branch maintained its own records, consolidating sales reports and monitoring overall business performance required additional time and effort. Manual data entry also increased the possibility of duplicate records, missing information, and human errors, which affected the accuracy and reliability of reports used by management. Point-of-Sale systems were designed to process and record sales transactions and maintain transaction information for business operations [1].

One of the major challenges encountered by the business was the lack of real-time access to sales information. Because transaction records from each branch were stored separately, management needed to collect reports from every location before preparing consolidated summaries. This process delayed the availability of important business information and made it difficult to respond quickly to operational concerns. The absence of a centralized reporting system also limited management's ability to monitor branch performance daily, making it more challenging to identify issues that required immediate attention.

The existing reporting process focused mainly on presenting sales figures without providing meaningful analysis of business performance. Although reports showed the total number of transactions and revenue generated by each branch, they did not provide sufficient information about customer purchasing behavior, product demand, or sales trends. Management found it difficult to determine which products performed well, which menu items required

improvement, and which periods generated the highest revenue. As a result, important business decisions often depended on manual analysis, which required additional time and effort before useful insights could be obtained.

Inventory management also became more challenging as business operations continued to expand. The consumption of ingredients depended directly on customer purchases, making accurate inventory monitoring essential to daily operations. Since sales transactions and inventory records were not fully integrated, monitoring stock movement required manual updates and regular verification. This process sometimes resulted in delays in replenishing supplies or maintaining excess inventory. These situations affected operational efficiency because management could not immediately determine the actual inventory status across different branches.

Employee scheduling presented another operational concern. Without detailed information about customer traffic and sales performance, work schedules were often prepared using previous experience instead of actual business data. During peak business hours, some branches experienced increased workloads due to insufficient staffing, while slower periods occasionally resulted in unnecessary labor costs. Reliable sales analysis could have supported better workforce planning by identifying customer demand patterns and helping management assign employees more efficiently.

The growing volume of operational data demonstrated the need for a more effective information management system. Instead of relying solely on manual reports, businesses increasingly adopted Business Intelligence (BI) technologies to organize, analyze, and present information in a more meaningful way. Business Intelligence transformed raw transactional data into visual reports, dashboards, and performance indicators that supported faster and more informed decision-making. By providing accurate and timely business insights, BI systems helped organizations monitor sales performance, identify revenue trends, evaluate product demand, and improve overall operational efficiency [2].

1.2 Statement of the Problem

Kape Tubong expanded its operations by opening several branches to serve more customers and reach a wider market. This growth led to an increase in sales transactions, inventory records, and day-to-day activities across the organization. While the additional branches created opportunities for the business, they also increased the amount and complexity of information that management needed to oversee. As operations became more distributed, maintaining a clear and consistent view of the organization's overall performance became more difficult.

Each branch maintained its own sales records and operational reports. This decentralized approach meant that information was collected and stored separately, using processes that could vary from one branch to another. Management therefore had to gather information from multiple sources before it could review performance across the organization. The lack of a consolidated view made it harder to access complete, accurate, and timely information, particularly when management needed to compare branches, identify operational concerns, or assess the business. As the volume of records increased, consolidating and reviewing data required progressively more time and coordination.

The existing process relied heavily on employees to collect, verify, organize, and consolidate branch records before preparing reports for management. These activities demanded substantial manual effort and depended on the accuracy and consistency of the information submitted by each branch. Differences in recording practices, delays in submitting records, and repeated handling of data could contribute to errors, duplicate entries, missing information, or

inconsistencies between reports. Employees also had to spend time checking and reconciling records, which could delay the preparation of reports and limit the time available for other operational responsibilities.

These reporting challenges affected management's ability to monitor the business effectively. Because branch records were maintained separately, reviewing overall sales performance and comparing branch operations required additional consolidation and verification. Managers could face delays in identifying changes in sales activity or responding to operational concerns because the relevant information was not immediately available in a unified form. When reports took longer to prepare, the information used for decision-making could also be less timely, reducing its usefulness for addressing current business needs.

Kape Tubong also had limited capabilities for analyzing its sales and operational data. Existing reports primarily presented numerical summaries, with insufficient dashboards and visual analytical tools to help users interpret the information. Without clear visual summaries, it could be more difficult to recognize sales trends, evaluate product performance, monitor inventory movement, and compare results among branches. Although historical sales records could provide useful evidence about past business activity, they were not fully used to identify patterns that might help management anticipate customer demand or plan inventory and workforce requirements.

The limited use of historical information also constrained planning. Trends across products, branches, and time periods could help management understand how business activity changes and where closer attention may be needed. However, when data remained fragmented and was presented mainly as unconnected numerical reports, drawing comparisons or identifying recurring patterns required additional manual analysis. This reduced the analytical support available for forecasting, performance evaluation, and decisions about future operations.

As Kape Tubong continued to expand, the effects of fragmented records, manual reporting procedures, and limited analytical capabilities became more significant. The organization needed to coordinate information across multiple branches while maintaining reliable records and providing management with a timely view of business performance. The existing process made this increasingly demanding, as employees had to devote considerable effort to consolidating information before it could be reviewed and used.

These conditions highlighted the need for an integrated approach to managing and analyzing sales and operational information across Kape Tubong's branches. A more unified process could support consistent record handling, reduce the effort required to consolidate data, and make information more readily available for management review. Business Intelligence capabilities could further help transform accumulated records into useful summaries and visual insights for monitoring sales, evaluating products and branches, tracking inventory movement, and supporting workforce and demand planning. Such an approach would strengthen the organization's ability to assess performance, respond to operational concerns, and make informed decisions as its operations continue to grow.

This study aimed to address the following problems:

1. The existing operations of Kape Tubong lacked a centralized Point of Sale (POS) system capable of consolidating sales transactions from multiple branches into a single database. This limitation resulted

in fragmented records and made it difficult to access complete and updated sales information across different locations.

2. The business had limited capabilities for analyzing and visualizing sales performance, product demand, inventory status, and branch performance. The absence of appropriate Business Intelligence tools made it difficult to identify sales trends, evaluate product performance, compare branch operations, and interpret historical business data.

3. Forecasting workforce requirements remained challenging because employee scheduling was not sufficiently supported by historical sales data and customer transaction patterns. As a result, staffing decisions relied primarily on experience, which could affect labor utilization, employee allocation, operational efficiency, and service quality during periods of varying customer demand.

4. Sales reporting depended on the manual preparation and consolidation of records from different branches. These procedures required considerable time and were vulnerable to errors, inconsistencies, duplicate entries, missing records, and reporting delays, which affected the efficiency and reliability of business reporting.

5. The business did not fully utilize Business Intelligence technologies to transform sales and operational data into meaningful information for forecasting, performance evaluation, and strategic planning. This limitation reduced the organization's ability to identify patterns, support data-driven decisions, improve operational efficiency, and develop strategies for sustainable business growth.

1.3 Objectives of the Study

1.3.1 General Objectives

The general objective of this study was to design and develop BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong, a web-based Point of Sale (POS) system integrated with Business Intelligence (BI) features. The system was developed to centralize sales transactions, operational records, and business-related information from all Kape Tubong branches into a unified database. Through this centralized approach, the study aimed to improve the organization, accessibility, and management of business data while addressing the limitations caused by fragmented records and manual reporting procedures. By integrating sales management and Business Intelligence capabilities into a single platform, the study sought to provide accurate, organized, and timely business information that supported improved operational monitoring, strategic planning, and decision-making processes.

The study aimed to transform raw sales transactions and operational records into meaningful and actionable business insights through the use of interactive dashboards, automated reports, and analytical visualizations. The developed system presented important business information, including revenue patterns, sales trends, branch performance, product performance, customer purchasing behavior, and inventory movement, in a clear and organized format. These analytical features enabled management to evaluate business performance more effectively, identify emerging trends, recognize operational concerns, and develop strategies based on actual business data rather than relying only on assumptions or personal experience. By providing centralized access to updated sales information and key

performance indicators, the platform improved the availability, accuracy, and usefulness of business data for daily operations and long-term planning.

The study also aimed to improve the efficiency, accuracy, and reliability of sales reporting by reducing dependence on manual preparation, consolidation, and verification of business records. The automated reporting process minimized repetitive administrative tasks, reduced the possibility of human errors, and improved the consistency of reports generated from different branches. Centralized data storage allowed the system to maintain complete and organized historical records that could be retrieved, reviewed, and analyzed whenever necessary. These improvements supported faster report generation, simplified branch performance comparison, and provided management with reliable operational information needed for effective evaluation and planning.

Another objective of the study was to support inventory management, workforce planning, and resource allocation through the analysis of historical sales data and operational patterns. The system provided insights regarding product demand, peak transaction periods, customer purchasing behavior, and branch performance, allowing management to make informed decisions regarding inventory replenishment, employee scheduling, promotional activities, and business strategies. These analytical capabilities helped improve operational planning, maximize resource utilization, enhance service quality, and support more efficient management of business activities across all Kape Tubong branches.

The study also aimed to strengthen the security, reliability, and integrity of business information through the implementation of secure user authentication, role-based access control, and controlled data management procedures. These security measures ensured that confidential sales records, financial information, and operational reports were accessed only by authorized personnel according to their assigned responsibilities. The system provided a protected environment for storing and managing business information while allowing authorized users to monitor operations and access necessary reports through a web-based platform. These features contributed to maintaining data confidentiality, preventing unauthorized access, and improving trust in the accuracy and reliability of the information generated by the system.

The study aimed to enhance operational efficiency, improve reporting accuracy, strengthen analytical capabilities, and support data-driven managerial decision-making through the development of a centralized Business Intelligence-based Point of Sale system. The developed BREWINSIGHT BI platform provided Kape Tubong with a scalable and adaptable technological solution that addressed existing challenges in sales monitoring, data management, reporting, and business analysis. Through the integration of POS and Business Intelligence features, the system supported more organized information management, improved accessibility of operational data, and enabled management to make informed decisions based on accurate and timely insights.

The implementation of the system contributed to improving the way Kape Tubong managed its business operations by reducing dependence on manual processes, increasing the efficiency of data analysis, and providing a more reliable method for monitoring business performance. The platform also established a foundation for future improvements by allowing the organization to accommodate increasing transaction volumes, additional branches, and changing business requirements. By providing a centralized and data-driven approach to managing operations, BREWINSIGHT BI

strengthened the company's ability to evaluate performance, respond to operational challenges, and develop strategies that supported continuous improvement and sustainable business growth.

1.3.2 Specific Objectives

Specifically, this study aimed to:

1. Develop a centralized web-based dashboard that consolidated and organized sales transactions from all Kape Tubong branches into a single platform to improve business monitoring, operational visibility, performance evaluation, and centralized data management. The dashboard was designed to provide management with real-time access to branch sales, allowing faster monitoring of business activities, easier comparison of branch performance, and more efficient management of sales information across all locations.

2. Automate the generation of daily, weekly, and monthly sales reports to reduce manual processing, minimize human errors, improve reporting accuracy, and provide timely access to reliable business information. The automated reporting feature was intended to simplify report preparation, standardize reporting procedures, maintain organized historical records, and support faster evaluation of business performance through consistent and accurate reports.

3. Integrate Business Intelligence features processed and visualized revenue patterns, sales trends, customer purchasing behavior, branch performance, peak operating hours, and top-selling products through interactive dashboards and analytical reports. These features were developed to transform raw sales data into meaningful business insights that supported forecasting, strategic planning, performance evaluation, and data-driven decision-making.

4. Provide inventory-related analytical insights based on sales performance data to assist management in monitoring product demand, identifying fast-moving and slow-moving products, improving inventory planning, reducing stock shortages and overstocking, and supporting efficient resource allocation across all Kape Tubong branches. These analytical capabilities also helped management align purchasing decisions with actual customer demand and improve inventory utilization

5. Implement secure authentication and role-based access control to protect business information while allowing authorized users to securely access system functions, reports, dashboards, and analytical data through a web-based platform. These security measures were intended to maintain the confidentiality, integrity, and availability of sales records while supporting safe and reliable system access for management and authorized personnel.

1.4 Scope and Delimitation

Scope

- This study focused on the design and development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong, a web-based Point of Sale (POS) system integrated with Business Intelligence (BI) features. The system was developed to centralize sales transactions from the main branch and all Kape Tubong kiosks into a single database to improve the management, organization, and analysis of business information. It served as a centralized platform that enabled management to monitor branch operations, evaluate sales performance, and generate analytical reports using data collected from daily business transactions. Through this approach, the study sought to improve operational efficiency, reduce dependence on fragmented reporting practices, and strengthen data-driven decision-making within the organization.

- The developed system included modules for transaction recording, product management, branch management, dashboard visualization, automated report generation, and Business Intelligence analytics. Sales data from all branches were consolidated into a centralized database, allowing management to access complete and updated information through a single web-based platform. The dashboard presented business performance using charts, graphs, summary panels, and comparative reports, making sales information easier to understand and interpret. The system also generated daily, weekly, and monthly reports that supported regular business monitoring and performance evaluation. By integrating these functions into one platform, the study improved the accessibility, consistency, and organization of sales information across all Kape Tubong branches.

- The Business Intelligence component of the system analyzed historical and current sales transactions to identify revenue patterns, product demand, customer purchasing behavior, peak operating hours, and branch performance. The analytical results were presented through interactive visualizations that enabled management to monitor business trends more efficiently. These features helped identify top-selling and slow-moving products, compare sales performance among branches, and evaluate revenue fluctuations across different periods. The generated insights supported data-driven planning, promotional activities, inventory management, and operational decision-making. The system transformed raw transaction records into meaningful information that improved the overall understanding of business performance.

- The study also included inventory-related analytical features derived from sales transactions. The system evaluated product demand and sales movement to identify fast-moving and slow-moving products and provide recommendations that supported inventory planning. These analytical outputs assisted management in maintaining appropriate stock levels, reducing overstocking, preventing stock shortages, and improving purchasing decisions based on historical sales performance. Branch-specific inventory insights also enabled management to recognize differences in product demand among locations and allocate inventory more effectively. However, the system functioned only as a decision-support tool and did not automate stock replenishment, warehouse management, or procurement processes.

- To improve operational monitoring, the system generated branch-level and overall business performance reports that allowed management to compare sales results among different locations. Comparative analysis enabled the identification of high-performing and underperforming branches based on actual sales records. These reports supported performance evaluation, operational assessment, and planning for future improvements. Management could also review historical sales information to monitor business growth and evaluate the effectiveness of implemented strategies over time. The centralized reporting structure reduced

the need for manual consolidation of branch reports and improved the accuracy and consistency of business information.

- The system also included security features to protect business information and maintain data integrity. User authentication and role-based access control were implemented to ensure that only authorized personnel could access system functions and confidential sales records. Different user roles were assigned specific access privileges according to their responsibilities within the organization. Authorized users could securely access the system using desktop or mobile web browsers with an internet connection. These security measures protected business information while allowing management to monitor operations remotely whenever necessary. Centralized storage of sales records also improved data reliability by reducing duplication, inconsistencies, and fragmented record management.

- The scope of the study was limited to the analysis of internal sales and transaction data generated by Kape Tubong. All dashboards, reports, and analytical outputs were based solely on information recorded through the developed system. The study covered sales transactions from the main branch and Kape Tubong kiosks and focused on revenue patterns, branch performance, product demand, customer purchasing behavior, inventory-related insights, and business performance monitoring.

Delimitation

- Several delimitations were established to maintain the focus and feasibility of the study. The developed system did not include physical Point of Sale hardware such as barcode scanners, receipt printers, cash drawers, card readers, or customer display devices. Payroll processing, accounting, taxation, budgeting, supplier management, procurement, warehouse management, customer relationship management, online payment gateways, and other Enterprise Resource Planning (ERP) functions were also excluded. The system focused only on centralized sales management, Business Intelligence, reporting, and revenue analysis.
- The study also excluded advanced technologies such as machine learning, artificial intelligence, predictive analytics, and automated forecasting models. The implemented Business Intelligence features were limited to descriptive analytics, historical trend analysis, dashboard visualization, comparative reporting, and revenue pattern analysis based on recorded sales transactions. Forecasting models and predictive algorithms were not included because they were beyond the scope and timeframe of the study. These features may be considered for future enhancement as business requirements continue to evolve.

- External factors, including competitor activities, customer demographics, economic conditions, weather changes, supplier performance, market trends, and other environmental influences, were excluded from the analytical process. As a result, the generated reports reflected only the operational performance of the business based on available transaction records.

- The effectiveness of the developed system depended on the accuracy and completeness of the sales transactions entered by authorized users. The study assumed that all business records were encoded correctly and consistently across all branches. Although the system included validation procedures and user authentication, it did not independently verify the accuracy of manually entered data beyond standard input

controls. Consequently, the quality of generated reports, dashboards, and analytical results depended on the reliability of the available sales records.

- The developed system was specifically designed to meet the operational requirements of Kape Tubong. The database structure, dashboard components, reports, and analytical features were customized according to the organization's existing business processes and reporting requirements. Although the system could be modified for implementation in other businesses, additional customization would be necessary to accommodate differences in operational workflows, product categories, reporting requirements, and organizational structures. Therefore, the study remained limited to the operational environment of Kape Tubong.

- The evaluation of the developed system was limited to functional and user acceptance testing conducted during the implementation period. Long-term operational performance, user adoption, infrastructure expansion, organizational change management, and future software enhancements were beyond the scope of this study. Despite these limitations, the developed BREWINSIGHT BI: Revenue Pattern Analytics System provided a centralized platform for sales management, automated reporting, revenue analysis, inventory-related insights, and data-driven decision-making within Kape Tubong.

1.5 Significance of the Study

The proposed system was expected to benefit the following:

- **Management:** The proposed system provided significant benefits to the management of Kape Tubong by bringing sales transactions and business information from all branches together in a centralized, web-based platform. Instead of gathering separate records and consolidating them manually, management could access information through a shared system and review reports from a single source. This helped make business data easier to retrieve and supported more consistent monitoring of operations across the organization.

  The system enabled management to track sales performance and compare results among branches. With access to updated reports and interactive dashboards, managers could review business activity across different locations and periods. These views helped turn detailed sales records into information that was easier to interpret, including summaries of sales trends, product performance, branch productivity, and revenue patterns. Automated reporting also reduced the effort involved in preparing routine summaries, allowing management to spend more time reviewing results and addressing business needs.

  The system’s Business Intelligence features supported the analysis of both current and historical sales information. Management could use reports and dashboards to recognize patterns in sales activity, assess which products performed well, and observe how results changed over time. Comparing information across branches and periods provided a clearer basis for evaluating performance and identifying areas that might require further attention. These analytical capabilities strengthened management’s ability to use available business data when assessing operational results.

The information provided by the system also supported planning and resource allocation. Sales trends, product performance, and branch comparisons could inform decisions about pricing and promotional activities. Historical records could help management anticipate potential inventory needs and consider workforce scheduling in relation to business activity. In addition, consolidated reports could provide useful context when evaluating future expansion opportunities. The system did not replace managerial judgment; it provided organized, timely information to support that judgment.

Centralized reporting improved the consistency of information available for management review. When branches used a common platform and reports followed standardized formats, management could compare performance using shared indicators. This helped reduce discrepancies arising from separate reporting practices and made it easier to review results across locations. Access to more timely information also reduced the delays associated with collecting and checking branch records before reports could be prepared.

The platform strengthened management oversight by making potential operational concerns easier to identify. Regularly reviewing branch results could help managers notice declining sales, differences in branch performance, or unusual transaction patterns sooner. Earlier visibility gave management more opportunity to investigate these changes and determine whether corrective action or additional support was needed. Branch comparisons could also help identify effective practices that might be considered for use in other locations.

Overall, the proposed system supported management by making business information more centralized, accessible, and useful for monitoring operations. Through consolidated records, standardized reports, dashboards, and analytical features, it improved the basis for evaluating performance and planning business activities. These capabilities strengthened operational control and supported more informed managerial decision-making across Kape Tubong's branches.

- **Staff:** The implementation of the proposed system benefited Kape Tubong staff by reducing the time and effort required to record sales, manage transaction information, and prepare reports. Under the previous process, employees often had to organize records, perform calculations, and compile information before submitting it for review. Centralized data processing and automated reporting simplified these tasks and reduced the amount of repetitive clerical work required at each branch.

  With less time spent preparing and consolidating reports, employees could devote more attention to customer service and other daily responsibilities. Staff could focus on assisting customers, supporting routine branch operations, and maintaining accurate transaction records while the system organized information for reporting. This helped make better use of employees' time, particularly during periods when sales and operational activity were high.

  The system also simplified the process of recording, retrieving, and reviewing sales information. A centralized platform made relevant records easier to locate, reducing the need to search through separate files or spreadsheets. Employees no longer had to manually combine information from multiple sources or calculate sales summaries in the same way as before. These changes made reporting activities more orderly and helped reduce the likelihood of errors associated with repeated data entry and manual calculations.

Standardized procedures improved consistency in how staff documented and handled sales information. When branch personnel followed a common process, reports were easier to review and compare, and there was less room for confusion caused by differing formats or recording practices. More consistent records also supported clearer coordination between employees at different branches and helped management obtain information in a form that was easier to understand and verify.

Centralized records strengthened communication between staff and management by making operational information more accessible. When questions arose about sales activity or transaction details, authorized users could refer to the same stored information. This made verification more straightforward and reduced the need to repeatedly request or recompile records. Historical information could also be retrieved more conveniently when staff or management needed to check previous transactions or review performance over time.

The platform supported accountability by keeping routine documentation more organized and consistent. Clearer records made it easier to review completed transactions and reporting activities, while standardized workflows helped employees understand the information they were expected to provide. This could reduce uncertainty about reporting requirements and make daily administrative tasks more predictable.

Overall, the proposed system improved staff productivity by simplifying routine reporting and record-management tasks. By reducing administrative workload, improving access to sales information, and promoting consistent documentation across branches, it supported a more organized and efficient working environment. Staff could spend less time on manual consolidation and devote more attention to customer service and the operational needs of their branch.

- **Organization:** The proposed system benefited Kape Tubong by improving how the organization managed sales information and coordinated operations across its branches. By storing branch sales records in a centralized database, the system provided a more consistent way to collect, organize, and access business information. This reduced reliance on fragmented reporting practices and gave the organization a clearer view of its activities across locations.

  Centralized records supported greater transparency and accountability. When information was maintained in a shared system, authorized users could review records through a common source, making it easier to verify reported figures and follow up on questions. Consistent documentation also improved coordination between branches and management by helping ensure that operational information was organized in a comparable format.

  The system's Business Intelligence features helped the organization make better use of its sales data. Management could analyze revenue patterns, compare branch performance, review product demand, and examine customer purchasing behavior based on available transaction records. These insights could help reveal changes in business activity and identify areas that might warrant closer review. By presenting information in organized reports and dashboards, the system made it easier to interpret data that might otherwise remain dispersed across individual branch records.

These analytical capabilities also supported organizational planning. Sales and product information could inform inventory decisions, while patterns in business activity could provide context for workforce allocation and promotional planning. Branch comparisons could help the organization assess where additional support might be needed or which practices could be considered for broader use. Historical sales data could also support the evaluation of business growth and help management review the results of organizational strategies over time.

The system promoted data-driven decision-making by making business information more accessible and timely. Rather than depending primarily on manual reporting and separate records, management could consult consolidated information when reviewing performance or responding to operational concerns. This improved the organization's ability to coordinate decisions across branches and strengthened its overall operational control.

The platform also provided a technological foundation that could accommodate continued business growth. As Kape Tubong's operations and information needs change, a centralized system can support more organized record management and reporting across existing branches. It can also provide a basis for considering future digital initiatives, subject to the organization's needs and resources.

Overall, the proposed system strengthened Kape Tubong's capacity to manage information, coordinate branch operations, and evaluate business performance. By improving record consistency, supporting analysis, and reducing dependence on manual reporting procedures, it contributed to greater operational efficiency and more informed planning. These capabilities can help the organization adapt to growth and support its long-term sustainability.

- **BSIT Students:** The study benefited Bachelor of Science in Information Technology (BSIT) students by providing a practical example of how software engineering, database management, web development, and Business Intelligence technologies were integrated to address real-world business problems. It demonstrated how concepts learned in the classroom could be applied in developing a functional web-based analytics platform that supported business operations and decision-making. The study also served as a useful reference for students conducting capstone projects related to Business Intelligence, decision support systems, dashboard development, sales analytics, and business process automation.

  By examining the implementation of the system, students gained a better understanding of the complete software development life cycle, including requirements gathering, system analysis, database design, interface development, implementation, testing, and deployment. The project also emphasized the importance of developing user-centered systems that addressed actual organizational needs. Using dashboard visualization, automated reporting, and centralized data management, students were introduced to practical methods of transforming business data into meaningful information that supported decision-making. Overall, the study strengthened students' technical knowledge, analytical skills, problem-solving abilities, and understanding modern Business Intelligence technologies, preparing them for future careers in the information technology profession.

- **Future Researchers:** The study served as a valuable reference for future researchers interested in Business Intelligence, sales analytics, decision support systems, digital transformation, and information systems development. It provided a practical framework for designing and implementing a centralized revenue analytics platform for a small and medium-sized enterprise in the food and beverage industry. The system architecture, development process, and analytical features could serve as a guide for future studies involving similar business environments.

  Future researchers could further improve the proposed system by integrating advanced technologies such as predictive analytics, machine learning, artificial intelligence, cloud computing, and mobile-based applications. They could also evaluate the effectiveness of these technologies in improving business performance, forecasting sales trends, and supporting organizational decision-making. In addition, the study could serve as a basis for comparative research involving different industries, organizations, or operational environments. Overall, the study contributed to the existing body of knowledge on Business Intelligence implementation and provided a strong academic foundation for future research involving data-driven business solutions and digital innovation.

1.6 Definition of Terms

- **Analytics:** Referred to the systematic process of collecting, organizing, examining, and interpreting data to identify meaningful patterns, trends, and relationships that supported business decision-making. It transformed raw business data into useful information that enabled organizations to evaluate performance, monitor operational activities, and improve strategic planning. Through analytics, businesses were able to examine historical and current data to understand customer purchasing behavior, evaluate sales performance, identify profitable products, and recognize recurring business trends. Analytics also assisted organizations in measuring operational efficiency, forecasting future demand, and allocating resources more effectively based on information instead of assumptions. By continuously analyzing business data, organizations were able to identify operational strengths, detect weaknesses, and implement appropriate improvements that contributed to business growth and sustainability. As digital technologies continued to develop, analytics became an essential component of modern business management because it provided measurable information that supported evidence-based decision-making. In this study, analytics referred to the systematic evaluation of Kape Tubong's sales transactions to identify revenue trends, customer purchasing behavior, product demand, branch performance, and peak operating periods through the BREWINSIGHT BI system [3].

- **Business Intelligence (BI):** Referred to the collection of technologies, processes, and analytical tools used to collect, organize, analyze, and present business information in a meaningful and understandable manner. It transformed raw operational and transactional data into useful insights through dashboards, reports, charts, graphs, and other visualization tools that supported managerial decision-making. Business Intelligence enables organizations to monitor daily operations, evaluate organizational performance, identify business trends, and improve planning based on reliable and accurate information. It also promoted operational consistency by integrating data from different sources into a centralized platform where information could be analyzed efficiently. Through Business Intelligence, organizations were able to improve productivity, strengthen operational control, enhance reporting accuracy, and support long-term business development. As businesses generated increasing amounts of digital information, Business Intelligence became an essential

organizational tool for improving efficiency, competitiveness, and strategic planning. In this study, Business Intelligence referred to the analytical framework integrated into the BREWINSIGHT BI platform for collecting, processing, analyzing, and visualizing Kape Tubong's sales and revenue data to support effective operational and managerial decision-making [2].

- **Dashboard:** Referred to a graphical user interface designed to present important business information through charts, graphs, tables, summary panels, and other visual components. It provided users with a clear and organized presentation of key performance indicators and operational metrics, making business information easier to understand and interpret. Dashboards simplified large volumes of transactional data by presenting summarized information that allowed managers to monitor organizational performance, compare operational results, and identify trends efficiently. Interactive dashboards also included features such as filtering, sorting, and drill-down analysis, enabling users to examine specific information in greater detail according to their analytical needs. Because information was presented visually, dashboards reduced the complexity of traditional reports and enabled both technical and non-technical users to understand business performance more quickly. They also improved communication among management by providing a standardized presentation of organizational data. In this study, the dashboard referred to the web-based interface of the BREWINSIGHT BI system that displayed Kape Tubong's sales performance, revenue patterns, branch comparisons, product performance, and other business analytics through interactive charts, graphs, and summarized reports [4].

- **Data-Driven Decision-Making:** Referred to the process of making business and operational decisions based on analyzed data rather than intuition, assumptions, or personal judgment. This approach relied on accurate, relevant, and timely information to support planning, problem-solving, and organizational improvement. By using factual evidence, organizations were able to reduce uncertainty, improve consistency, and develop more effective business strategies. Data-driven decision-making enabled management to evaluate historical performance, identify operational issues, monitor organizational progress, and implement appropriate corrective actions based on measurable business information. It also strengthened accountability because business decisions were supported by documented evidence instead of subjective opinions. Furthermore, this approach improved strategic planning by helping organizations identify opportunities for growth, respond to market changes, and continuously improve operational performance. As analytical technologies became more accessible, data-driven decision-making became increasingly important for organizations seeking to improve efficiency and competitiveness. In this study, data-driven decision-making referred to the use of analytical reports, dashboards, and sales insights generated by the BREWINSIGHT BI system to assist Kape Tubong's management in making informed operational and strategic decisions based on actual sales performance and revenue analysis [5].

- **Kiosk:** Referred to a small retail outlet or satellite sales location established to provide products and services in areas with high customer traffic. Compared with full-sized branches, kiosks occupied smaller operational spaces while offering similar products and services to customers. Businesses commonly established kiosks in shopping malls, transportation terminals, schools, commercial centers, and other strategic locations to improve customer accessibility and expand market coverage. Although smaller in size, kiosks generated valuable transactional information that could be analyzed to evaluate business performance, customer demand, and sales trends. Sales records collected from multiple kiosks could be consolidated into a

centralized information system, allowing management to monitor operations, compare branch performance, and make more informed business decisions. Kiosks also supported business expansion by increasing market presence while maintaining centralized operational monitoring. In this study, kiosk referred to the satellite sales locations of Kape Tubong where sales transactions were recorded and consolidated into the BREWINSIGHT BI platform for centralized monitoring, branch comparison, and revenue analysis.

- **Point of Sale (POS):** Referred to the hardware and software system used to process, record, and manage sales transactions between a business and its customers. A POS system captured important transaction details such as products sold, quantity purchased, selling price, payment method, transaction date, and transaction time. It improved transaction accuracy by minimizing manual recording errors while maintaining organized digital records that supported reporting and business analysis. Modern POS systems also included features such as inventory monitoring, transaction history, sales reporting, and customer management, making them essential tools for business operations. In organizations operating multiple branches, POS systems promoted standardized transaction recording and generated reliable business data that could be integrated into analytical platforms. The information collected through POS systems served as the primary source for generating dashboards, reports, and performance evaluations. In this study, Point of Sale (POS) referred to the transaction-recording system used by Kape Tubong branches to capture sales information that was integrated into the BREWINSIGHT BI platform for centralized reporting, visualization, and revenue pattern analysis [1].

- **Revenue Pattern:** Referred to the observable trend or behavior of business income over a specific period based on recorded sales transactions. It reflected how revenue changed according to factors such as time, branch location, product demand, customer purchasing behavior, seasonal conditions, and promotional activities. Revenue pattern analysis enabled organizations to identify peak sales periods, slow business intervals, seasonal fluctuations, and products that contributed significantly to business performance. It also allowed management to compare current performance with previous periods to evaluate the effectiveness of operational strategies, marketing campaigns, and pricing decisions. Understanding revenue patterns supported better planning, inventory management, staffing decisions, and resource allocation. Continuous monitoring of revenue behavior also helped organizations evaluate business growth, recognize emerging opportunities, and identify operational weaknesses requiring improvement. In this study, revenue pattern referred to the sales trends and income behaviors identified by the BREWINSIGHT BI system through the analysis of Kape Tubong's transactional sales records from the main branch and its kiosks.

- **Web-Based System:** Referred to a software application that operated on a web server and was accessed through a web browser using an internet or local network connection. Unlike traditional desktop applications, a web-based system did not require installation on individual computers because authorized users could access its features through supported web browsers using different devices. Web-based systems provided greater flexibility by allowing users to retrieve information from various locations using desktop computers, laptops, tablets, or mobile devices connected to the internet. They also supported centralized data storage, ensuring that users accessed consistent and updated information while reducing data duplication across the organization. In addition, web-based systems simplified software maintenance because updates were performed on the server rather than on individual devices. Security mechanisms such as user authentication and role-based access control also protected business information from unauthorized access. In this study,

the web-based system referred to the online architecture of the BREWINSIGHT BI platform that enabled authorized Kape Tubong personnel to securely access sales reports, dashboards, revenue analytics, and business performance information through internet-connected devices [6].

REFERENCES

[1] Oracle, “What is a retail POS (Point of Sale) system?,” Oracle. [Online]. Available: https://www.oracle.com/apac/retail/pos-systems/what-is-retail-pos/

[2] IBM, “What is business intelligence (BI)?,” IBM Think. [Online]. Available: https://www.ibm.com/think/topics/business-intelligence

[3] IBM, “What is business analytics?,” IBM Think. [Online]. Available: https://www.ibm.com/think/topics/business-analytics

[4] Microsoft, “Introduction to dashboards for Power BI designers,” Microsoft Learn. [Online]. Available: https://learn.microsoft.com/en-us/power-bi/create-reports/service-dashboards

[5] T. Mucci, “What is data-driven decision-making?,” IBM Think. [Online]. Available: https://www.ibm.com/think/topics/data-driven-decision-making

[6] Amazon Web Services, “What is a web application?,” AWS. [Online]. Available: https://aws.amazon.com/what-is/web-application/

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Kumar and Singh presented the study entitled “Data Visualization in Retail Analytics: Impact on Decision-Making,” which investigated the role of interactive visualization technologies in improving managerial decision-making within retail and food-service organizations [7]. The study focused on how dashboard-driven analytical tools assisted businesses in interpreting large amounts of transactional and operational information more effectively. The researchers found that organizations utilizing visual analytics platforms demonstrated improved efficiency in strategic decision-making compared with those that depended primarily on conventional spreadsheets and text-based reports. The study emphasized that interactive dashboards allowed managers to examine business information through visual components such as graphs, charts, heatmaps, and trend indicators, making complex data easier to understand and evaluate.

Through these visual presentations, decision-makers were able to recognize important patterns and relationships within business records more efficiently than through manual examination of numerical reports.

The researchers explained that data visualization improved the ability of organizations to monitor important business indicators, including revenue movement, sales performance, product results, and operational changes across different periods. By converting large amounts of numerical information into organized visual displays, analytical systems reduced the difficulty of interpreting business data and allowed users to evaluate information with greater speed and accuracy. Visual dashboards enabled managers to identify unusual sales behaviors, declining performance, and operational concerns at an earlier stage, allowing immediate actions and adjustments to be implemented. This capability supported proactive management because organizations could respond to possible issues before they created significant effects on business operations.

The study also highlighted that dashboard-based visualization improved communication and collaboration among management teams by providing a shared platform for analyzing business performance. Since visual reports presented information in a clear and standardized manner, different users could evaluate organizational conditions using the same data references and performance indicators. This reduced differences in interpretation and improved coordination during planning and decision-making activities. The availability of self-service analytics also allowed managers to explore information independently without requiring continuous assistance from technical personnel. As a result, organizations became more flexible in comparing branch performance, analyzing product results, and reviewing operational trends based on their current requirements.

Kumar and Singh [7] further identified the importance of visualization tools in supporting business forecasting and strategic planning. Through the analysis of historical information, organizations were able to recognize recurring sales patterns, seasonal variations, and changes in customer behavior. These insights provided valuable support for activities such as inventory preparation, employee scheduling, promotional planning, and resource allocation. The researchers emphasized that adopting integrated visualization platforms contributed to digital transformation efforts in retail and food-service industries because these technologies improved the ability of businesses to use available information effectively. Organizations with stronger analytical capabilities became more prepared to handle market changes, shifting customer preferences, and operational challenges.

The study also revealed that visualization-based analytics increased managerial confidence by presenting information in a transparent, structured, and understandable format. Clear graphical representations helped reduce uncertainty during decision-making because managers could easily interpret performance conditions and evaluate possible actions. Dashboard systems improved accessibility by placing important business measurements in a centralized environment where users could review essential information without requiring advanced knowledge of data analysis. This encouraged organizations to rely more on evidence-based strategies and strengthened the use of analytical information in regular management practices.

Another important finding of the research involved the value of continuous monitoring through real-time or regularly updated business information. Unlike traditional reporting approaches that provided information only after manual preparation, dashboard systems allowed organizations to observe operational conditions more consistently. Continuous access to updated sales and performance data enabled managers to respond quickly to changes in demand,

revenue performance, and resource requirements. This capability supported more effective adjustments in staffing decisions, inventory management, and promotional activities. The researchers concluded that continuous monitoring improved operational awareness and allowed businesses to manage daily activities more efficiently.

The findings of Kumar and Singh [7] were highly applicable to the development of BREWINSIGHT BI: Revenue Pattern Analytics System because they supported the integration of interactive dashboards and analytical visualization features within the proposed platform. Kape Tubong experienced difficulties caused by separated sales records, limited centralized monitoring, and challenges in evaluating the performance of multiple branches. These limitations affected the ability of management to obtain complete information regarding revenue patterns and operational performance. Through the implementation of dashboard visualization, BREWINSIGHT BI converted collected sales transactions into organized analytical displays that allowed managers to identify successful products, compare branch performance, analyze sales periods, and recognize changes in revenue trends.

The application of self-service analytics and comparative monitoring also supported the objectives of BREWINSIGHT BI. The system enabled Kape Tubong administrators and managers to review operational information without depending on manual consolidation of reports from different branches. The analytical capabilities discussed in the study aligned with the purpose of the proposed system to identify sales patterns, support resource planning, and improve managerial decision-making. Through centralized dashboards, management gained access to a unified platform that simplified monitoring of different locations and improved the ability to respond quickly to business concerns.

The findings regarding reduced dependence on technical personnel from Kumar and Singh [7] also supported the usability goals of BREWINSIGHT BI. The study demonstrated that Business Intelligence platforms with clear interfaces and accessible analytical tools allowed users with limited technical experience to understand and utilize business information effectively. This concept was suitable for small and medium enterprises where dedicated data specialists might not always be available. By providing understandable dashboards, organized reports, and simplified analytical features, BREWINSIGHT BI supported easier adoption among Kape Tubong personnel and allowed users to obtain valuable insights without requiring advanced technical skills.

The research of Kumar and Singh [7] provided significant support for integrating dashboard visualization and analytical reporting functions into BREWINSIGHT BI. Their findings showed that visual analytics improved the interpretation of business information, increased decision-making efficiency, strengthened operational monitoring, and enhanced organizational responsiveness. The study served as an important basis for recognizing the value of Business Intelligence features in improving the ability of retail and food-service businesses to manage information and develop effective strategies.

Chen Li and Maria Gonzalez conducted the study entitled “Business Intelligence Systems and Organizational Performance in Small and Medium Enterprises,” which examined how Business Intelligence adoption influenced operational efficiency, decision-making quality, and organizational performance among small and medium-sized businesses [8]. The research focused on the contribution of centralized BI platforms in improving access to information, increasing reporting accuracy, and supporting faster managerial responses compared with organizations that relied on traditional manual reporting procedures. The researchers explained that centralized Business Intelligence systems

allowed companies to integrate various types of operational information, including sales transactions, inventory records, and performance reports, into a single analytical environment. This improved visibility across departments and locations by providing management with a complete overview of business conditions.

The study emphasized that Business Intelligence solutions improved strategic decision-making by converting large amounts of raw business data into useful and actionable information. Through features such as dashboards, automated reports, and trend analysis, managers were able to evaluate sales performance, understand customer purchasing behavior, and identify operational weaknesses more effectively. Organizations that adopted BI systems were able to respond more quickly to changes in business conditions, including declining sales, inventory shortages, and shifting customer demands. The automation of reporting processes also reduced the time required for manual data preparation while improving the consistency and accuracy of information presented to decisionmakers.

The researchers further identified that Business Intelligence implementation improved organizational coordination by establishing standardized reporting methods and consistent performance measurements. Through centralized analytical systems, managers and supervisors from different locations could evaluate business activities using similar indicators and reporting structures. This improved the reliability of comparisons among branches, products, and operational periods. In addition, user-friendly BI interfaces allowed employees with limited technical backgrounds to access and understand analytical information independently, reducing the need for continuous assistance from information technology personnel when preparing routine reports.

The study also demonstrated that organizations using Business Intelligence technologies developed stronger capabilities in forecasting and operational planning. Historical business records provided valuable information for identifying recurring sales behaviors, seasonal demand changes, and customer preferences. These insights supported better decisions related to inventory management, workforce scheduling, promotional activities, and allocation of organizational resources. The researchers highlighted that scalable BI solutions provided important advantages for small and medium enterprises because they allowed growing businesses to improve competitiveness and adapt more effectively to changing market conditions.

The findings of Chen Li and Maria Gonzalez [8] were relevant to the development of BREWINSIGHT BI: Revenue Pattern Analytics System because they supported the importance of creating a centralized Business Intelligence platform for improving business monitoring and decision-making. Kape Tubong encountered operational challenges caused by fragmented sales records, manual report preparation, and limited visibility regarding the performance of its different branches. These conditions affected the ability of management to analyze business performance accurately and obtain timely information for planning.

BREWINSIGHT BI addressed these challenges by combining Point of Sale capabilities with Business Intelligence features in a unified platform. The system centralized sales information from different branches and transformed collected records into meaningful analytical outputs through dashboards, reports, and performance evaluations. Through revenue trend analysis, product performance monitoring, branch comparison features, and automated reporting functions, the platform supported improved operational management and provided management with more reliable information for decision-making.

The study of Chen Li and Maria Gonzalez [8] strengthened the foundation of BREWINSIGHT BI by demonstrating that Business Intelligence technologies could provide practical benefits for small and medium enterprises. Their findings confirmed that centralized data management, automated reporting, and analytical tools could improve efficiency, increase information accessibility, and support evidence-based planning. These concepts aligned with the objectives of the proposed system, which aimed to enhance Kape Tubong's ability to monitor operations, analyze revenue patterns, and make informed decisions through the effective use of business data.

Johnson and Lee conducted the study entitled "Business Intelligence Dashboards and Organizational Performance in Small Enterprises" published in the ***Journal of Information Systems and Technology Management*** [9]. The research examined the influence of dashboard-based Business Intelligence systems on operational efficiency and managerial decision-making among small and medium enterprises. The study focused on how centralized reporting platforms and visual analytics could improve organizational performance, particularly among businesses with limited technological resources and workforce capacity. The researchers investigated small retail and food-service organizations that implemented dashboard solutions to monitor daily operations and evaluate business performance. Data were gathered through surveys, interviews, and comparisons of operational reports before and after the adoption of Business Intelligence dashboards.

The findings revealed that centralized dashboards improved the accuracy and accessibility of information related to sales performance, customer activities, and inventory operations compared with traditional manual reporting approaches. By presenting data through visual elements such as graphs, scorecards, and performance indicators, dashboards simplified the interpretation of complex business information. Managers were able to monitor operational conditions more efficiently and identify issues such as declining sales, inventory shortages, and inconsistencies among branches without manually reviewing large amounts of spreadsheet-based records. Centralized information access improved communication among management personnel and supported faster coordination when addressing operational concerns. As a result, organizations were able to make more timely and data-supported decisions in response to changing business conditions.

The research further emphasized that Business Intelligence dashboards strengthened forecasting and planning activities through historical data analysis. Managers were able to recognize recurring sales patterns, seasonal changes, and customer demand trends that supported improved operational preparation. These insights contributed to better inventory allocation, workforce scheduling, and promotional planning. Predictive capabilities within analytical dashboards helped organizations anticipate periods of increased demand and minimize possible operational disruptions. By understanding product performance and resource requirements, businesses were able to improve productivity, control unnecessary expenses, and allocate resources more effectively.

The study also highlighted that dashboard systems reduced dependence on technical personnel by allowing managers and business owners to access and analyze operational information independently. User-friendly interfaces enabled non-technical users to explore reports, monitor performance indicators, and generate insights without requiring specialized analytical skills. Automated reporting processes reduced errors associated with manual data entry and report preparation while decreasing the time required to compile business information. Through real-time access to

updated performance data, organizations were able to recognize changes in sales conditions more quickly and implement corrective actions when necessary. This improved operational responsiveness and allowed employees to focus more on service improvement and business development activities.

Business Intelligence dashboards also contributed to improved customer service and strategic planning by providing deeper understanding of consumer behavior. Analysis of transaction records allowed organizations to identify purchasing patterns, preferred products, and peak customer activity periods. These insights supported the development of more effective marketing strategies, promotional activities, and product offerings aligned with customer expectations. Continuous monitoring of performance indicators allowed businesses to respond proactively to market changes and operational challenges. The integration of dashboard technologies therefore supported long-term organizational sustainability and competitiveness.

The research further explained that visualized business information encouraged collaboration among different organizational units. Since performance reports were generated from a centralized data source, managers and employees could evaluate objectives using consistent information. Shared access to performance dashboards improved transparency, strengthened accountability, and supported coordinated decision-making among administrators and operational teams. The use of Business Intelligence technologies promoted a culture where business decisions were based on accurate and timely information rather than assumptions or delayed manual reports.

The findings of Johnson and Lee [9] were highly relevant to the development of BREWINSIGHT BI: Revenue Pattern Analytics System. The study supported the integration of centralized dashboards and self-service analytics features to address the operational challenges experienced by Kape Tubong. Problems related to branch data consolidation, sales monitoring, and performance evaluation were similar to the concerns discussed in the research. By implementing dashboard-based analytics, BREWINSIGHT BI improved access to business information, strengthened sales monitoring, and provided management with clearer visibility across different kiosk locations. Automated reporting and centralized data processing reduced delays in information preparation while improving the reliability of sales records.

The visual analytics features of BREWINSIGHT BI supported the identification of revenue trends, customer purchasing behaviors, and inventory requirements. These capabilities helped managers evaluate branch productivity, compare sales performance, and recognize operational concerns requiring immediate attention. Forecasting and trend analysis features provided valuable support for inventory distribution, employee scheduling, and promotional decisions based on historical demand patterns. Through improved planning and resource allocation, the system contributed to reducing operational inefficiencies, preventing stock-related issues, and enhancing customer service quality.

The concept of self-service analytics presented in the study also aligned with the objectives of BREWINSIGHT BI. The system allowed branch administrators and authorized users to access dashboards and interpret business information without requiring advanced technical knowledge. Reducing dependence on manual reporting improved operational efficiency and accelerated access to important business insights. The adoption of a centralized Business Intelligence platform strengthened communication among branches, improved management coordination, and

supported more effective strategic planning. Through these capabilities, BREWINSIGHT BI provided Kape Tubong with a technological solution that enhanced operational management and supported sustainable business growth.

Garcia and Torres conducted the study entitled "Sales Analytics Systems for Food Service Enterprises" published in the *International Journal of Hospitality and Food Service Management* [10]. The research examined the role of sales analytics systems in improving operational performance and decision-making among food and beverage enterprises. The study focused on how digital analytics platforms support businesses in understanding customer purchasing behavior, monitoring sales activities, and improving operational efficiency. The researchers evaluated the effectiveness of analytics-based reporting tools among restaurants, cafés, and beverage establishments that utilized digital systems for tracking transactions, customer activities, and inventory movement. Data were collected through surveys, interviews, and performance evaluations conducted before and after the implementation of analytics solutions.

The findings revealed that businesses using sales analytics systems achieved improved monitoring of revenue performance and faster recognition of changes in product demand. Through visual analytics and automated reporting features, managers were able to identify high-performing products, less profitable offerings, and peak sales periods more effectively compared with traditional manual reporting methods. Interactive dashboards provided a clearer understanding of sales conditions by presenting important business information in organized visual formats. These capabilities reduced the time required to prepare and analyze reports while supporting better decisions regarding product management, pricing strategies, promotional activities, and overall operational planning.

The study further explained that sales analytics systems improved customer service by providing deeper insights into consumer purchasing behaviors and preferences. Transaction data analysis allowed businesses to identify frequently purchased products, preferred purchasing schedules, and customer spending patterns. These insights supported the development of targeted promotions and improved product offerings based on actual customer demand. By understanding customer behavior more accurately, businesses were able to strengthen customer satisfaction, improve retention strategies, and respond more effectively to changing market expectations.

Sales analytics platforms also increased organizational responsiveness by providing direct access to updated operational information and performance indicators. Businesses were able to recognize changes in customer demand, seasonal variations, and unexpected sales fluctuations more quickly. Real-time monitoring allowed managers to evaluate promotional effectiveness, identify declining sales performance, and implement corrective actions without significant delays. This improved the ability of organizations to maintain consistent service quality and adapt to changing operational conditions, especially during periods of high customer activity.

The research highlighted the importance of analytics systems in improving inventory management and reducing operational waste. By analyzing historical sales records and product movement patterns, businesses were able to estimate inventory requirements more accurately and minimize problems related to excess stock or product shortages. Digital platforms improved monitoring of consumption rates and purchasing activities while reducing errors commonly associated with manual recordkeeping. As a result, organizations experienced improved reliability of operational data, better resource utilization, and increased profitability through more efficient inventory practices.

The study also emphasized that digital reporting platforms supported long-term strategic planning by providing management with continuous access to relevant business information. Analysis of historical sales patterns enabled organizations to identify future demand trends and improve decisions related to resource allocation, product planning, and operational strategies. Sales analytics encouraged the development of a data-driven organizational culture where decisions were supported by measurable performance results rather than assumptions. Centralized access to business information also improved communication and coordination among departments by providing a consistent source of operational data.

The findings of Garcia and Torres [10] were directly relevant to the development of BREWINSIGHT BI: Revenue Pattern Analytics System. The study supported the integration of sales analytics, visual reporting, and centralized monitoring features to address the operational requirements of Kape Tubong. Since the business manages multiple branches with separate sales records, the ability to consolidate transaction information into a unified platform is essential for improving performance monitoring and decision-making. Through dashboard-based analytics, BREWINSIGHT BI provides management with clearer visibility of branch performance, product demand, revenue patterns, and operational trends.

The real-time monitoring capabilities discussed in the study aligned with the objectives of BREWINSIGHT BI. Automated dashboards and analytical reports allow Kape Tubong administrators to monitor daily transactions, evaluate product performance, and identify possible operational concerns more efficiently. Forecasting features support better planning of inventory distribution, employee scheduling, and promotional activities based on historical sales information. These capabilities reduce operational inefficiencies, minimize resource waste, and contribute to improved service quality across different branches.

The study's emphasis on customer behavior analysis also supported the design of BREWINSIGHT BI. Understanding purchasing patterns enables management to identify popular products, determine peak transaction periods, and evaluate the effectiveness of business strategies. Centralized data access allows managers and administrators to coordinate operations using accurate and consistent information from all locations. The research confirmed the importance of adopting analytics-driven Business Intelligence solutions in food and beverage enterprises. Through the implementation of BREWINSIGHT BI, Kape Tubong can strengthen operational management, improve decision-making processes, and support sustainable business growth through effective use of sales data.

Martinez and Cooper conducted the study entitled "Interactive Business Intelligence Systems and Decision Support in Food Retail Operations" published in the *International Journal of Digital Commerce and Analytics* [11]. The research examined how interactive Business Intelligence systems contribute to improved operational management and decision-making processes within food retail and beverage organizations. The study focused on the ability of dashboard-based platforms to assist managers in interpreting business information, monitoring operational activities, and responding effectively to changing market conditions. The researchers evaluated organizations that utilized centralized dashboards for tracking sales performance, customer behavior, and branch-level operations. Findings showed that businesses implementing interactive dashboard systems achieved improved efficiency in operational monitoring and reporting compared with organizations that relied primarily on spreadsheet-based processes. Centralized dashboards improved access to business information by converting complex operational records into understandable visual representations.

The study highlighted that interactive dashboards supported deeper business analysis through various visualization features, including graphs, trend indicators, heatmaps, comparative charts, and performance metrics. These analytical tools allowed managers to identify high-performing products, declining sales areas, customer demand changes, and operational concerns more effectively than traditional reporting methods. Visual presentation of information improved the interpretation of business data by organizing complex records into clear and meaningful formats. Interactive features such as filtering and drill-down analysis enabled users to examine performance according to specific branches, product categories, and time periods. This flexibility improved the accuracy and depth of business evaluation while allowing organizations to obtain more valuable insights from existing operational data.

The findings further demonstrated that organizations using interactive Business Intelligence platforms developed stronger responsiveness toward operational challenges and market changes. Near-real-time updates of performance indicators allowed managers to quickly recognize sales declines, changes in customer purchasing behavior, and possible inefficiencies among branches. Continuous access to updated information supported proactive management by enabling decision-makers to address concerns before they significantly affected business performance. The ability to monitor operations consistently improved organizational adaptability and allowed businesses to implement corrective strategies more efficiently.

The research also emphasized the role of dashboard systems in improving communication and collaboration among management teams. Interactive reports provided a common source of information for evaluating branch performance, discussing operational issues, and developing business strategies. Since reports were generated from centralized data, differences in interpretation caused by manually prepared documents were reduced. Improved transparency allowed managers to monitor activities through a unified platform while strengthening coordination among different organizational units. Centralized oversight contributed to more consistent performance evaluation and better alignment of operational objectives.

Another important finding of the study was the contribution of self-service analytics in reducing dependence on technical personnel. User-friendly dashboard interfaces allowed managers with limited technical backgrounds to generate reports, explore business information, and analyze performance independently. This capability provided significant advantages for small and medium enterprises where dedicated data analysts or information technology specialists might not always be available. Simplified access to analytics encouraged regular system utilization and strengthened confidence among users in applying data-based insights during daily decision-making activities.

Interactive Business Intelligence systems also supported strategic forecasting and long-term business planning. By analyzing historical transaction records and demand patterns, organizations were able to improve decisions related to inventory management, workforce scheduling, promotional activities, and product development. Businesses using dashboard platforms were better prepared to anticipate seasonal changes and adjust operational strategies based on measurable trends. Improved resource planning reduced issues related to excessive inventory, insufficient staffing, and ineffective promotional timing. The adoption of centralized analytical platforms supported digital transformation initiatives within food retail and beverage industries.

The findings of Martinez and Cooper [11] were highly relevant to the development of BREWINSIGHT BI: Revenue Pattern Analytics System. The study provided support for incorporating interactive dashboard analytics into the

proposed platform for Kape Tubong. Existing challenges involving separated sales records, delayed report preparation, and limited visibility across branches were consistent with the operational issues addressed in the research. Through interactive dashboards, BREWINSIGHT BI transformed raw transaction records into organized visual insights that improved revenue monitoring and business evaluation. The system enabled management to identify high-performing products, compare branch performance, analyze sales movements, and recognize operational concerns through centralized reporting features.

The self-service analytics capabilities discussed in the study also aligned with the usability objectives of BREWINSIGHT BI. The platform was designed to allow authorized users and administrators to access reports, evaluate performance trends, and generate insights without requiring advanced technical knowledge. Centralized access supported remote monitoring of multiple Kape Tubong branches, allowing management to oversee operations even when they were not physically present at specific locations. This improved coordination, strengthened supervision, and allowed faster responses to branch-level concerns.

The findings of Martinez and Cooper [11] provided significant theoretical and practical support for the implementation of BREWINSIGHT BI. Interactive Business Intelligence technologies improved operational monitoring, reporting efficiency, forecasting capability, and evidence-based decision-making within food retail environments. The study served as an important reference confirming the relevance of integrating dashboard analytics, centralized reporting, and decision-support features into the proposed system. Through these capabilities, BREWINSIGHT BI can provide Kape Tubong with a scalable solution for improving business oversight, operational efficiency, and future organizational growth.

2.2 Local Studies

Reyes and Santos conducted the study entitled “Digital Transformation in Philippine Micro-Enterprises: A Study on BI Adoption in the F&B Sector” published in the *Philippine Review of Business and Economics* [12]. The research examined the effectiveness of Business Intelligence (BI) dashboards compared with traditional manual approaches for monitoring sales performance and operational activities among food and beverage enterprises in the Philippines. The study focused on how dashboard-based BI solutions contribute to improved efficiency, accuracy, and responsiveness in business decision-making. Data were gathered from micro, small, and medium food service enterprises that implemented digital analytics platforms for monitoring sales transactions, product performance, and customer-related activities. The findings showed that organizations adopting BI dashboards significantly reduced the time required for analyzing sales and operational information compared with businesses that relied mainly on manual recordkeeping and spreadsheet-based reporting.

The study revealed that dashboard-based systems improved the ability of food and beverage businesses to monitor daily sales activities, evaluate product performance, and respond to changes in customer demand. Through centralized visual reporting, managers gained faster access to important operational information without requiring extensive manual data consolidation. Dashboard features such as charts, trend indicators, and comparative visualizations helped business owners recognize sales movements, identify high-performing products, and evaluate performance differences among branches or business locations. The simplified presentation of information allowed users with limited technical

knowledge to interpret business data more effectively and make timely decisions based on current operational conditions.

The researchers further explained that centralized dashboards improved management efficiency by consolidating different sources of business information into a single platform. Instead of depending on administrative personnel to prepare reports, managers were able to independently access and analyze performance data through user-friendly interfaces. Standardized reporting structures reduced inconsistencies commonly associated with manual report preparation and improved transparency within organizations. By maintaining consistent performance indicators, businesses were able to evaluate operations more accurately, strengthen accountability, and improve coordination among management teams.

The study also emphasized that BI dashboards increased organizational adaptability in competitive business environments. Immediate access to analytical information allowed decision-makers to evaluate the effectiveness of pricing strategies, promotional activities, and operational adjustments. Businesses were able to respond more effectively to changing customer preferences, seasonal demand variations, and market pressures because decisions were supported by updated performance data rather than assumptions. The availability of historical sales records within BI platforms also strengthened forecasting capabilities by allowing organizations to identify recurring patterns and prepare better strategies for future business conditions.

Another significant finding of the research was the contribution of BI systems toward digital transformation among Philippine micro, small, and medium enterprises. The authors highlighted that dashboard technologies provide accessible solutions for organizations seeking to improve operations through data-driven management practices. User-friendly interfaces and automated analytical functions allow businesses to utilize advanced reporting capabilities without requiring highly specialized technical personnel or complex infrastructure. The adoption of BI systems therefore supports not only operational improvement but also long-term competitiveness and organizational sustainability.

The researchers further identified the importance of BI dashboards in preserving historical records, improving organizational transparency, and strengthening communication among business teams. Structured storage of sales information allowed organizations to analyze past performance, identify trends, and support future planning activities. Consistent measurement of business indicators improved comparisons among branches and departments, while accessible information sharing encouraged more objective discussions and evidence-based decisions. These capabilities contributed to stronger coordination, improved accountability, and a more effective decision-making environment.

The findings of Reyes and Santos [12] provided strong local support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. Similar to the enterprises examined in the study, Kape Tubong experiences challenges related to separated sales records, delayed reporting processes, and limited visibility into branch-level performance. The proposed system addresses these concerns by integrating sales transactions from different kiosks into a centralized dashboard platform. Through this approach, management can monitor revenue performance, analyze product results, compare branch contributions, and access operational information more efficiently.

The study also supported the inclusion of visualization tools, historical analysis, and automated reporting features within BREWINSIGHT BI. Through dashboard-based analytics, Kape Tubong management can identify revenue trends, determine high-performing products, evaluate branch productivity, and make decisions using accurate and timely information. The forecasting capabilities discussed in the research align with the objectives of the proposed system by supporting inventory planning, workforce scheduling, and promotional strategies based on historical sales patterns and customer demand.

The research of Reyes and Santos [12] served as valuable local evidence demonstrating the practicality and effectiveness of Business Intelligence solutions for food and beverage enterprises in the Philippines. Their findings reinforced the importance of implementing BREWINSIGHT BI as a technological solution for improving operational efficiency, strengthening data-driven decision-making, and supporting the continued growth and sustainability of Kape Tubong.

Villanueva, in the study entitled "From Intuition to Evidence: The Digital Maturity of Food Service MSMEs in Western Visayas" published in the *Philippine Journal of Science and Technology Management*, examined the transition of food service micro, small, and medium enterprises (MSMEs) from intuition-based management practices toward data-driven decision-making through digital technologies [13]. The study emphasized that many Filipino food service entrepreneurs possess strong knowledge of local markets and customer preferences; however, business expansion creates challenges in maintaining direct managerial supervision across multiple branches. These challenges include maintaining quality control, ensuring inventory consistency, and monitoring revenue performance effectively.

The findings indicated that businesses adopting Business Intelligence (BI) tools experienced improved operational consistency, particularly among enterprises managing multiple locations. Villanueva identified localized data awareness as an important factor in business growth, referring to the ability of managers to understand branch-specific customer preferences, purchasing behaviors, and peak operating periods. Instead of applying identical strategies across all branches, businesses using localized analytics were able to adjust their operations based on the specific needs and behaviors of each market area. This capability was particularly valuable in the competitive and diverse Philippine food service industry.

The study also discussed the concept of operational leakage, which refers to potential losses caused by inaccurate recording of inventory, sales transactions, and cash-related activities commonly associated with manual and paper-based systems. Villanueva explained that centralized dashboards improve transparency and accountability by enabling continuous monitoring of operational information. Through digital tracking of sales and inventory records, businesses can improve data accuracy, strengthen compliance with procedures, and reduce inconsistencies associated with manual reporting practices.

The research further highlighted that many MSMEs in Western Visayas remain in the early stages of digital maturity, with several organizations still depending on handwritten records, spreadsheets, and verbal reporting methods. While these approaches may be effective for small single-location businesses, they become increasingly inefficient as enterprises expand. The absence of integrated digital systems may result in delayed communication, inconsistent reporting, and limited managerial visibility, making it more difficult for business owners to maintain service quality and monitor branch performance.

Villanueva further explained that Business Intelligence systems support not only data consolidation but also organizational learning and strategic adaptability. Through dashboard visualizations and automated reporting tools, managers can identify patterns and trends that may not be immediately visible through traditional reporting methods. Visual analytics transform large volumes of transactional information into meaningful insights that support product evaluation, demand analysis, promotional assessment, and operational planning. Historical sales records also provide valuable information for forecasting activities related to inventory management, workforce scheduling, and promotional strategies.

Another significant finding emphasized the importance of real-time operational visibility in improving managerial responsiveness. Delayed or incomplete reports may prevent organizations from addressing operational concerns before they become more serious. Centralized dashboards serve as monitoring tools by providing access to sales trends, branch performance indicators, and inventory movements. Organizations using integrated BI systems demonstrated improved ability to respond to operational changes and maintain consistency during periods of business expansion. Increased transparency also encouraged stronger accountability among employees by promoting standardized monitoring procedures.

The study also emphasized that digital transformation should be viewed as an organizational process rather than simply a technological improvement. Successful BI adoption requires managers and employees to develop a data-driven culture where decisions are supported by analytical information rather than relying only on assumptions or personal experience. Businesses that incorporated analytics into their daily operations showed improved adaptability, resilience, and competitiveness in responding to changing customer preferences and market conditions.

The findings of Villanueva [13] provide valuable local support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. The study is closely aligned with the objectives of the proposed system, particularly in centralized sales monitoring, branch performance evaluation, operational transparency, and data-driven decision-making. It supports the integration of Point-of-Sale (POS) systems with Business Intelligence dashboards to improve managerial oversight and operational efficiency. The research also reinforces the importance of automated reporting and visualization tools in transforming transactional data into meaningful business insights, highlighting the value of adopting a centralized analytics platform to support sustainable growth and long-term competitiveness within the Philippine food service industry.

Garcia Navarro and Cruz, in their study entitled “Implementation of Business Intelligence Systems in Philippine Small Food Enterprises” published in the *Asian Journal of Information Technology and Business Management*, examined how Business Intelligence (BI) solutions influence operational performance, sales monitoring, and managerial decision-making among small and medium-sized food service enterprises in the Philippines [14]. The study focused on how centralized BI platforms improve the management of sales transactions, inventory information, and operational performance indicators. The findings revealed that businesses adopting BI dashboards experienced improvements in the accuracy, accessibility, and efficiency of sales reporting compared with organizations that continued using manual procedures and spreadsheet-based methods. By integrating sales records, inventory information, and performance data from multiple sources into a centralized platform, businesses gained better visibility of their overall operations and reduced delays caused by manual data consolidation.

The study emphasized that BI dashboards improved the ability of managers to identify sales trends, monitor product demand, and evaluate branch performance through visual analytics and automated reporting. The use of charts, graphs, and summarized performance indicators allowed decision-makers to understand complex business information more effectively and recognize operational concerns such as declining sales, inventory shortages, and inconsistent branch performance. Through immediate access to organized information, managers were able to respond more quickly to operational challenges and make decisions based on reliable business data. The researchers also highlighted that automated reporting reduced administrative workload by minimizing repetitive data preparation tasks and decreasing the possibility of errors associated with manual report generation.

The researchers further explained that centralized BI systems strengthened transparency and accountability within organizations by establishing standardized performance measurements across branches and departments. Consistent reporting structures allowed managers to compare business performance more accurately and evaluate operational results using objective information. This improved coordination among different business units and supported more effective resource allocation. The study also found that organizations using BI platforms demonstrated greater adaptability because they were able to analyze customer purchasing behavior, sales history, and operational trends when developing strategies and responding to market changes.

The study also highlighted the role of Business Intelligence systems in supporting digital transformation among Philippine small and medium enterprises. Through historical data analysis, organizations were able to identify recurring sales patterns, forecast future demand, and improve inventory planning. The researchers emphasized that affordable and user-friendly BI solutions provide practical opportunities for growing businesses to modernize their operations without requiring highly complex technological infrastructure. Centralized dashboards were identified as valuable tools for improving managerial control, operational monitoring, and strategic planning, especially among businesses operating in multiple locations.

The findings of Garcia Navarro and Cruz [14] provided significant support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. Similar to the enterprises discussed in the study, Kape Tubong experiences challenges related to fragmented sales records, delayed report preparation, and limited visibility into kiosk-level performance. The proposed system addresses these concerns by consolidating sales transactions and operational information from different branches into a centralized Business Intelligence platform. Through dashboard visualization, automated reporting, and analytical features, BREWINSIGHT BI enables management to monitor revenue trends, evaluate product performance, compare branch operations, and make more informed business decisions.

The study further supports the integration of BI dashboards, historical trend analysis, and centralized reporting features within BREWINSIGHT BI. By transforming raw transactional data into meaningful business insights, the system can assist Kape Tubong in improving operational efficiency, strengthening transparency, and supporting data-driven decision-making. The research demonstrates the relevance of adopting Business Intelligence solutions for Philippine food service enterprises seeking to improve management practices, enhance business performance, and achieve sustainable growth.

Lopez and Ramirez, in their study entitled "Centralized Sales Monitoring Systems and Business Performance among Philippine Café Enterprises" published in the *Journal of Philippine Management and Information Systems*, examined the effects of centralized sales analytics platforms on operational productivity and managerial decision-making among café and beverage businesses in the Philippines [15]. The research focused on how dashboard-based monitoring systems improve sales management practices, performance evaluation, and organizational adaptability in response to changing customer preferences and market conditions. The study investigated the role of centralized information systems in improving access to sales data, strengthening operational monitoring, and supporting more effective business decisions among small and medium-sized enterprises.

The findings revealed that businesses adopting centralized dashboard systems experienced improvements in reporting efficiency and daily operational monitoring compared with organizations that depended on manual reporting methods. Digital platforms reduced the time and effort required to collect, organize, and consolidate sales information while improving the consistency and reliability of performance evaluation across different locations. By providing managers with access to sales information through a unified interface, centralized systems allowed businesses to monitor branch activities more effectively, identify operational concerns, and recognize opportunities for improvement in a timely manner.

The study emphasized the importance of dashboard visualization in improving the interpretation of business information. Interactive charts, trend indicators, and summarized analytics enabled managers to identify sales patterns, monitor product demand, and recognize peak customer activity periods more efficiently than traditional reports. These visual tools simplified complex sales information and allowed decision-makers to develop better strategies regarding inventory management, employee scheduling, promotional activities, and daily operational adjustments. Through improved understanding of business performance, organizations were able to make decisions based on accurate and organized data.

The researchers further explained that centralized sales monitoring systems improved organizational responsiveness and coordination. Access to updated sales and operational information allowed managers to respond quickly to unexpected revenue changes, inventory concerns, and branch performance issues. The use of standardized performance indicators also strengthened communication between branch supervisors and management teams by ensuring that all departments relied on consistent measurements and reporting structures. This improved alignment between operational activities and organizational goals.

Lopez and **Ramirez** also highlighted that centralized analytics platforms encouraged the adoption of data-driven management practices. Instead of relying primarily on assumptions or personal judgment, businesses were able to use historical sales information and analytical reports to support strategic planning. The availability of stored transaction records and forecasting capabilities helped organizations identify recurring trends, evaluate promotional effectiveness, and anticipate future customer demand. The researchers noted that user-friendly analytics platforms were particularly beneficial for small and medium enterprises because they allowed managers with limited technical knowledge to utilize advanced reporting and analytical tools effectively.

The findings of Lopez and **Ramirez** [15] provided strong support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. Similar to the businesses examined in their research, Kape

Tubong experiences challenges related to separated sales records, delayed data consolidation, and limited visibility into individual kiosk performance. The proposed system addresses these concerns by integrating transactional and operational information from multiple branches into a centralized Business Intelligence platform. Through dashboard monitoring, automated reporting, and analytical features, BREWINSIGHT BI enables management to evaluate revenue trends, compare branch performance, monitor product demand, and access timely business information.

The study also supports the inclusion of visualization tools and forecasting features within BREWINSIGHT BI. By transforming raw sales records into meaningful analytical insights, the system assists Kape Tubong management in improving operational efficiency, optimizing resource allocation, and making evidence-based decisions. The research demonstrates the value of centralized sales monitoring systems in Philippine café and beverage enterprises and reinforces the importance of implementing a Business Intelligence solution to support business growth, operational control, and long-term strategic planning.

Fernandez and Gutierrez, in their study entitled "Revenue Analytics and Dashboard Utilization among Philippine Small Food Businesses" published in the *Philippine Journal of Information Technology and Business Analytics*, examined the effects of revenue analytics systems and digital dashboard platforms on operational management and strategic decision-making among small and medium-sized food enterprises in the Philippines [16]. The study focused on food businesses such as coffee shops, milk tea establishments, snack centers, and food kiosks that adopted digital analytics solutions to replace traditional manual sales monitoring practices. The research aimed to determine how centralized dashboard systems improve revenue tracking, product performance evaluation, branch monitoring, and overall business management among small and medium enterprises.

The findings revealed that businesses utilizing dashboard-based analytics systems experienced significant improvements in operational visibility and management efficiency. By automatically consolidating transaction records from different business locations into a centralized platform, these systems reduced delays associated with manual data collection and report preparation. The availability of organized and accurate sales information allowed managers to monitor overall business performance more effectively and make operational decisions based on reliable data rather than incomplete records or assumptions. Centralized analytics improved the consistency of sales monitoring and strengthened the reliability of business reports.

The researchers emphasized the importance of visual analytics in helping managers understand revenue patterns and customer purchasing behavior. Through dashboards containing graphs, trend analysis, and interactive reports, decisionmakers were able to identify changes in sales performance, evaluate product demand, recognize peak operating periods, and compare branch productivity more efficiently. These visual tools simplified the interpretation of complex transactional information and allowed businesses to respond more quickly to operational concerns and market changes. The improved accessibility of analytical information supported faster and more informed decision-making.

The study further highlighted that dashboard systems strengthened branch monitoring and organizational accountability through standardized performance indicators. By applying consistent measurements across different locations, managers were able to compare branch results objectively and evaluate operational performance more accurately. Real-time access to revenue information also improved business responsiveness by allowing management

to make timely adjustments related to inventory management, employee scheduling, pricing strategies, and promotional activities. This capability helped organizations maximize resource utilization and improve daily operational control.

Fernandez and Gutierrez also explained that revenue analytics systems supported strategic planning through the analysis of historical business data. By identifying recurring sales patterns and customer demand trends, organizations were able to improve forecasting, inventory preparation, workforce allocation, and marketing decisions. The researchers noted that user-friendly dashboard interfaces allowed managers with limited technical knowledge to utilize Business Intelligence tools effectively, making these systems practical solutions for small and medium-sized enterprises seeking to improve their operations through data-driven management.

The findings of Fernandez and **Gutierrez** [16] provided valuable support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. Like the businesses examined in the study, Kape Tubong experiences challenges related to scattered sales records, delayed report preparation, and limited visibility into kiosk performance. The proposed system addresses these limitations by centralizing transactional and operational information into a unified Business Intelligence platform. Through dashboard visualization, automated reporting, and analytical features, BREWINSIGHT BI enables management to monitor revenue trends, evaluate product performance, compare branch results, and analyze future business patterns.

The study also validates the integration of visual reporting, historical trend analysis, and automated analytics within BREWINSIGHT BI. By transforming raw sales transactions into meaningful business insights, the system supports improved decision-making, operational efficiency, and strategic planning for Kape Tubong. The research demonstrates that centralized revenue analytics platforms are practical and beneficial solutions for Philippine food enterprises seeking to strengthen management control, improve performance monitoring, and support sustainable business growth.

2.3 Related Literature

Existing literature in the fields of business operations, information systems, and food and beverage management consistently emphasized the importance of data-driven decision-making in improving organizational efficiency and performance. Researchers have highlighted that businesses, particularly small and medium enterprises (SMEs), benefit from technologies that provide accurate and timely access to information regarding sales activities, customer behavior, and operational conditions. Unlike traditional management approaches that rely primarily on experience, assumptions, or manual evaluation, data-driven systems enable organizations to make decisions based on measurable evidence. Through the integration of historical and current business data, organizations can improve forecasting accuracy, optimize resource allocation, and develop more effective strategies. Studies further indicated that businesses utilizing analytical technologies are better prepared to respond to changing market conditions, shifting consumer preferences, and increasing competitive pressures. In industries such as food and beverage, where customer demand can change rapidly, the ability to access reliable operational insights is essential for maintaining efficiency and sustainability. Business Intelligence (BI) technologies support this process by transforming raw transactional information into meaningful insights that assist managers in evaluating performance and identifying improvement opportunities.

Researchers also emphasized that data-driven management contributes to stronger organizational transparency and accountability by establishing measurable performance indicators. Through analytical systems, organizations can monitor operational results, evaluate productivity, and identify areas requiring improvement. The availability of structured business information allows managers to detect potential problems at an earlier stage and implement corrective actions before these issues significantly affect performance. Literature suggested that organizations adopting analytical approaches develop a culture of continuous improvement because decisions and strategies are regularly evaluated using objective performance measurements. The increasing availability of cloud computing and digital analytics platforms has further encouraged the adoption of BI solutions among SMEs by making advanced analytical capabilities more accessible and affordable. As a result, business intelligence systems are increasingly recognized not merely as technological enhancements but as essential management tools that support operational control, strategic planning, and long-term competitiveness.

A frequently discussed concept in related literature is the use of interactive dashboards as a method for presenting complex business information in a clear and understandable manner. Dashboard-based BI systems allow organizations to visualize important performance indicators through charts, graphs, summaries, and other graphical representations. Researchers have noted that these visual tools simplify the interpretation of large amounts of data by presenting information in a format that is easier for users to understand compared with traditional reports and spreadsheets. Dashboards are particularly valuable for managers who require immediate access to business information but may not have advanced technical knowledge or data analysis expertise. By providing centralized access to important metrics, dashboards allow decision-makers to monitor operational conditions, identify trends, and evaluate performance more efficiently.

Studies further revealed that dashboard systems improve managerial responsiveness by enabling real-time monitoring of critical business activities. Instead of waiting for manually prepared reports, managers can directly access updated information regarding sales performance, inventory movement, customer activity, and operational outcomes. This capability reduces delays in decision-making and allows organizations to respond quickly to emerging concerns. Interactive dashboards also support deeper analysis through features such as filtering, comparisons, and detailed performance views, allowing users to examine specific areas of business operations. Researchers have also emphasized that dashboards improve communication among stakeholders because they provide standardized information presented through consistent measurements and visual formats. This shared understanding reduces inconsistencies in reporting and strengthens coordination between different levels of management. Consequently, literature identifies dashboards as important components of modern BI systems because they improve information accessibility, operational visibility, and decision-making efficiency.

Another important area discussed in existing literature is sales and revenue analytics, which focuses on analyzing business transactions to understand revenue patterns and financial performance. Researchers have identified revenue analytics as a valuable practice for organizations seeking to evaluate sales trends, monitor income generation, and improve operational planning. Through analytical systems, businesses can examine sales records according to different periods, locations, products, and customer behaviors. These insights allow management to determine factors affecting revenue changes and develop strategies based on actual business performance rather than assumptions. In food and beverage enterprises, revenue analytics is particularly useful because customer demand can be influenced by seasonal changes, promotional activities, local events, and consumer preferences.

Historical sales analysis has also been recognized as an essential function of revenue analytics because it allows businesses to identify recurring patterns and predict future demand. By examining previous transaction records, organizations can determine periods of increased customer activity, evaluate the effectiveness of promotional strategies, and prepare appropriate operational responses. Historical information supports better decisions regarding inventory purchasing, workforce scheduling, and marketing activities. Researchers further explained that automated revenue analysis reduces the limitations associated with manual reporting, such as delayed information gathering, calculation errors, and inconsistent data preparation. By automatically processing sales information and presenting meaningful results, BI systems allow managers to focus more on strategic analysis rather than administrative tasks. Therefore, revenue analytics plays an important role in improving financial visibility, forecasting accuracy, and overall business performance.

Product performance analysis is another significant concept emphasized in business intelligence and food service management literature. Researchers have identified product-level analytics as an important method for evaluating customer demand, profitability, and the overall contribution of individual products to business performance. In food and beverage businesses, understanding product performance is essential because customer preferences frequently change due to market trends, seasonal factors, pricing strategies, and promotional activities. Through detailed sales analysis, organizations can identify best-selling products, low-performing items, and products that generate significant revenue. These insights allow management to make informed decisions regarding menu improvements, product development, pricing adjustments, and promotional planning.

Literature further indicated that product analytics supports more effective inventory and resource management by helping businesses determine which items require greater attention and investment. Products with consistently high demand can be prioritized during procurement planning, while slow-moving items can be evaluated to reduce unnecessary costs and minimize waste. Researchers also suggested that analyzing product performance patterns can reveal customer purchasing behaviors, including frequently purchased combinations and seasonal preferences. These insights may support strategies such as product bundling, targeted promotions, and improved customer engagement. In competitive food service environments, continuous monitoring of product performance provides organizations with the ability to adapt quickly to changing consumer expectations and maintain profitability. Therefore, product analytics is considered a valuable component of business intelligence systems designed to support operational improvement and strategic decision-making.

Another area frequently discussed in related literature is peak-hour analysis, which focuses on identifying periods of increased and decreased customer activity. Researchers emphasized that understanding customer traffic patterns is important for improving workforce management, service efficiency, and operational planning. In food and beverage establishments, customer volume often varies throughout the day depending on factors such as work schedules, school activities, holidays, and local events. By analyzing peak periods, businesses can allocate employees more effectively, prepare sufficient inventory, and organize workflows to maintain service quality during high-demand situations.

Studies indicated that peak-hour analysis contributes to improved resource utilization by allowing organizations to align operational activities with actual customer demand. Proper staffing during busy periods can reduce waiting times, improve customer satisfaction, and prevent service delays. Conversely, identifying low-demand periods enables businesses to reduce unnecessary operational expenses by adjusting workforce allocation and scheduling activities

such as maintenance, restocking, or administrative tasks during slower hours. Researchers also noted that analyzing customer activity patterns across different branches allows organizations to compare operational performance and determine location-specific demand behaviors. This information is valuable for businesses managing multiple outlets because it supports more accurate planning and enables management to implement strategies based on actual branch conditions rather than general assumptions.

Inventory monitoring and management systems have also been widely examined in literature because of their significant role in improving operational efficiency, particularly in food service environments. Researchers emphasized that effective inventory management helps organizations maintain appropriate stock levels, minimize waste, and ensure the continuous availability of products and ingredients. This is especially important for cafés, restaurants, and food kiosks where many materials are perishable and require careful monitoring. Traditional manual inventory processes are often associated with delays, inaccurate records, and difficulty in tracking consumption patterns. In contrast, digital inventory systems provide more accurate and timely information regarding stock movement, usage rates, and replenishment requirements.

Literature indicated that automated inventory monitoring reduces human error by minimizing dependence on manual counting and spreadsheet-based tracking. By maintaining organized records of inventory transactions, businesses can identify purchasing inefficiencies, detect unusual stock changes, and improve supplier coordination. Researchers further explained that inventory analytics supports better decision-making by allowing organizations to predict future stock requirements based on historical consumption patterns and sales performance. This capability helps prevent both overstocking and stock shortages, which directly affect operational costs and customer satisfaction. In addition, inventory monitoring improves accountability by creating transparent records of stock movement and reducing the possibility of unauthorized usage or inaccurate reporting. Consequently, inventory management systems are recognized as essential tools for improving cost control, operational reliability, and service continuity in food and beverage businesses.

The integration of inventory analytics with other business intelligence functions further strengthens organizational decision-making. When inventory information is combined with sales, product, and revenue data, managers gain a more complete understanding of how operational resources contribute to business performance. This integrated perspective allows organizations to determine which products generate higher demand, which ingredients require frequent replenishment, and how purchasing decisions should be adjusted according to actual sales patterns. Researchers emphasized that combining different operational data sources through analytical platforms enables businesses to move from reactive management toward proactive planning. Therefore, inventory monitoring within a BI environment serves not only as a stock control mechanism but also as a strategic tool for improving efficiency and supporting sustainable business growth.

Literature on automation and cloud-based technologies has also emphasized their importance in improving the efficiency, accessibility, and reliability of modern business information systems. Researchers have explained that automated processes reduce the time and effort required for repetitive administrative activities, particularly those involving data collection, report preparation, and information consolidation. In traditional business environments, manual encoding and spreadsheet-based reporting often require significant human involvement and may result in delays, inconsistencies, and data entry errors. Through automation, organizations can generate reports more efficiently,

improve data accuracy, and allow managers to focus more on analysis and strategic decision-making rather than routine administrative tasks.

Cloud-based platforms have further contributed to the accessibility and scalability of business intelligence systems. Unlike traditional systems that depend heavily on physical infrastructure and localized data storage, cloud-based solutions allow users to access business information through internet-enabled devices from different locations. Researchers have highlighted that this capability is particularly beneficial for organizations managing multiple branches because managers can monitor operations without being physically present at each location. Cloud-based systems also support organizational growth by allowing additional users, branches, and functions to be integrated without requiring major system restructuring. This flexibility makes digital analytical platforms more practical for small and medium enterprises seeking to improve operational control while managing limited technological resources.

Studies have also shown that cloud-based business systems strengthen collaboration by ensuring that users access consistent and updated information. Centralized storage allows different departments and management levels to work with the same set of records, reducing discrepancies caused by separate files or independently maintained reports. In addition, automated backup features and improved data security mechanisms reduce the risks associated with information loss and unauthorized access. Researchers further noted that cloud technologies support digital transformation by reducing dependence on paper-based processes and encouraging organizations to adopt more organized and efficient methods of managing information. As a result, automation and cloud integration are increasingly recognized as important drivers of productivity, scalability, and operational modernization.

Another significant concept discussed in related literature is centralized data management, which refers to the consolidation of business information from multiple sources into a unified system. Researchers emphasized that centralized data environments improve organizational control by ensuring that all operational units rely on consistent and standardized information. In businesses with multiple branches or locations, decentralized records and separate spreadsheets often create difficulties in comparing performance, verifying information accuracy, and generating comprehensive reports. Centralized systems address these challenges by maintaining structured databases where transactional records, performance indicators, and operational information can be accessed from a single platform.

Centralized data management improves reporting efficiency by reducing duplication, minimizing inconsistencies, and simplifying the retrieval of historical information. Researchers noted that organizations using centralized systems are better able to conduct comparative analysis because information from different branches can be evaluated using the same performance standards. This capability is particularly valuable for businesses operating multiple outlets because management can identify differences in revenue contribution, operational efficiency, and customer demand among locations. Furthermore, centralized databases strengthen data governance by supporting standardized procedures for data entry, validation, and reporting, resulting in more reliable organizational information.

Literature also highlighted that centralized systems improve communication and coordination among management teams. When all users have access to synchronized operational information, discussions and decisions become more objective because they are based on shared evidence rather than conflicting reports. Researchers suggested that centralized data environments promote greater accountability because business outcomes can be measured through clearly defined indicators. Additionally, these systems support scalability by allowing organizations to incorporate

additional branches, departments, or operational processes without significantly changing the existing structure. Therefore, centralized data management is widely considered a foundational element of effective business intelligence implementation.

Branch performance evaluation is another important topic emphasized in studies involving multi-location businesses. Researchers identified branch-level analytics as a valuable approach for comparing operational performance, revenue generation, and productivity across different locations. Through branch performance monitoring, organizations can determine which locations are achieving expected results and which require improvement strategies. This information allows management to implement targeted interventions based on specific branch conditions rather than applying generalized solutions across the entire organization.

In food and beverage enterprises, branch analytics can reveal differences in customer behavior, sales patterns, operational efficiency, and resource utilization among locations. Researchers explained that comparative branch analysis supports better decision-making by allowing managers to identify successful practices from high-performing branches and apply them to other locations. It also helps determine factors that contribute to branch success, such as customer traffic, product demand, staffing effectiveness, and promotional performance. Through measurable performance indicators, branch evaluation systems strengthen transparency and accountability by providing clear standards for assessing operational outcomes.

Furthermore, branch performance analytics support strategic planning by providing evidence for expansion decisions and resource allocation. Organizations can analyze branch profitability, identify growth opportunities, and determine areas requiring additional support or adjustment. Researchers emphasized that businesses with access to branch-level insights are better equipped to maintain consistency across multiple locations and respond effectively to operational challenges. Consequently, branch performance evaluation is considered an essential analytical capability for organizations seeking to improve coordination, control, and long-term operational sustainability.

Within the Philippine business environment, literature has increasingly emphasized the importance of localized analytics in understanding consumer behavior and responding to market conditions. Researchers have recognized that businesses, particularly those in the food and beverage sector, operate within communities where purchasing patterns may be influenced by local events, cultural practices, seasonal activities, and consumer preferences. Because of these contextual factors, organizations require analytical systems capable of identifying localized trends rather than relying solely on general performance measurements. Localized analytics enables businesses to understand variations in customer demand across different areas and develop strategies that are more aligned with the specific characteristics of their target markets.

Studies indicated that local factors such as holidays, festivals, school schedules, tourism activities, and community events can significantly influence sales performance in food service establishments. By analyzing these influences, businesses can prepare more effectively through adjustments in inventory planning, staffing arrangements, and promotional strategies. Researchers explained that organizations capable of identifying localized demand patterns are better positioned to respond to sudden changes in customer behavior and maintain operational efficiency. This is particularly relevant for community-based cafés and food kiosks, where customer preferences are often closely connected to local conditions and daily activities.

Literature further suggested that localized analytics contribute to stronger market responsiveness by allowing businesses to move beyond standardized decision-making approaches. Instead of applying identical strategies across all locations, organizations can use analytical insights to customize operations according to branch-specific conditions. This capability improves customer engagement and allows businesses to develop more effective strategies for different market environments. Therefore, incorporating localized and contextual analysis into business intelligence systems provides organizations with a competitive advantage, especially in industries where customer behavior is highly influenced by local factors.

Another important concept emphasized in related literature is real-time reporting, which focuses on providing immediate access to updated operational information. Researchers have identified real-time reporting as a critical feature of modern business intelligence systems because it allows managers to monitor business activities as they occur. Traditional reporting methods often involve delays caused by manual data collection, consolidation, and verification processes. In contrast, real-time reporting enables organizations to access current information regarding sales performance, inventory levels, and operational conditions without waiting for scheduled reports.

Studies demonstrated that real-time visibility improves managerial responsiveness by allowing organizations to identify and address issues immediately. In fast-paced industries such as food and beverage, where customer demand can change within short periods, timely access to information is essential. Real-time reporting allows managers to respond quickly to unexpected sales changes, product shortages, operational disruptions, and variations in customer activity. Researchers further noted that immediate access to accurate information improves decision confidence because actions are based on current business conditions rather than outdated records.

Real-time analytics also strengthen operational control by improving awareness across different business locations. For organizations managing multiple branches, centralized access to live information enables management teams to monitor performance consistently and coordinate responses more effectively. Researchers emphasized that this capability reduces dependence on delayed summaries and supports proactive management practices. As a result, real-time reporting has become an important component of business intelligence systems designed to improve agility, efficiency, and decision-making quality.

Historical trend analysis is another widely discussed concept in literature because of its importance in forecasting and long-term planning. Researchers explained that historical analysis allows organizations to examine previous performance data to identify recurring patterns, seasonal fluctuations, and long-term changes in business behavior. By understanding historical trends, businesses can make more accurate predictions regarding future demand and prepare appropriate operational strategies.

In food and beverage operations, historical sales information provides valuable insights into recurring customer behaviors, high-demand periods, and the effectiveness of previous business strategies. These insights support decisions related to inventory purchasing, workforce scheduling, promotional planning, and resource allocation. Researchers also noted that historical trend analysis allows organizations to evaluate whether business performance is improving, declining, or remaining stable over time. This provides management with measurable evidence for assessing progress and identifying areas requiring adjustment.

Furthermore, historical analytics support organizational learning by allowing businesses to evaluate the outcomes of previous decisions and strategies. By examining past sales results, promotional activities, and operational changes, managers can determine which approaches produced positive results and which require improvement. Literature suggested that organizations with access to historical analytical information are better prepared to engage in proactive management because they can anticipate possible challenges and opportunities before they occur.

Finally, related literature highlighted the role of decision-support systems in improving managerial effectiveness and reducing uncertainty in organizational planning. Decision-support systems combine analytical tools, reporting capabilities, and structured information presentation to assist managers in evaluating alternatives and selecting appropriate actions. Researchers emphasized that these systems reduce dependence on intuition-based management by providing objective evidence that supports business decisions.

Decision-support technologies are particularly valuable for small and medium enterprises because managers often handle multiple responsibilities and require efficient access to summarized operational information. Through integrated analytical platforms, managers can evaluate sales performance, identify operational issues, and develop strategies using reliable information. Studies indicated that decision-support systems improve planning consistency because decisions can be supported by measurable indicators and documented evidence.

Researchers further explained that decision-support systems enhance organizational coordination by ensuring that different departments and management levels operate using the same information foundation. This reduces communication gaps and prevents conflicting interpretations of business performance. Additionally, these systems support faster problem-solving by allowing managers to identify, analyze, and address operational concerns using integrated reports and analytical insights.

Overall, literature has consistently recognized business intelligence, analytical dashboards, centralized data management, and decision-support technologies as essential tools for improving organizational performance. These technologies enhance operational visibility, strengthen forecasting capabilities, reduce dependence on manual processes, and support evidence-based decision-making. In food and beverage businesses, where responsiveness and efficiency are critical, analytical systems provide valuable capabilities for monitoring sales, evaluating performance, managing resources, and adapting to changing market conditions.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies, together with the related literature, collectively provide strong theoretical and practical support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. Across the reviewed sources, a common theme emerged regarding the importance of Business Intelligence (BI), data analytics, and digital technologies in improving organizational performance, operational efficiency, and managerial decision-making. Researchers consistently emphasized that organizations which utilized centralized analytical systems were better equipped to monitor business operations, evaluate performance, and respond effectively to changing market conditions. These findings demonstrate that transforming transactional data into meaningful information enables businesses to make informed decisions based on evidence rather than intuition.

The reviewed literature highlighted the significance of dashboard-based Business Intelligence systems in presenting complex operational data through graphs, charts, and summarized reports. Both foreign and local studies agreed that interactive dashboards improve data accessibility, simplify the interpretation of business information, and allow managers to monitor key performance indicators in real time. Through visual analytics, organizations can easily identify revenue trends, sales fluctuations, product demand, and operational issues that require immediate attention. These capabilities improve managerial responsiveness, strengthen organizational transparency, and support faster, more accurate decision-making.

Another recurring theme across the reviewed sources was the value of centralized data management and automated reporting. Previous studies consistently demonstrated that consolidating information from multiple branches into a single database improves reporting accuracy, minimizes data duplication, and reduces delays associated with manual report preparation. Researchers further explained that centralized systems strengthen accountability by standardizing reporting practices and providing management with consistent and reliable operational information. Such findings strongly support the proposed system's centralized architecture, which integrates sales records from different Kape Tubong kiosks into one analytical platform.

The literature also emphasized the importance of sales analytics, historical trend analysis, and product performance evaluation in supporting strategic business planning. Researchers agreed that analyzing historical sales records allows organizations to recognize recurring demand patterns, evaluate promotional effectiveness, forecast future sales, and optimize inventory management. Product-level analytics likewise enable businesses to identify best-selling and underperforming items, allowing management to improve pricing strategies, promotional activities, and procurement decisions. These findings validate the inclusion of revenue trend analysis, product performance monitoring, and historical reporting as core features of the BREWINSIGHT BI system.

The reviewed studies consistently highlighted the operational benefits of branch performance evaluation and real-time monitoring. Comparative branch analytics enable organizations to measure productivity across multiple locations, identify operational strengths and weaknesses, and implement branch-specific improvements based on factual performance data. Real-time reporting further enhances organizational agility by allowing management to respond immediately to changes in sales performance, inventory levels, and customer demand. These capabilities are particularly valuable for Kape Tubong, where effective monitoring of multiple kiosks is essential for maintaining operational consistency and service quality.

Both foreign and local literature also emphasized that Business Intelligence systems function not only as reporting tools but as decision-support systems that strengthen organizational planning and governance. Researchers consistently argued that analytical platforms reduce reliance on intuition by providing measurable evidence to support operational and strategic decisions. Organizations utilizing BI technologies demonstrated improved forecasting, stronger accountability, enhanced collaboration, and greater adaptability to changing business environments. Furthermore, the reviewed studies recognized that cloud-based and web-based technologies increase accessibility, scalability, and operational flexibility, making Business Intelligence solutions practical even for small and medium enterprises.

Within the Philippine context, the reviewed local studies confirmed that many food service MSMEs continue to experience challenges associated with manual reporting, fragmented sales records, delayed information processing, and limited visibility across multiple branches. Local researchers consistently found that centralized Business Intelligence platforms effectively address these challenges by improving operational transparency, accelerating report generation, and supporting evidence-based managerial decisions. These findings demonstrate that BI implementation is both feasible and beneficial for Philippine food service enterprises, particularly those experiencing business growth and expansion.

The reviewed literature establishes a strong foundation for the development of the BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. The collective findings consistently support the integration of centralized data management, dashboard visualization, revenue analytics, historical trend analysis, branch performance monitoring, automated reporting, and decision-support functionalities within a single Business Intelligence platform. The literature further confirms that adopting these technologies enhances operational efficiency, strengthens managerial oversight, improves strategic planning, and promotes data-driven decision-making. Consequently, the reviewed studies and literature provide substantial justification for the proposed system and affirm its potential to improve the operational performance, competitiveness, and long-term sustainability of Kape Tubong.

REFERENCES

[7] R. Kumar and P. Singh, “Data visualization in retail analytics: Impact on decision-making,” International Journal of Business Intelligence, vol. 14, no. 2, pp. 45–58, 2022. [Online]. Available: https://doi.org/10.11114/ijbi.v14i2.25472

[8] C. Li and M. Gonzalez, “Business intelligence systems and organizational performance in small and medium enterprises,” Journal of Information Systems and Technology Management, vol. 18, no. 3, pp. 112–128, 2021. [Online]. Available: https://doi.org/10.4301/jistem.2021.18.3.8

[9] T. Johnson and M. Lee, “Business intelligence dashboards and organizational performance in small enterprises,” Journal of Information Systems and Technology Management, vol. 18, no. 3, pp. 102–118, 2021. [Online]. Available: https://doi.org/10.4301/jistm.2021.18.3.102

[10] L. Garcia and A. Torres, “Sales analytics systems for food service enterprises,” International Journal of Hospitality and Food Service Management, vol. 9, no. 1, pp. 55–70, 2020. [Online]. Available: https://doi.org/10.2139/ijhfsm.2020.9.1.55

[11] S. Martinez and J. Cooper, “Interactive business intelligence systems and decision support in food retail operations,” International Journal of Digital Commerce and Analytics, vol. 13, no. 2, pp. 67–83, 2021. [Online]. Available: https://doi.org/10.2245/ijdca.2021.13.2.67

[12] M. Reyes and J. Santos, “Digital transformation in Philippine micro-enterprises: A study on BI adoption in the F&B sector,” Philippine Review of Business and Economics, vol. 60, no. 1, pp. 12–29, 2023. [Online]. Available: https://doi.org/10.11114/prbe.v60i1.3321

[13] E. Villanueva, “From intuition to evidence: The digital maturity of food service MSMEs in Western Visayas,” Philippine Journal of Science and Technology Management, vol. 19, no. 2, pp. 88–105, 2024. [Online]. Available: https://doi.org/10.53899/pjstm.v19i2.441

[14] G. Navarro and E. Cruz, “Implementation of business intelligence systems in Philippine small food enterprises,” Asian Journal of Information Technology and Business Management, vol. 19, no. 2, pp. 88–104, 2022. [Online]. Available: https://doi.org/10.1080/ajitbm.2022.19.2.104

[15] A. Lopez and C. Ramirez, “Centralized sales monitoring systems and business performance among Philippine café enterprises,” Journal of Philippine Management and Information Systems, vol. 15, no. 3, pp. 77–96, 2021. [Online]. Available: https://doi.org/10.4587/jpmis.2021.15.3.77

[16] L. Fernandez and M. Gutierrez, “Revenue analytics and dashboard utilization among Philippine small food businesses,” Philippine Journal of Information Technology and Business Analytics, vol. 10, no. 2, pp. 54–76, 2022. [Online]. Available: https://doi.org/10.6215/pjitba.2022.10.2.54

.

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CHAPTER III

METHODOLOGY

3.1 Research Design

The study used a Developmental Research Design in developing BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. The research design focused on examining the existing sales recording, monitoring, and reporting processes of Kape Tubong and using the information gathered as the basis for developing the proposed system. The researchers examined the handling of sales records from the kiosks and main branch, the preparation of reports, and the process of reviewing sales information. Interviews, observations, and discussions with the business owner and selected operational staff were conducted to understand how sales information was recorded, collected, organized, and reported. These activities also helped identify problems involving scattered sales records, manual consolidation, delayed reporting, and the lack of a centralized environment for monitoring sales performance.

The developmental approach allowed the researchers to transform the identified requirements into a working web-based Business Intelligence system. The development focused on creating a centralized database where sales information from the kiosks and main branch could be stored and retrieved. The system was designed to provide dashboards and analytical features for revenue pattern analysis, product performance, branch comparison, peak sales period identification, and automated reporting. The researchers used the information gathered from the existing business operations to determine the system requirements and features that needed to be included in BREWINSIGHT BI.

Testing and evaluation were conducted throughout the development process to determine whether the system performed according to the identified requirements. Individual features were tested to verify their functionality, accuracy, and reliability. The researchers checked calculations, database operations, dashboard information, reports,

and analytical results to ensure that the system produced information consistent with the stored sales records. User feedback was also considered during the development and evaluation process. The business owner and selected personnel reviewed the developed features and provided feedback regarding functionality and usability. User Acceptance Testing was conducted to determine whether the system could support actual business activities and whether the information presented through dashboards and reports was understandable and useful to the intended users.

The system was evaluated based on functionality, usability, reliability, efficiency, and security. These criteria were used to determine whether the developed system performed its intended functions, provided an understandable interface, operated consistently, responded efficiently, and protected business information. The criteria were aligned with the software product quality characteristics described in ISO/IEC 25010 [17]. The Developmental Research Design allowed the researchers to connect the identified business problems with the development and evaluation of the proposed solution. The process moved from examining existing operations, identifying requirements, designing the system, implementing its features, testing the developed functions, gathering user feedback, and improving the system according to the identified sales monitoring, reporting, and analytical needs of Kape Tubong.

3.2 System Development Methodology

The researchers adopted the Agile System Development Methodology to guide the development of BREWINSIGHT BI. Agile was selected because it supports an iterative approach in which the system can be planned, developed, reviewed, and refined in manageable increments. Rather than treating development as a single sequence that ends only after all features are complete, the methodology allows the researchers to revisit requirements, assess completed work, and make adjustments as the project progresses. This was suited to the project because the system needed to reflect Kape Tubong's sales monitoring and reporting requirements while incorporating feedback from people familiar with its operations.

Agile emphasizes collaboration, the delivery of working software, stakeholder involvement, and responsiveness to changing or clarified requirements [18]. These principles shaped how the researchers organized development activities and communicated with the business owner and selected personnel. Stakeholders could review completed functions and provide feedback on their relevance and usability. The researchers could then consider that feedback when refining existing features or planning subsequent work. This helped maintain a connection between the system's development and the needs it was intended to address.

The development process was organized into seven phases: Planning, Analysis, Design, Implementation, Testing, Deployment, and Maintenance. Each phase had a distinct focus while contributing to the development and continued support of the complete system.

During Planning, the researchers established the project's direction and priorities. They identified the business needs that BREWINSIGHT BI was intended to address, set initial objectives, considered the technical resources required, and organized development activities according to the available schedule and resources. This phase also provided an initial basis for involving the business owner and selected personnel throughout the project.

During Analysis, the researchers examined Kape Tubong's existing sales recording, reporting, and monitoring processes in greater detail. They gathered information about how sales records were created and handled, identified the information users needed, and documented the system's functional and non-functional requirements. The analysis also considered user roles, business rules, and the movement of information through the proposed system. Its findings clarified what the system needed to do and what qualities it needed to support.

During Design, the researchers translated the requirements into a plan for the system's structure and user experience. This included planning the application architecture, database, user interface, navigation, dashboards, analytical functions, security controls, and processing logic. Models and design documentation were prepared to guide implementation and show how information would be stored, processed, and presented to users.

During Implementation, the researchers developed BREWINSIGHT BI's major functions in increments. Front-end and back-end components were built and connected to the centralized database. Development included sales data management, dashboards, reporting, and Business Intelligence features. Completed work was reviewed and refined as development continued, and feedback from stakeholders was considered when appropriate.

During Testing, the researchers evaluated whether the system's functions operated according to the identified requirements. They examined individual features, calculations, database operations, component integration, interface behavior, security controls, and the accuracy of information presented by the system. Intended users also participated in acceptance activities, allowing the researchers to gather feedback on the system's usability and relevance to business tasks. Issues identified during testing could be addressed and checked again.

During Deployment, the researchers prepared the completed system for operational use. Activities included preparing the server environment, configuring database access, organizing relevant records, setting up user accounts and permissions, and orienting intended users. Initial use was monitored so that performance and user feedback could be considered.

During Maintenance, the researchers addressed issues and changes arising after deployment. This phase included monitoring system performance, correcting errors, reviewing security and database practices, considering user feedback, and assessing possible improvements as Kape Tubong's operational or reporting needs evolved.

The phases provided an organized way to manage the project, with the results of each phase informing the work that followed. For example, requirements identified during Analysis guided the system designs, while those designs guided Implementation and provided a reference for Testing. At the same time, Agile's iterative approach allowed the researchers to address issues and feedback throughout development rather than treating each phase as entirely isolated. This made it possible to refine the system as its functions were developed and reviewed.

Through this methodology, the researchers worked toward a system aligned with Kape Tubong's needs for centralized sales information, business monitoring, analysis, and reporting. The combination of defined phases and iterative review provided a structured process for developing BREWINSIGHT BI and preparing it for use.

Planning Phase

The Planning Phase established the direction, scope, and initial priorities for developing BREWINSIGHT BI. At this stage, the researchers examined Kape Tubong's needs and identified the main difficulties associated with its existing sales recording, monitoring, and reporting procedures. The work provided an initial basis for determining what the proposed system should accomplish, which people should participate in its development, and what resources would be needed.

The researchers coordinated with the business owner, Bryan Tacayon, and selected operational staff to learn how sales information was handled in practice. Discussions focused on how transactions were recorded, how sales records were maintained, how information was collected from different locations, and how reports were prepared for review. The researchers also examined the methods used to record sales information from the kiosks and the main branch. This helped clarify how transactions were documented at each location and how records moved from daily operations into business reports.

Reviewing these procedures helped the researchers understand the challenges associated with manual sales monitoring and reporting. Because branch records were handled separately, management could face difficulty

obtaining a complete view of business performance. Information had to be collected and organized before it could be consolidated and reviewed, which added time and effort to report preparation. Separate records could also make comparisons across locations more demanding and limit how quickly management could access information for operational review.

Based on these identified needs, the researchers established the objectives of BREWINSIGHT BI. The system was intended to centralize sales information, provide a dashboard for monitoring sales, analyze revenue patterns, evaluate product performance, compare branch results, identify peak sales periods, and generate automated reports. These objectives defined the main areas of work for the proposed system and helped guide decisions about which features should receive priority during development.

The researchers also identified the system's preliminary functional requirements. These included user authentication and management, sales data management, branch and product information management, dashboard visualization, revenue analysis, product performance analysis, branch comparison, peak sales period analysis, and report generation. Defining these functions at the planning stage helped establish the expected capabilities of the system and provided a starting point for more detailed requirements work in the following phases.

Non-functional requirements were considered alongside the planned features. Usability was important so that intended users could navigate the application and access business information. Reliability and efficiency were relevant to consistent handling of sales records and timely access to summaries and reports. Security and maintainability were also considered, as well as the system's ability to handle the expected volume of sales information. These considerations helped frame the qualities BREWINSIGHT BI would need to support in addition to its specific functions.

The researchers identified the technical resources required to develop the system. The planned technologies and tools included Python and Flask for back-end development; HTML, CSS, JavaScript, and Bootstrap for the front end; MySQL for data storage; and Chart.js for data visualization. XAMPP, Visual Studio Code, Git, and Google Chrome were also identified as tools and software environments relevant to development and system access. Establishing these resources early helped the researchers plan the technical work needed to build the application.

Development activities were organized according to the available schedule and resources. Features related to sales monitoring, data centralization, analytics, dashboards, and reporting were given priority because they aligned closely with the problems identified in the existing process. Planning these priorities helped provide a practical sequence for development and established which system capabilities were central to the project's objectives.

The researchers also planned for the participation of the business owner and selected personnel throughout development. Their familiarity with Kape Tubong's daily operations could help clarify requirements, review proposed features, and provide feedback as the system took shape. Including these stakeholders in the development process was intended to keep the planned functions connected to actual business activities and user needs.

Documentation was maintained to record the problems identified, the system objectives, preliminary requirements, technical resources, planned development activities, and proposed features. These records provided a reference for the work that followed and helped preserve the decisions made during planning.

The Planning Phase concluded with an established direction for developing BREWINSIGHT BI. It defined the main business needs the project would address, the system's initial objectives and priorities, the resources required, and the expected involvement of stakeholders. This foundation informed the Analysis Phase, where the researchers examined the existing processes and requirements in greater detail.

Analysis Phase

The Analysis Phase focused on developing a detailed understanding of Kape Tubong's existing operations and identifying the information and system capabilities needed to support them. The researchers examined how sales transactions were recorded, how branch information was maintained, how reports were prepared, and how management monitored business performance. The purpose of this work was to identify the challenges the proposed BREWINSIGHT BI system needed to address and to define requirements that reflected the organization's operating context.

To gather information, the researchers consulted the business owner, Bryan Tacayon, branch managers, administrative personnel, and selected operational staff. These discussions covered the tasks employees performed when recording transactions, maintaining sales information, preparing reports, and reviewing business results. Participants also shared how branch performance was monitored, what information management needed, and where delays or difficulties arose in the existing process. Consulting people with different responsibilities helped the researchers consider both day-to-day data handling and management's reporting needs.

The researchers examined the existing sales recording and reporting processes to understand how information moved from a branch transaction to a report used for review. This included considering how employees documented sales, organized records, and provided information for consolidation. The analysis also considered the effort involved in collecting and verifying records from multiple branches, along with the challenges that could arise when information was stored or prepared separately. These findings helped establish the need for a centralized approach to sales information and reporting.

The system's data requirements were identified by determining which information would be needed to support transaction management, analysis, and reporting. Required data included sales transaction details, product information, quantity sold, prices, sales totals, dates, branch information, and other relevant sales details. The researchers considered how each item would be used by system functions, such as calculating revenue, summarizing sales by period, reviewing product performance, or comparing branch results.

The quality of sales information was also considered. Incomplete, duplicated, inconsistent, or incorrect entries could affect the reliability of summaries and analyses. The researchers therefore examined the types of information that needed to be captured and the importance of maintaining consistent records. This analysis informed later decisions about data organization, input validation, and the handling of sales information within the proposed system.

User roles and responsibilities were identified as another part of the requirements analysis. The researchers considered the needs of the business owner, administrative personnel, branch personnel, and other operational users. Because users may perform different tasks, the analysis examined which system functions were relevant to each role and what information each group needed to access. These findings provided a basis for defining appropriate permissions and connecting access to users' responsibilities.

Functional requirements were analyzed to specify what BREWINSIGHT BI needed to do. These requirements included authentication, sales data management, dashboard generation, data filtering, revenue calculation, product analysis, branch comparison, report generation, and account management. The researchers considered the user activities associated with each function and how system responses should support those activities. For example, users would need to access sales information, select relevant criteria, review calculated summaries, and obtain reports for business monitoring.

The analysis also included non-functional requirements, which describe qualities the system should support in addition to its specific functions. Usability was considered in relation to how easily intended users could navigate the application and understand its information. Reliability and efficiency were relevant to the system's ability to process and present sales information consistently. Security requirements informed the need for authentication, access controls, and appropriate handling of submitted data. Responsiveness, maintainability, and scalability were

also considered so the system could remain usable, support future changes, and accommodate growth in business information.

The researchers examined the expected flow of information through the system. They considered how sales information would enter the application, how it would be processed and stored, and how summaries and analytical results would be presented to users. Process models and Data Flow Diagrams were considered to represent the movement of information among users, system processes, and data storage. Describing these flows helped clarify the system's expected behavior and identify the information needed at each stage.

The feasibility of the proposed system was assessed in relation to the resources available to Kape Tubong and the project. This included consideration of hardware, software, development tools, technical resources, and existing business operations. The researchers considered whether the planned system could be developed and used in the organization's context and whether its requirements aligned with available resources. This assessment helped inform the scope and direction of the proposed solution.

Business rules were established to guide how the system would handle key information and operations. These rules covered sales transactions, product information, branch information, user access, calculations, and reports. Defining them helped ensure that system functions would follow consistent expectations when information was entered, processed, or presented. The rules also provided a reference for later system design and implementation.

Requirements related to centralized sales monitoring, dashboard visualization, revenue analysis, and reporting were prioritized because they addressed the organization's identified information and monitoring needs. The researchers also considered future scalability, including the possibility of additional sales records, products, users, and branches. This consideration helped ensure that the proposed requirements accounted for business growth as well as current operations.

The problems identified, functional and non-functional requirements, data requirements, business rules, user roles, and process flows were documented at the conclusion of the Analysis Phase. These findings provided the basis for the Design Phase, where the researchers defined how BREWINSIGHT BI would be structured and how its planned functions would be implemented.

Design Phase

The Design Phase translated the requirements identified during the Analysis Phase into a structured plan for BREWINSIGHT BI. It defined how the system would be organized, how users would interact with its functions, how sales information would be stored and processed, and how the results would be presented for review. The researchers designed the system architecture, database, user interface, navigation structure, dashboards, analytical features, security controls, and processing logic. These design decisions provided a guide for the subsequent implementation and testing activities.

The system architecture was planned around communication among the client, server, and database. Users would access BREWINSIGHT BI through a web browser, which would serve as the client interface. When users performed an action—such as opening a dashboard, filtering records, or requesting a report—the application server would receive the request, apply the relevant business logic, and communicate with the database as needed. The database would store the centralized sales information used by the system. The server would then return the requested information for display in the browser. This structure established how user actions, application processing, and stored records would work together.

The database design was based on the information and business processes identified during analysis. The researchers determined which entities and attributes the system needed to represent, including users, branches, products, sales transactions, and other required information. They considered how each type of information would be stored and

how records would relate to one another. For example, transaction records needed to be associated with relevant branch and product information so that the system could organize sales by location and product.

Primary and foreign keys were considered when establishing relationships among database tables. Primary keys provided a way to identify individual records, while foreign keys supported connections between related information. The researchers also considered appropriate constraints to help maintain valid relationships and organized records. Entity Relationship Diagrams (ERDs) were prepared to represent the planned entities, attributes, and relationships. These diagrams helped document the database structure before it was implemented.

Data Flow Diagrams (DFDs) were used to describe how information would move through the system. They represented interactions among users, system processes, and data storage, including how information would enter the application, be processed, and be retrieved for display or reporting. Together, the ERDs and DFDs helped describe both the structure of the stored data and the movement of information through BREWINSIGHT BI.

The user interface was designed to give users clear access to the system's functions. The researchers prepared layouts and wireframes for the main pages, dashboards, forms, reports, and management screens. These designs helped establish where information and controls would appear and how users would move between related tasks. Forms were planned for entering or managing information, while tables and reports were designed to present records in an organized format.

The dashboard received particular attention because it was planned as one of the main features of BREWINSIGHT BI. The researchers considered how charts, tables, numerical summaries, and other visualizations could present sales information in a way that supported review and comparison. Planned dashboard content included revenue trends, product performance, branch comparisons, and peak sales periods. The selection and arrangement of these elements were intended to help users interpret business information without having to examine every underlying transaction individually.

Navigation was organized to group related functions and make important features accessible. The researchers considered how users would move between sales records, dashboards, reports, analytical views, and management functions. Clear grouping and consistent page layouts were intended to help users understand where to find information and how to complete common tasks. Responsive design was also considered so that pages and visual elements could adapt to different screen sizes.

The design included plans for authentication and authorization to protect functions that required controlled access. User roles and permissions were based on responsibilities identified during the Analysis Phase. The researchers considered which functions should be available to each role and how the system should distinguish between authorized and unauthorized access. These decisions informed the planned access controls and helped connect security requirements with the system's navigation and features.

Processing logic was designed for the system's sales and analytical functions. The researchers planned how sales totals would be calculated, how records could be filtered and sorted, and how summaries would be grouped by time period, branch, or product. They also defined the intended logic for generating product performance results, branch comparisons, and revenue summaries. This planning helped establish how raw sales records would be transformed into information suitable for dashboards and reports.

Performance, scalability, usability, and future expansion were considered throughout the design process. The researchers considered how the system would handle its expected information and how its structure could support possible growth in users, branches, products, and sales records. Usability considerations informed the layout, navigation, and presentation of information, while performance considerations informed the planned interactions among the application, its processing logic, and the database.

The proposed design was reviewed with the business owner and selected personnel. Their feedback provided an opportunity to assess whether the planned architecture, data structure, interface, and analytical features aligned with the organization's needs. Comments from the review were considered before implementation so that relevant adjustments could be reflected in the design.

Documentation was maintained for the architecture, database structure, interface layouts, process flows, security controls, and analytical functions. These materials served as references during development and provided a shared description of how the system was intended to work. By the end of the Design Phase, the researchers had established a detailed plan for implementing and evaluating BREWINSIGHT BI.

Implementation Phase

The Implementation Phase focused on developing BREWINSIGHT BI in accordance with the requirements and system design established during the preceding phases. During this stage, the researchers translated the planned features, workflows, and interface designs into a functional application. The work involved building the system's user-facing pages, implementing its underlying application logic, connecting it to a centralized database, and developing its sales management, reporting, and Business Intelligence capabilities.

Development proceeded incrementally through Agile iterations. Each iteration addressed a group of related functions, allowing the researchers to develop and review system components progressively. Rather than waiting until every feature had been built before examining the system, the team could evaluate completed functions, identify issues, and make refinements as development continued. This process also gave stakeholders opportunities to review work and provide feedback while there was still time to incorporate it into later iterations. The approach supported Agile principles such as delivering working software, collaborating with stakeholders, and responding to changing or clarified requirements [18].

The system's front-end and back-end components were developed to serve different but connected purposes. The front end was built using HTML, CSS, JavaScript, and Bootstrap. HTML provided the structure of the pages, while CSS and Bootstrap supported their visual presentation and responsive layouts. JavaScript was used to support interactive interface behavior. Together, these technologies were used to create the navigation, forms, tables, dashboards, and other elements through which users interacted with the system.

The interface was organized to help users locate and use the system's functions. Navigation elements provided access to the relevant pages, while forms gave users a consistent way to enter or manage information. Tables presented records in an organized format, and dashboards displayed sales information for review. Responsive layouts helped the interface adjust to different screen sizes. These elements were designed to make information and common functions accessible without requiring users to work directly with the system's underlying database.

The back end was developed using Python Flask to process application requests and manage the system's operations. It handled application logic, user authentication, data processing, database communication, analytical calculations, and report generation. When a user interacted with a function in the interface, the back end processed the request, performed the required operation, and returned the appropriate response or information for display. This division of responsibilities allowed the presentation of information to remain separate from the processes used to retrieve, calculate, and manage it.

Authentication and role-based access were implemented to regulate access to the system. Users were required to authenticate before accessing protected functions, and permissions were assigned according to their roles. This helped ensure that users could use the features relevant to their responsibilities. Input validation was also incorporated to check information submitted through the application, while protected database operations helped ensure that data was handled through the application's defined processes.

A substantial part of implementation involved developing the sales data management and Business Intelligence features. These functions used sales information in the centralized database to produce summaries and analyses for business monitoring. The system included revenue analysis, product performance analysis, branch comparison, and peak sales period analysis. It also provided daily, weekly, monthly, and annual sales summaries based on the information available in the records.

The sales summaries were intended to help users review activity over different time periods. Daily summaries provided a view of sales recorded for a particular day, while weekly and monthly summaries supported review over longer intervals. Annual summaries provided a broader view of recorded sales activity. These time-based views made it possible to examine business results at different levels and compare activity across periods, subject to the sales information stored in the system.

Product performance analysis processed transaction information to calculate sales quantities and revenue contributions. This allowed users to review how individual products contributed to recorded sales and to compare products using the information available in the database. Branch analysis organized sales information by Kape Tubong location, giving management a consolidated basis for comparing branch results. Peak sales period analysis examined sales activity by period to help identify intervals with higher recorded activity. Together, these analytical functions were intended to make sales records easier to interpret and more useful for operational review.

Automated reporting was implemented to generate sales summaries and reports from the centralized database. By using the stored information directly, the reporting functions reduced the need to manually combine records from separate sources. Generating reports through the application also provided a more consistent way to present sales information for review. Dashboards and reports displayed processed data in formats intended to help users examine sales results and make comparisons across products, branches, and time periods.

The application was integrated with the database so that its features relied on a centralized source of sales information. Database operations supported the storage, retrieval, and processing of records required by the management, analytical, and reporting functions. This integration allowed the application to use the same organized source of information when displaying records, calculating summaries, and generating reports. It also supported consistency between the data used by different parts of the system.

Validation and testing were performed throughout implementation to check whether developed functions operated as intended. The researchers reviewed application behavior, calculations, database operations, interface responses, and analytical outputs. These checks helped identify issues such as incorrect calculations, unexpected behavior, or difficulties retrieving and processing records. When a problem was identified, the researchers debugged the affected function and reviewed it again after making corrections. Performing these checks as development progressed helped address issues before they could affect additional features.

Stakeholders participated in sprint reviews and provided feedback on completed functions. Their comments helped the researchers assess whether system features were understandable and relevant to the needs of intended users. Feedback could lead to adjustments to the interface, workflows, reports, or analytical features. Reviewing completed work with stakeholders throughout implementation helped maintain a connection between the system being developed and Kape Tubong’s operational requirements.

The researchers also performed code optimization and refactoring as the application developed. These activities helped organize the source code, improve its maintainability, and support corrections to identified problems. Version control was used to manage changes to the code throughout development. This provided a structured way to track revisions and coordinate changes to different system components. Documentation was maintained to record relevant system functions and implementation activities.

By the end of the Implementation Phase, the approved requirements and designs had been translated into the major working functions of BREWINSIGHT BI. Incremental development, integration with the centralized database, ongoing validation, stakeholder feedback, and source-code maintenance supported the creation of the system's core sales management, reporting, and Business Intelligence capabilities. The completed functions were then ready for subsequent evaluation and preparation for deployment.

Testing Phase

The Testing Phase was conducted to evaluate whether BREWINSIGHT BI functioned according to the requirements identified for the system. It examined how individual features operated, whether connected components exchanged information correctly, and whether the information presented by the system corresponded with the underlying sales records. Testing also covered analytical accuracy, performance, browser and screen compatibility, security, error handling, data integrity, and user acceptance. The activities were guided by recognized software testing concepts [19].

The researchers began by checking the system's individual features using inputs and expected outcomes. This included authentication, user management, sales data management, dashboards, filtering, report generation, and the system's analytical functions. For each feature, they provided relevant inputs, followed the expected workflow, and observed how the application responded. These checks helped determine whether functions could be accessed and used as intended, whether the system displayed appropriate results, and whether it responded properly to different types of input.

Authentication and user management were examined to verify that users could log in and access functions appropriate to their assigned roles. The researchers checked how the system responded to valid and invalid login information and reviewed whether access restrictions were applied to protected functions. Sales data management was tested by entering, viewing, and managing sales information. The researchers checked whether the system accepted valid information, displayed stored records, and handled incomplete or invalid entries appropriately.

Dashboard and filtering tests examined whether users could view sales information in the available summaries and narrow results using the system's filters. The researchers checked whether displayed information changed appropriately in response to selected criteria and whether the resulting summaries remained consistent with the relevant sales records. Report generation was also examined to determine whether users could request reports and whether the resulting information reflected the data selected for review.

The accuracy of analytical calculations received particular attention because management decisions may rely on the information presented by the system. The researchers compared revenue totals and sales summaries with the underlying transaction records. They also checked product performance results against recorded product quantities and revenue contributions, and reviewed branch comparisons against the sales information associated with each location. Peak sales period information was similarly compared with the relevant sales records to verify that the periods shown were supported by the available data. These comparisons helped identify discrepancies between stored records, calculations, and displayed results.

Integration testing examined how the system's components worked together. The researchers reviewed the interactions among the front end, back end, database, authentication, dashboards, analytical functions, and reporting features. For example, they checked whether sales information entered or modified through the application was stored and whether that information was subsequently reflected in the relevant tables, dashboard summaries, analyses, and reports. They also examined whether application requests reached the appropriate back-end functions and whether those functions retrieved or processed information correctly.

Performance and load testing focused on the system's responsiveness during common user activities. The researchers observed page loading, dashboard generation, database queries, filtering, and report generation to

identify possible delays or other performance concerns. These observations provided a basis for reviewing how the system behaved while users accessed its main functions. The testing also considered whether processing and displaying sales information remained practical for the intended business activities.

Compatibility testing examined the system in the supported browser environment and across different screen sizes. The researchers reviewed whether pages, navigation elements, forms, tables, dashboards, and reports remained usable and displayed as expected. This helped identify interface issues that could affect users accessing the system through different screen dimensions or supported browsers.

Security testing examined authentication, authorization, access control, and input validation. The researchers reviewed whether protected features required appropriate access and whether role restrictions were applied. They also checked how the system handled submitted information, including whether input validation helped prevent invalid data from being processed. Error handling was evaluated by providing invalid or incomplete information and observing whether the system responded in a controlled and understandable manner.

Data integrity testing focused on consistency between database records and information presented through the application. The researchers compared stored information with corresponding values in tables, dashboards, analytical results, and reports. This helped determine whether records were accurately retrieved and displayed and whether data changes were reflected in the relevant parts of the system. Maintaining this consistency was important so that users could rely on the information shown during business review.

User Acceptance Testing was conducted with intended users, including Bryan Tacayon and selected staff, branch representatives, administrative users, and operational personnel. Participants carried out actual or simulated business activities using the system. These activities included logging in, reviewing dashboards, examining sales trends, checking product performance, comparing branch information, identifying peak sales periods, and generating reports. The exercises allowed users to evaluate the system in relation to the tasks they were expected to perform.

Feedback was collected through observation, interviews, discussions, and evaluation forms. The researchers considered comments about the system's functions, interface, reports, and usefulness for the intended activities. When feedback or testing identified an issue or an opportunity for improvement, the relevant feature was revised and tested again. Regression testing was then conducted to check whether a modification had affected functions that had previously been evaluated.

Testing documentation was maintained throughout the phase to record the functions examined, the activities performed, observed issues, feedback, and subsequent revisions. This documentation supported the review of testing progress and the tracking of improvements made during evaluation. The Testing Phase concluded after the major system functions and user activities had been evaluated and identified improvements had undergone further testing.

Deployment Phase

The Deployment Phase focused on preparing BREWINSIGHT BI for use in Kape Tubong's operations. The work moved the system from development and testing into an environment where authorized users could access its sales management, monitoring, analysis, and reporting functions. Before deployment, the researchers reviewed the completed application against the requirements and confirmed that its major functions had been implemented and tested. This review provided an opportunity to check that the system components were ready to be configured for operational use.

The researchers prepared the server environment by configuring the application environment, database connection, and relevant server settings. They checked that the application could communicate with the MySQL database and that the services required to run the system were operating correctly. These checks helped confirm that users would be able to access the application and that its functions could retrieve and process the information stored in the database.

Database preparation was another important part of deployment. The researchers organized the sales information needed by the system and entered or migrated relevant historical records when applicable. They reviewed the prepared information to determine whether it followed a structure that the system could process and use in its summaries, dashboards, analyses, and reports. Preparing the records in this way helped establish a usable information base for the system's sales monitoring and analytical features.

User accounts and permissions were configured according to the roles identified during the analysis and design phases. The researchers set up access for authorized users and assigned privileges appropriate to their responsibilities. Credentials were provided to the intended users so they could log in and use the functions available to them. Configuring access before operational use also helped users begin with a clear understanding of how they would enter the system and which features they could access.

The researchers conducted user orientation and training to familiarize participants with BREWINSIGHT BI. The orientation covered the system interface, login process, dashboard, sales information, analytical features, reports, and other relevant functions. Users were introduced to the steps involved in accessing information and carrying out common activities, such as reviewing sales results, examining branch or product information, and generating reports. This helped users understand how the system could support their assigned operational or management tasks.

Backup procedures were also considered as part of preparing the system for use. Maintaining copies of important database information can help protect business records in the event of data loss or other problems affecting stored information. The researchers documented the relevant backup procedures so the organization would have guidance for preserving its records as part of ongoing system use.

After deployment, the researchers monitored the system during its initial operational use. They observed its performance and checked whether users could complete their tasks without encountering problems. This monitoring provided an opportunity to identify issues that might not have appeared during development or controlled testing, including difficulties related to navigation, access, data display, or report use. Observations from initial use could then inform troubleshooting or later refinements.

The researchers collected feedback from users as they began working with the system. Comments concerning usability, dashboard content, reports, navigation, and functionality were reviewed to identify areas that might benefit from improvement. User feedback provided practical context about how the system performed in day-to-day activities and whether its presentation and workflows aligned with users' needs.

Deployment documentation was maintained to record the system configuration, database preparation, user setup, deployment activities, and relevant procedures. This documentation provided a reference for the activities performed during deployment and supported future operation and maintenance of the system.

The Deployment Phase concluded with BREWINSIGHT BI prepared for operational use by authorized Kape Tubong users. The system provided a centralized web-based environment for accessing sales information, monitoring business activity, reviewing analytical results, and generating reports across the organization's operations.

Maintenance Phase

The Maintenance Phase focused on supporting BREWINSIGHT BI after deployment and helping ensure that it continued to meet Kape Tubong's operational requirements. Once users began relying on the system, maintenance activities addressed technical issues, changes in the operating environment, and evolving business needs. These activities included technical support, performance monitoring, error correction, security updates, database management, review of user feedback, and consideration of system improvements.

The researchers monitored the application to identify issues that could interfere with normal use. Areas reviewed included database connections, page loading, calculations, report generation, authentication, and other system functions. If a function did not behave as expected, the issue could be investigated, corrected, and checked again. This ongoing attention helped keep the system's core capabilities available to users and provided a way to address problems reported during day-to-day operations.

Performance monitoring considered how quickly the system responded to common user activities. The researchers reviewed response times, database operations, dashboard loading, filtering, and report generation for signs of delays or inefficient processing. If performance concerns arose, database queries or application processes could be examined and optimized as appropriate. These reviews were especially relevant as users accessed more records or performed more frequent searches and reporting activities.

Maintenance also considered the software components on which the system depended. Updates to dependencies or other system components may be needed to address security concerns, maintain compatibility, or support continued operation. Before applying a change, its effect on the application's functions and database interactions could be considered. This helped ensure that necessary updates did not introduce problems into authentication, data processing, dashboards, or reporting.

Database management formed another part of ongoing maintenance. The researchers monitored database operations and considered optimization when queries or data processing became less efficient. Backup and recovery procedures were maintained to help protect important sales information and provide a means of restoring records if data were lost or affected by a technical problem. These procedures supported the continued availability and preservation of business information.

Security reviews were also considered during the maintenance period. User permissions, authentication, input handling, and data protection procedures could be reviewed to identify weaknesses or changes that might be needed. Such reviews helped ensure that access continued to reflect users' roles and that information submitted to the system was handled appropriately. Security improvements could be introduced as system components, dependencies, or operational requirements changed.

User feedback provided a practical basis for identifying areas that needed attention. Users could report technical problems, difficulties completing tasks, or changes in their information needs. The researchers considered this feedback alongside system monitoring to determine whether a concern required a correction, clarification, or broader improvement. For example, users might have difficulty locating a function, interpreting dashboard information, or understanding labels used in the interface.

Interface improvements could be made in response to actual user experience. Adjustments to labels, navigation, page organization, or the presentation of dashboard information could make common tasks easier to complete and results easier to interpret. Compatibility with supported browsers and devices was also monitored so that key pages and functions remained usable in the intended environments.

The maintenance process also accounted for changes in Kape Tubong's operations. As the business developed, its reporting requirements, branch activities, products, sales processes, and management needs could change. The researchers considered these changes when reviewing possible system adjustments. For example, an additional branch, a larger volume of sales records, more users, or a broader product range could affect how records were organized, how reports were used, and what information management needed to review.

Scalability was therefore considered as the volume and variety of business information increased. The system might need adjustments to accommodate additional branches, products, users, and sales records while continuing to support its core functions. Reviews of database performance, reporting needs, and application processes could help identify where future changes would be useful as Kape Tubong expanded its operations.

Potential enhancements were also considered as part of planning for the system's continued development. These could include predictive analytics, sales forecasting, more advanced reports, and additional visualization techniques. Such enhancements would require consideration of the organization's needs and the availability and quality of the data needed to support them. They were treated as possible future developments rather than as replacements for the system's existing sales monitoring and reporting functions.

Maintenance documentation was updated to record corrections, updates, configuration changes, and enhancement decisions. Keeping these records provided a reference for understanding how the system had changed and why particular adjustments were made. It also supported more organized technical support and future maintenance activities.

Through monitoring, corrective work, security and database reviews, user feedback, and consideration of changing business requirements, the Maintenance Phase supported the continued usefulness of BREWINSIGHT BI. These activities helped the system remain aligned with Kape Tubong's needs as its operations and information requirements evolved.

3.3 System Architecture

BREWINSIGHT BI used a client-server architecture consisting of three major layers: the client layer, server layer, and database layer. The architecture organized the system into interconnected components responsible for user interaction, application processing, and data storage. This structure allowed the system to centralize Kape Tubong's sales information while providing users with access through a web browser. A web application commonly follows a client-server structure in which the client handles user interaction while the server processes requests and returns the appropriate response [20]. The client layer served as the main point of interaction between users and the system. Users accessed BREWINSIGHT BI through web browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge. HTML, CSS, JavaScript, and Bootstrap were used to create the system pages, forms, navigation elements, tables, charts, and dashboard components. The interface provided users with access to dashboards, reports, sales information, analytical functions, and other authorized features.

The server layer handled the main application logic of BREWINSIGHT BI. It processed requests from the client, applied business rules, communicated with the database, performed calculations, managed authentication, generated reports, and returned the appropriate information to the user. The system was developed using Python Flask and operated within the development environment supported by XAMPP and Apache. The server components were organized into modules responsible for authentication, user management, sales management, analytics, reporting, and database communication. The database layer served as the centralized storage location for the system's sales and operational information. MySQL was used as the relational database management system to store information such as users, branches, products, sales transactions, and other required records.

The database used relationships among related tables to support organized storage and retrieval of information. Primary keys were used to identify individual records, while foreign keys established relationships between related tables. MySQL supports foreign keys for cross-referencing related data and maintaining consistency between related records [21]. SQL operations were used to filter, sort, aggregate, retrieve, and compare sales information. These operations supported revenue summaries, product performance analysis, branch comparisons, and historical trend analysis. The request-response process connected the client, server, and database layers. When a user performed an action through the browser, the request was sent to the server, which processed the request and, when necessary, retrieved or modified information in the database before returning the resulting information to the user.

The architecture supported centralized monitoring of Kape Tubong's kiosks and main branch. Instead of requiring users to review separate records individually, sales information could be stored in one centralized database and accessed through the web-based application. This reduced the need for manual consolidation because stored sales information could be processed for dashboards, analytical views, and reports. The architecture also supported centralized system updates because changes to the application could be applied to the server and accessed by users through their web browsers. In addition, the system allowed possible future expansion by accommodating additional branches, products, users, sales records, and analytical features as business requirements increased. Database backup and management were also considered because the database served as the central repository of important business information. Through the connection of user access, application processing, data storage, and analytical reporting, BREWINSIGHT BI provided Kape Tubong with a centralized environment for sales monitoring and Business Intelligence activities.

3.4 Tools and Technologies Used

- The development of BREWINSIGHT BI used several tools and technologies to support the creation of the web-based Business Intelligence system. HTML, CSS, JavaScript, and Bootstrap 5 were used for front-end development. These technologies were used to create the structure, design, interaction, navigation, and responsive layout of the system.

- Python Flask was used for back-end development. It handled application routes, server-side processing, authentication, database communication, business logic, and other application operations required by BREWINSIGHT BI.

- MySQL was used as a database management system. It stored sales information, user records, branch information, product data, transaction details, and other information required by the application.

- Visual Studio Code was used as the primary development environment for writing and organizing the system source code. XAMPP was used to support the local development and server environment.

- Google Chrome was used for browser-based system access and testing. Chart.js was used for the presentation of analytical information through charts and visualizations. Git was used for version control and development tracking.

3.5 Data Gathering Techniques

- The researchers used structured interviews, observation, document analysis, and surveys to gather the information required for the development of BREWINSIGHT BI. These techniques allowed the researchers to obtain information about the existing business processes, sales records, reporting practices, user needs, and technology preferences of Kape Tubong.

- Structured interviews were conducted with the business owner, Bryan Tacayon, and selected personnel. The interviews focused on sales recording procedures, reporting requirements, branch monitoring practices, problems encountered in handling sales information, and expectations for the proposed system.

- Observation was conducted to examine the actual sales processes at the kiosks and main shop. The researchers observed how transactions were recorded, how sales information was handled, and how records were maintained.

- Document analysis was used to examine available sales records, receipts, manual logs, and other relevant business documents. These records helped the researchers identify the types of sales information that needed to be stored in the database and provided references for understanding the existing sales records.

- Surveys were also used to gather information from staff regarding their familiarity with technology, preferred reporting methods, and expectations regarding the proposed system. The information gathered through these techniques was organized and reviewed by the researchers and used during the analysis and design phases.

3.6 Testing Procedures

The testing procedures were conducted to determine whether BREWINSIGHT BI met its functional, technical, and operational requirements. The testing process included Unit Testing, System Testing, and User Acceptance Testing, together with checks for performance, reliability, security, compatibility, data accuracy, and error handling. The testing activities were conducted progressively to identify errors and make necessary corrections during development [19].

The researchers tested the different components of the system to determine whether each function performed according to its intended purpose. The testing process covered authentication, database connections, dashboard components, data entry forms, reports, account management, analytical functions, and other major features of BREWINSIGHT BI. The researchers also examined the interaction among the system components to ensure that information was properly processed, stored, and displayed.

Performance, compatibility, security, data accuracy, and error handling were also considered throughout the testing process. The researchers tested different inputs and operational conditions to identify possible errors or unexpected behaviors. The results of the testing activities were documented and used as a basis for making necessary corrections and improvements before the system was deployed.

Unit Testing

Unit Testing was conducted to examine the individual modules of BREWINSIGHT BI. Each module was tested independently to determine whether it performed its intended function. The researchers tested authentication, database connections, dashboard components, data entry forms, reports, account management, and analytical functions. The authentication module was tested using valid and invalid credentials to determine whether authorized users could successfully access the system and whether invalid credentials were rejected. The logout function was also tested to verify that user sessions could be properly terminated.

Database connection functions were tested to determine whether the application could connect to MySQL, retrieve information, insert records, and update information when authorized. The researchers also examined how the system

responded to database-related errors. Dashboard functions were tested to determine whether displayed summaries and visualizations corresponded to the stored sales information. Filters and selections were tested using different dates, branches, products, and available parameters. Report generation was also tested by checking whether reports contained correct information based on selected criteria and comparing the generated results with the corresponding database records.

Account management functions were tested to determine whether authorized users could manage accounts according to their assigned permissions. Restricted functions were also checked to ensure that they remained unavailable to unauthorized users. Forms were tested using valid, incomplete, incorrect, and boundary-value inputs to determine whether required fields were properly validated and inappropriate values were rejected. Analytical functions were tested to verify the accuracy of revenue calculations, product performance results, branch comparisons, and peak sales analysis by comparing system-generated results with verified values based on the stored sales records.

Error handling was tested using invalid and unexpected inputs to determine whether the system responded appropriately without causing system failure or loss of information. Performance was considered by observing the response of individual functions, especially those requiring significant processing. Compatibility was also considered by checking whether interface elements and functions behaved properly within the supported browser environment and different screen sizes. Edge cases such as empty search results, zero sales values, missing information, large data sets, and invalid selections were also tested. Regression testing was performed after modifications to determine whether changes affected previously tested functions. The researchers documented the identified errors, corrections, testing results, and observations throughout Unit Testing.

System Testing

System Testing was conducted after the major components had been integrated. The researchers tested the complete application to determine whether the front-end, back-end, database, dashboard, analytics, and reporting functions worked together correctly. Operational scenarios were used to simulate actual business activities, including user login, sales data entry, data retrieval, dashboard viewing, filtering, branch comparison, analytical review, and report generation.

Synchronization between system components was tested to determine whether information entered into the database could be retrieved correctly by the dashboard and reports. Analytical accuracy was also evaluated at the system level by comparing the complete data flow from database storage to analytical processing and dashboard presentation. The researchers checked whether revenue information, product performance results, branch comparisons, and other analytical outputs remained consistent throughout the system.

System stability and performance were also evaluated through repeated use and different operations. The researchers observed page loading, dashboard generation, database queries, filtering, and report generation to identify possible delays and determine areas for optimization. Concurrent access was also considered to determine whether the system could handle access by more than one authorized user without causing inconsistent information or significant system problems. Compatibility and responsiveness were reviewed across the supported browser environment and different screen sizes.

Security testing was conducted to examine authentication, authorization, input validation, and access restrictions. The researchers checked whether protected functions could be accessed only by authorized users. Error recovery was also tested by examining the system's response to invalid operations and selected unexpected conditions. Data integrity was verified by comparing stored information with the information displayed through the system. Testing documentation was maintained throughout System Testing to record identified errors, corrections, retesting activities, and results.

User Acceptance Testing

User Acceptance Testing was conducted with the intended users of BREWINSIGHT BI. Bryan Tacayon and selected staff, branch representatives, administrative users, and operational personnel participated in the evaluation based on their relevance to the system functions. The users performed actual or simulated business tasks using the system, including logging in, viewing the dashboard, checking sales information, examining revenue trends, reviewing product performance, comparing branch information, identifying peak sales periods, and generating reports.

The researchers observed how users navigated the system and interacted with its functions. They considered whether users could understand labels, locate features, interpret dashboard information, and complete common tasks without unnecessary difficulty. The clarity of the dashboard was also evaluated by allowing users to review charts, tables, summaries, and indicators and provide feedback regarding whether the information was understandable and useful for their business activities.

The learning process was considered by observing whether users could become familiar with the system after orientation and perform the required tasks. User feedback was gathered through observations, interviews, discussions, and evaluation forms. The feedback focused on usability, functionality, information presentation, navigation, reporting, and analytical features.

The researchers reviewed the feedback and identified areas that needed improvement. Necessary changes were implemented and tested again to determine whether the modifications addressed the identified concerns. Training and orientation were also considered during User Acceptance Testing, where users were guided through the basic system operations and provided with the information needed to use the dashboard, reports, and other system functions.

Testing Conclusion

The testing procedures provided a systematic way to examine the functionality and reliability of BREWINSIGHT BI. Unit Testing allowed the researchers to identify problems within individual modules, while System Testing verified the interaction among the different components of the complete application. These testing activities helped the researchers identify errors before the system was used in its intended operational environment.

The analytical testing procedures helped verify that revenue calculations, product performance information, branch comparisons, and peak sales results were consistent with the stored sales records. Security, compatibility, performance, and error-handling tests also provided additional information regarding the technical behavior of the system. These

tests helped identify possible weaknesses and allowed the researchers to make necessary adjustments before deployment.

User Acceptance Testing provided feedback from the intended users and helped determine whether the system was understandable and practical for actual business activities. The researchers considered user observations, interviews, discussions, and evaluation results when refining the interface, dashboard, reporting features, and other system components.

Overall, the testing procedures provided the basis for determining whether the major functions of BREWINSIGHT BI had undergone technical and user-based evaluation before deployment. The results from Unit Testing, System Testing, and User Acceptance Testing were used to identify necessary improvements and ensure that the developed system addressed the identified functional, technical, and operational requirements.

3.7 Evaluation Criteria

The developed BREWINSIGHT BI system was evaluated using five major criteria: functionality, usability, reliability, efficiency, and security. These criteria were selected because they represented important aspects of software quality and were relevant to the intended purpose of the system. The criteria were also aligned with the software quality characteristics described in ISO/IEC 25010 [17]. The evaluation focused on determining whether the system performed its intended functions, was easy to use, operated consistently, responded efficiently, and protected business information.

Functionality

Functionality evaluation examined whether BREWINSIGHT BI could perform the tasks for which it was developed and whether its features supported the requirements identified during the earlier phases. The researchers reviewed the system's sales management, data processing, dashboard visualization, revenue analysis, product performance analysis, branch comparison, peak sales period analysis, and reporting capabilities. The evaluation considered both whether each function was available and whether it produced an appropriate result when users performed the related task.

The researchers examined common user activities, including logging in, managing sales information, opening dashboards, applying filters, reviewing analytical results, and generating reports. For each activity, they considered whether users could complete the expected steps and whether the system responded with the information or outcome required. This helped assess the system's functions from the perspective of the tasks users would perform when monitoring sales and reviewing business information.

The sales management functions were evaluated to determine whether the system could store, retrieve, and process sales information within the centralized environment. The researchers considered whether relevant information from different business locations could be managed through the system and used by its other features. Since dashboards,

analyses, and reports depend on stored records, the evaluation also considered whether data processing connected properly with the system's sales information.

Dashboard functionality was reviewed to determine whether the system could present sales information in a form users could examine. The researchers checked whether dashboard views displayed the intended summaries and whether filters allowed users to focus on relevant records or periods. They also considered whether the resulting views helped users examine sales information across products, branches, or time periods.

Analytical functions were assessed by reviewing the outputs for revenue analysis, product performance, branch comparisons, and peak sales periods. The researchers compared calculations and summaries with the underlying sales data to check whether the outputs were consistent with the records being analyzed. Revenue totals and sales summaries were reviewed for accuracy, while product performance results were considered in relation to recorded quantities and revenue contributions. Branch comparisons were checked against the information associated with each location, and peak period results were reviewed against the relevant sales activity.

Completeness was also considered. The researchers examined whether BREWINSIGHT BI included the major functions identified during requirements analysis and whether these features were available for their intended uses. This included checking that the system supported the expected sales management, analytical, dashboard, and reporting activities. Reviewing completeness helped determine whether important requirements had been represented in the developed system.

Automation was evaluated by examining whether the system could produce sales summaries and reports from centralized records without requiring users to manually consolidate individual branch information. The researchers considered whether report generation used the available sales data and whether users could obtain the information required for review through reports or dashboard views. This evaluation addressed the system's intended role in reducing manual preparation and providing a more consistent way to access sales information.

Access control was evaluated by checking whether users could access functions appropriate to their assigned roles. The researchers reviewed whether authentication was required for protected areas and whether restricted functions were controlled according to configured permissions. This helped determine whether the system's available functions were presented and protected in line with the user roles identified during requirements analysis.

Scalability was considered in relation to the system's ability to accommodate additional sales records, users, products, and branches. The researchers considered whether the system's functional structure could support growth in the volume of information and the number of entities represented in its records. This was relevant to the system's continued use as Kape Tubong's operations expanded.

Usability

Usability evaluation examined how easily intended users could understand, navigate, and operate BREWINSIGHT BI when carrying out their tasks. The researchers considered whether the interface made the system's functions

understandable, whether information was organized in a logical way, and whether users could learn the steps needed to complete common activities. The evaluation covered the clarity of the interface, navigation, readability, consistency, ease of learning, system feedback, and the practical effort required to use the system.

The clarity of the interface was assessed by reviewing the purpose and presentation of its pages and controls. The researchers considered whether labels, buttons, menus, forms, and dashboard components communicated their intended use. Clear wording and recognizable controls can help users understand what actions are available and what will happen when they select them. The researchers also considered whether page content was arranged so that users could distinguish between data entry, record review, analysis, and reporting functions.

The layout and navigation were evaluated by examining how the system organized its information and features. Related functions were grouped together, and frequently used features were made accessible through the main navigation. The researchers considered whether the available menu options reflected users' expected tasks and whether the names and placement of those options made their purpose clear. They also examined whether users could move between pages without losing track of where they were or having difficulty returning to a relevant function.

The ease of locating functions was an important part of the evaluation. Users needed to find areas for reviewing sales information, opening dashboards, examining analytical results, applying filters, and generating reports. The researchers considered whether these functions were accessible through the page structure and navigation provided. A well-organized interface could reduce the time users spent searching for a feature and help them focus on the business information they needed.

Consistency was reviewed across the system. Similar buttons, controls, labels, tables, and navigation elements were examined for consistent appearance and behavior. When comparable elements work in similar ways, users can apply what they have learned on one page to other areas of the system. Consistency can also help reduce uncertainty when users encounter familiar controls in different functions.

Readability and visual organization were evaluated particularly in the dashboard and reports. The researchers examined charts, tables, labels, summaries, and indicators to determine whether information was presented clearly. They considered whether users could identify the information being shown, understand the labels and categories, and distinguish relevant values or trends. The organization of dashboard components was also considered, including whether summaries and visualizations were arranged in a way that supported review of sales activity.

The researchers observed users after orientation to see whether they could perform basic tasks with the system. These tasks included navigating to relevant pages, reviewing sales information, using dashboard elements, and accessing reports. Observation helped reveal difficulties that might not be apparent from reviewing the interface alone, such as uncertainty about where to start, confusion about a control, or difficulty interpreting a displayed result.

System feedback was also evaluated. The researchers considered the messages presented after successful actions, invalid inputs, or errors. Feedback needed to help users understand whether an action had been completed or whether the system required further input. When an error occurred, the researchers considered whether the message explained

the problem clearly enough for users to decide what to do next. Helpful feedback can reduce uncertainty and support recovery from mistakes.

The evaluation also considered whether users could recover from errors or incorrect selections. For example, the researchers examined whether a user could revise an entry, change a filter, or return to the relevant page after an unsuccessful action. The interface's ability to make correction steps understandable was relevant to helping users continue their work without unnecessary frustration.

User productivity and role-related needs were considered as well. The researchers examined whether the system supported the activities expected of different user roles and whether users could review sales information, monitor branch results, and generate reports without unnecessary manual steps. The aim was to assess whether the organization and presentation of the system's functions helped users complete their responsibilities efficiently.

Reliability

Reliability evaluation examined whether BREWINSIGHT BI could operate consistently during normal use and continue to produce correct results across repeated tasks. The researchers considered whether the application remained available, whether its functions behaved predictably, and whether business information remained accurate as it moved between the database, dashboards, and reports. The evaluation also considered how the system responded to conditions that could interrupt or complicate routine use.

The stability of the application was assessed by performing common tasks repeatedly. These activities included accessing the system, reviewing sales information, applying filters, viewing dashboards, using analytical functions, and generating reports. The researchers observed whether the system completed each activity without unexpected failures and whether repeated actions produced consistent results. This helped determine whether the application behaved reliably over the course of routine work rather than only during a single successful use.

The researchers also examined system behavior when processing different amounts of sales information. They considered whether records could be retrieved and processed consistently as the volume of information changed, and whether dashboards and reports continued to display the expected results. This was relevant to understanding how the system might behave as additional transactions and historical records were included.

Output consistency was evaluated by repeating selected analytical and reporting functions and comparing their results. The researchers reviewed whether revenue totals, sales summaries, product performance information, branch comparisons, and peak sales period results remained consistent when the same underlying records and criteria were used. When results varied unexpectedly, the relevant calculation or data-processing steps could be examined to identify the source of the difference.

The evaluation also considered how the system responded to unexpected or incomplete conditions. These included invalid inputs, missing information, empty search results, and other situations that could affect normal processing. The researchers observed whether the application handled these cases in a controlled manner and whether it displayed

suitable feedback instead of producing misleading results or failing unexpectedly. This helped assess whether the system could continue to support users when the information provided did not match the expected format or produced no records.

Database integrity was examined by comparing stored records with the corresponding information displayed in the application. The researchers considered whether sales information remained accurate when retrieved for tables, dashboards, analytical functions, and reports. They also reviewed whether information changes were reflected in the parts of the system that relied on those records. These checks helped assess whether users could rely on the displayed information as a faithful representation of the stored data.

System availability was considered by checking whether the application and database services could be accessed when required. The researchers reviewed whether users could reach the system and whether the application could communicate with the database to retrieve and process information. Problems with either service could affect users' ability to review records or generate reports, so their availability formed part of the reliability evaluation.

Concurrent access was also considered. The researchers examined whether multiple authorized users could access the system without causing inconsistent information or significant problems in its operation. This included considering whether users could work with the system at the same time and whether shared records remained consistent during their activities.

Data synchronization between the database, dashboards, and reports was evaluated to determine whether updated information appeared in relevant system views. The researchers considered whether the data used for summaries and analyses reflected the records stored in the database and whether changes were carried through to the appropriate outputs. Consistent synchronization helped ensure that users reviewing different parts of the system were working with compatible information.

Backup and recovery were considered because sales information is important to Kape Tubong's operations. The researchers reviewed the role of backup procedures in protecting stored records from possible loss and considered whether recovery procedures could support restoration of information if needed. These measures were part of maintaining the availability and integrity of business data.

Efficiency

Efficiency evaluation examined how effectively BREWINSIGHT BI used available system resources and how quickly it responded to user operations. The researchers considered the time required to access common functions, retrieve and process sales information, update interface elements, and produce analytical outputs. They also reviewed whether the system reduced unnecessary manual work by automating calculations and report preparation.

Response time was observed during common activities, including opening pages, loading dashboards, retrieving sales records, applying filters, and generating reports. The researchers considered whether the system returned information promptly enough for users to continue their work without undue delay. They also examined whether response times

differed among functions, since a page that displays a simple record list may require different processing from a dashboard that calculates and presents several summaries.

Database performance was evaluated by examining the queries used to retrieve, filter, sort, and summarize sales information. These operations affect how quickly the system can present useful records and results. The researchers considered whether queries returned the requested information efficiently and whether database processing remained practical as the amount of stored sales information increased.

The movement of information between the database, server, and client interface was also reviewed. Data synchronization was considered in relation to how efficiently the system transferred records and updates between these components. Delays in communication could affect how quickly users see information or how promptly system functions complete, so the researchers examined this flow as part of the overall efficiency evaluation.

Analytical processing was assessed by observing the time needed to generate revenue summaries, product performance results, branch comparisons, and other dashboard information. These functions may require the system to retrieve and calculate information across multiple records. The researchers considered whether calculations completed in a reasonable period and whether the results could be displayed without unnecessary delay.

The responsiveness of the user interface was reviewed through common controls and interactions. The researchers examined buttons, forms, navigation elements, filters, and dashboard components to determine whether they responded promptly to user actions. This included considering whether pages and interface elements remained usable while information was being processed or retrieved.

Multiple-user access was considered to understand whether system performance remained acceptable when several authorized users accessed the application. The researchers reviewed whether simultaneous activity affected access to pages, database operations, dashboards, or reports. This was relevant to the system's intended use across Kape Tubong's branches and by users with different responsibilities.

System initialization was evaluated by observing how long the application took to load its primary components and establish required connections. This included access to the application and the connections needed for the system to communicate with its database. A smooth start-up process can help users begin their work without avoidable waiting.

Data storage and retrieval efficiency were also examined. The researchers observed how quickly sales information could be stored and later retrieved for review or processing. These operations underpin the system's management, dashboard, analytical, and reporting functions, so delays in handling records could affect several parts of the application.

Automation was considered as a way to reduce the manual work required to review sales information. Automated dashboard calculations and reports could reduce the need for users to combine records or calculate summaries themselves. The researchers considered whether these functions generated the needed information through the application and whether the automated process reduced steps in routine reporting activities.

Resource use was reviewed in relation to the available server, database, and browser resources. The researchers considered whether the system's operations made practical use of these resources while performing its intended functions. This included attention to the work required to load pages, process queries, calculate summaries, and display dashboards or reports.

Security

Security evaluation examined the measures used to protect BREWINSIGHT BI, its functions, and the business information stored in the system. Because the application handled sales records and user account information, the researchers considered whether access was limited to authorized users and whether information was handled through appropriate system controls. The evaluation covered authentication, authorization, role-based access, input validation, data protection, activity monitoring, backups, database protection, and user activity during concurrent access.

Authentication was evaluated to determine whether users needed valid credentials to access protected areas of the application. The researchers tested successful and unsuccessful login attempts and reviewed how the system responded when credentials were invalid or incomplete. These checks helped determine whether the login process prevented users without valid credentials from entering protected parts of the system and whether the response to unsuccessful attempts was handled appropriately.

Authorization and role-based access control were examined after login. The researchers checked whether users could access functions that matched their assigned roles and whether restricted functions remained unavailable to users without the necessary permissions. This included considering how access to sales information, management features, reports, and account-related functions was controlled. The purpose was to ensure that users could perform their assigned tasks without receiving broader access than their responsibilities required.

Input validation was evaluated by submitting different kinds of information through system forms. The researchers considered how the application handled valid entries, incomplete information, invalid values, and unexpected input. These checks helped assess whether the system identified information that did not meet expected requirements before processing or storing it. Appropriate validation can reduce incorrect records and help protect system processes from inputs that could cause unexpected behavior.

Data protection was considered because BREWINSIGHT BI stored sales records and user account information. The researchers examined whether stored information was accessed through authorized system operations and whether access to records was consistent with user permissions. They also considered how database protection supported the confidentiality and integrity of business information, including whether system functions handled data in a controlled manner.

Activity monitoring was included in the broader security review as a way to support oversight of system use. The researchers considered whether relevant user activities could be reviewed when investigating questions about access or operations. Such monitoring can help provide context for reviewing how the system was used and whether actions were performed through authorized accounts.

Backup procedures were also considered as part of protecting business information. Maintaining copies of important records can help reduce the impact of data loss and support recovery if stored information becomes unavailable or damaged. The researchers reviewed backup practices in relation to the need to preserve sales and account information.

Concurrent access was examined from a security perspective to determine whether users could perform only the operations allowed by their accounts while other authorized users were also using the system. This included considering whether permissions continued to apply during simultaneous access and whether one user's activity could provide access to functions restricted for another user.

The researchers also considered database protection as part of the system's overall security. This involved reviewing how the application communicated with the database and whether database operations were carried out through defined system functions. Protecting the database was important because it contained the records used by sales management, analysis, dashboards, and reports.

Evaluation Conclusion

The evaluation of BREWINSIGHT BI was organized around five criteria: functionality, usability, reliability, efficiency, and security. Together, these criteria provided a structured way to examine whether the system met its intended requirements and was suitable for use in Kape Tubong's sales monitoring, analysis, and reporting activities. Each criterion focused on a different aspect of system quality, allowing the researchers to consider both what the system could do and how it performed for its intended users.

Functionality focused on whether the system provided the required business and analytical functions and whether those functions produced appropriate results. The evaluation considered sales management, dashboard visualization, revenue analysis, product performance analysis, branch comparison, peak sales period analysis, and report generation. It also examined whether users could complete expected activities, such as logging in, managing sales information, applying filters, reviewing summaries, and generating reports. Accuracy and completeness were considered by comparing calculations and analytical outputs with the underlying sales records and by checking whether the major functions identified during requirements analysis were available.

Usability examined whether intended users could understand and operate the system effectively. The researchers considered the clarity of the interface, organization of information, navigation, readability, and consistency of controls and page layouts. They also considered whether users could locate functions and complete common tasks after orientation, including reviewing dashboards, examining sales information, and accessing reports. System messages and error feedback were reviewed to determine whether users could understand the outcome of their actions and respond appropriately when information was incomplete or invalid.

Reliability focused on whether BREWINSIGHT BI operated consistently and maintained the integrity of business information. The evaluation considered how the system behaved during repeated use, whether analytical and reporting functions produced consistent results, and how the application handled invalid inputs, missing information, and empty results. It also examined whether records stored in the database corresponded with information displayed in tables, dashboards, and reports. Availability, data synchronization, concurrent access, and backup and recovery practices were considered as part of the system's ability to support dependable use.

Efficiency examined how promptly and effectively the system handled user operations and used available resources. The researchers considered response time during page loading, dashboard generation, data retrieval, filtering, analysis, and report generation. Database processing and the movement of information between the database, server,

and interface were also reviewed. In addition, the evaluation considered whether automated calculations and reports reduced the manual effort required to consolidate records and prepare sales summaries.

Security focused on controls designed to protect user accounts, system functions, and business information. The evaluation considered whether users needed valid credentials to access protected areas, whether permissions matched assigned roles, and whether restricted functions were properly controlled. Input validation and database protection were also examined, along with procedures related to activity monitoring and backups. These considerations helped assess whether access to sales records and account information was appropriately limited and managed.

Using these criteria gave the researchers a consistent basis for reviewing different aspects of BREWINSIGHT BI. The criteria connected the evaluation to the system's requirements: functionality assessed whether required features were provided; usability considered whether users could work with them; reliability examined consistency and data integrity; efficiency considered responsiveness and processing; and security assessed the protection of access and information.

The evaluation results could also help identify areas that needed correction or further development. Findings related to a particular criterion could point to issues in the corresponding system functions, interface, processing, or controls. This provided a basis for reviewing the system's operational suitability and determining possible improvements.

REFERENCES

[17] ISO/IEC 25010:2023, *Systems and software engineering—Systems and software Quality Requirements and Evaluation (SQuaRE)—Product quality model*, 2nd ed., 2023. [Online]. Available: https://www.iso.org/standard/78176.html

[18] K. Beck, M. Beedle, A. van Bennekum, A. Cockburn, W. Cunningham, M. Fowler, J. Grenning, J. Highsmith, A. Hunt, R. Jeffries, J. Kern, B. Marick, R. C. Martin, S. Mellor, K. Schwaber, J. Sutherland, and D. Thomas, "Manifesto for Agile Software Development," Agile Manifesto, 2001. [Online]. Available: https://agilemanifesto.org/

[19] ISO/IEC/IEEE 29119-1:2022, *Software and systems engineering—Software testing—Part 1: General concepts*, 2nd ed., 2022. [Online]. Available: https://www.iso.org/standard/81291.html

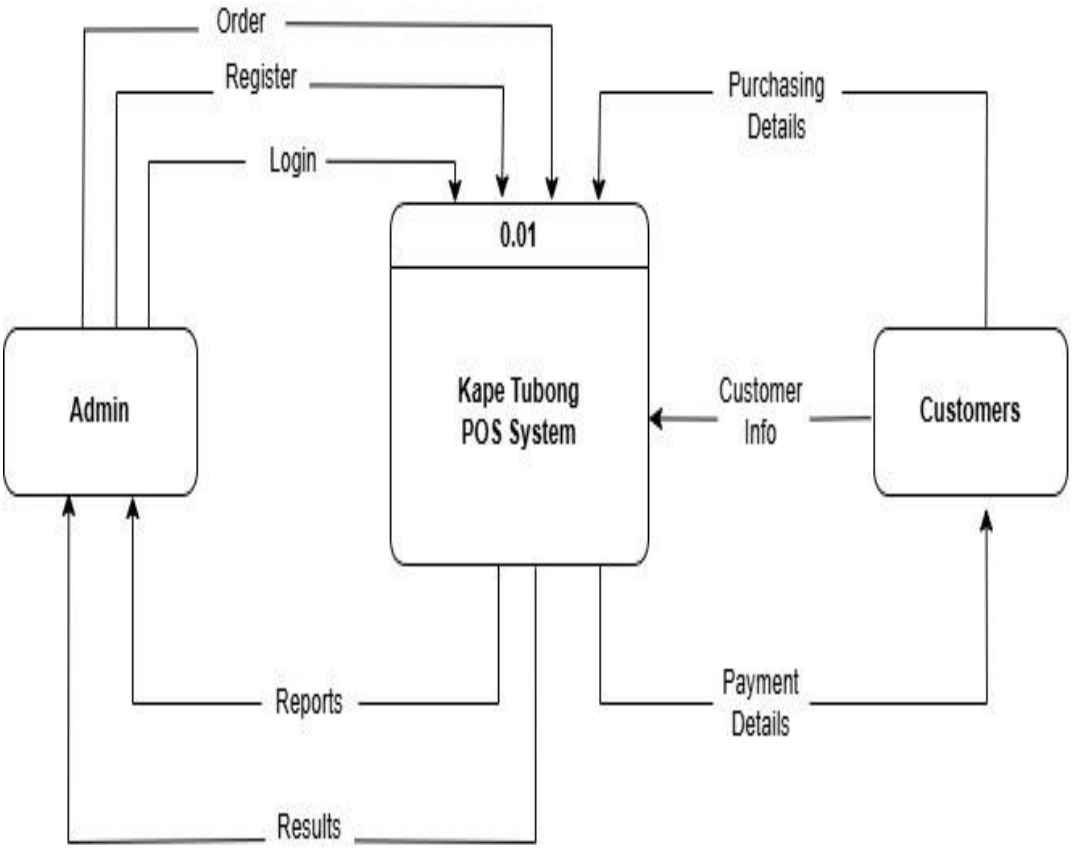
[20] Amazon Web Services, "What is a web application?," AWS. [Online]. Available: https://aws.amazon.com/what-is/web-application/

[21] Oracle Corporation, "MySQL 8.4 Reference Manual: FOREIGN KEY Constraints," MySQL. [Online]. Available: https://dev.mysql.com/doc/refman/8.4/en/create-table-foreign-keys.html

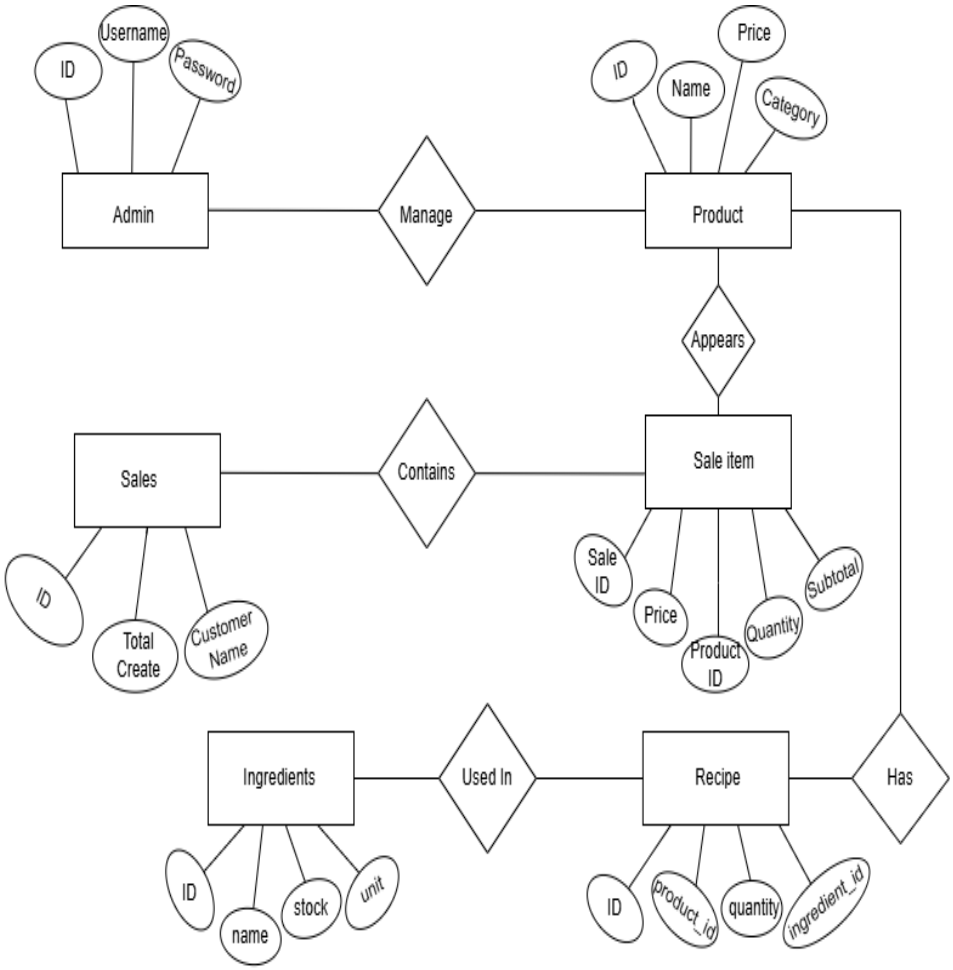
SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

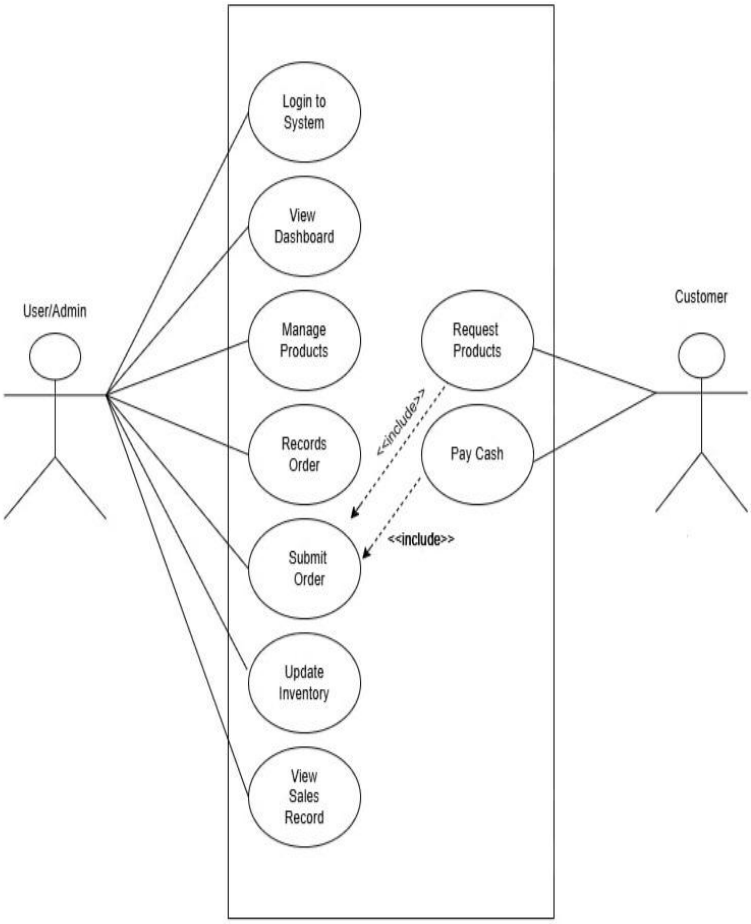
Admin → Pos System→ Customers



Entity Relationship Diagram (ERD)



Use Case Diagram


Login to System
View Dashboard
Manage Products
Records Order
Submit Order
Update Inventory
View Sales Record
Request Products
Pay Cash
<<include>>
<<include>>
User/Admin
Customer

CHAPTER IV

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong was developed as a web-based Point-of-Sale (POS), inventory management, sales reporting, and business analytics system to improve the daily operations of Kape Tubong. It was designed to replace the manual process of recording sales, managing products, monitoring ingredient inventory, and preparing reports. The system provided a faster and more organized way of handling business transactions while reducing repetitive manual tasks and possible errors in record keeping.

Before the system was developed, sales transactions and inventory updates were recorded manually. This process took more time and sometimes resulted in inaccurate records, delayed inventory updates, and difficulties in monitoring product sales and available ingredients. Preparing reports also required checking transaction records individually, making the process less efficient. These limitations created a need for a centralized system that could record, process, store, and present business information in an organized manner.

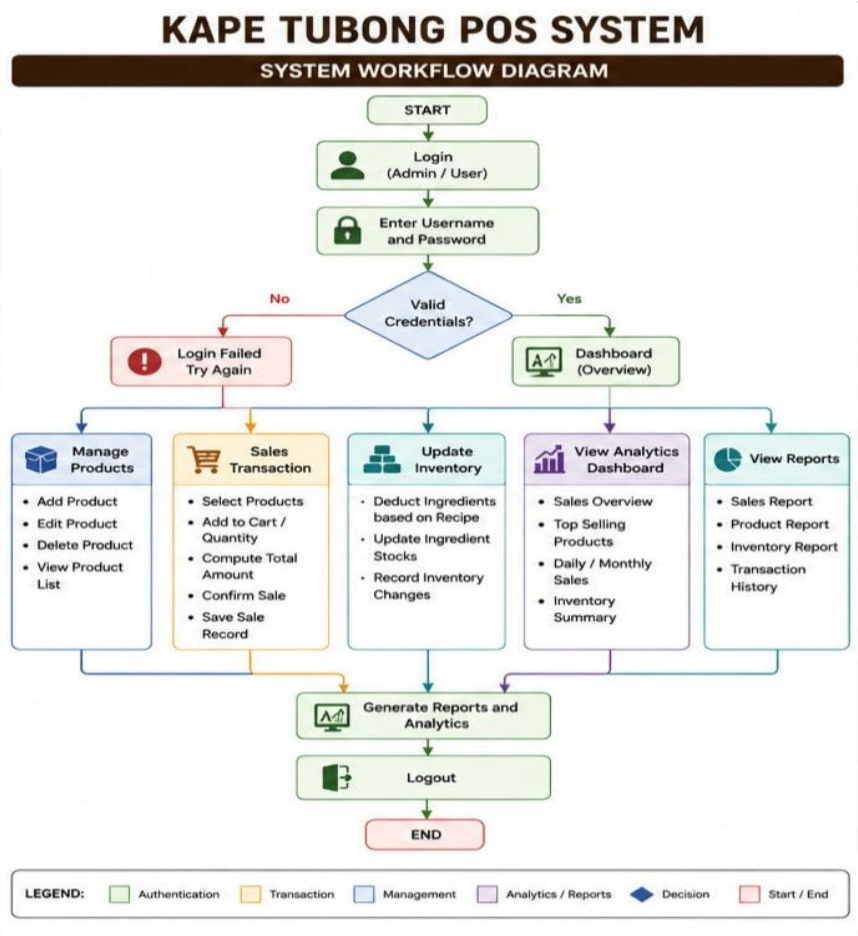
The developed system addressed these problems by providing secure user authentication and centralized access to the system. After entering a valid username and password, authorized users were directed to the dashboard where they could access the different features of the system. If incorrect login credentials were entered, the system displayed a login failed message and required the user to enter valid credentials before access was granted.

The dashboard served as the main page of the system and allowed users to perform different business operations. The system included the following modules:

- **User Authentication –** verified the username and password of administrators and staff before granting access to the system.
- **Dashboard –** displayed an overview of sales, inventory, and other important business information.
- **Product Management –** allowed authorized users to add, edit, delete, and view available products.
- **Sales Transaction –** allowed users to select products, add items to the cart, calculate the total amount, confirm the sale, and save transaction records.
- **Inventory Management –** automatically deducted ingredient stocks based on product recipes, updated inventory levels, and recorded inventory changes after every sale.
- **Analytics Dashboard –** displayed the sales overview, top-selling products, daily and monthly sales, and inventory summary.
- **Reports Module –** generated sales reports, product reports, inventory reports, and transaction history.

All of these modules worked together to support the overall operations of Kape Tubong. Every time a sale was completed, the transaction was saved automatically, the corresponding ingredient inventory was updated, and the necessary information became available for reporting and analysis. Because of this integration, business operations became more organized, record keeping became more consistent, and less time was spent on repetitive manual tasks. The system also made it easier for the business owner and authorized users to monitor sales, inventory, and overall business performance.

Figure 4.1 illustrates the overall workflow of the developed system.



The workflow connected the administrator and staff users through a centralized system, allowing product management, sales transactions, inventory monitoring, report generation, and sales analytics to be completed within one platform. This process made the daily operations of Kape Tubong more organized and efficient by reducing manual work and improving access to business records.

4.3 System Development Methodology

The development of the Kape Tubong Point-of-Sale (POS) System followed the Agile Software Development Methodology. Agile provided an iterative and flexible framework that allowed the researchers to build the system in manageable increments, review completed work, and make refinements as development progressed. This approach

suited a project in which the system needed to support the business owner's operational requirements and could benefit from feedback gathered while functions were being developed.

Agile emphasizes collaboration, working software, and responsiveness to change during software development [18]. These principles shaped how the researchers organized the work and involved the business owner. Instead of waiting until the entire system had been completed to seek feedback, the researchers could present work at different stages, discuss whether it supported the intended business processes, and use relevant comments to guide subsequent development. This helped maintain alignment between the system's features and Kape Tubong's needs.

The project was organized around a Product Backlog and a series of related development activities. The Product Backlog served as a structured list of proposed system features, requirements, and improvements. Items were considered and prioritized according to their importance to the POS system and the business activities it was expected to support. This provided a reference for selecting work to be addressed during each development iteration.

During Planning, the researchers determined which backlog items to address and organized the work according to the project's objectives, schedule, and available resources. Planning helped define the focus of each iteration and clarify the expected outcome of the work. It also provided a basis for coordinating development activities and identifying which requirements needed further discussion.

In Requirements Analysis, the researchers examined Kape Tubong's business processes and clarified what the POS system needed to do. This activity involved identifying user needs, expected system functions, relevant data, and any constraints that could affect implementation. Requirements could be reviewed as new information or feedback emerged, helping the researchers refine the system's scope and priorities.

The Design activity translated requirements into plans for the system's structure and user experience. The researchers considered how system components would be organized, how information would be stored and processed, and how users would interact with the POS features. Design work provided guidance for development and helped ensure that the planned functions were connected to the identified requirements.

During Development, the researchers implemented selected system features. Work was organized so that functions could be built incrementally and integrated into the developing application. This allowed the researchers to produce working portions of the system for review and testing, rather than postponing evaluation until the end of the project.

Testing was carried out to examine whether implemented features operated as expected and met the relevant requirements. The researchers checked system behavior, input handling, calculations, and interactions among components as appropriate to the features developed. Identified issues could be corrected and the affected functions reviewed again before they were considered ready for further evaluation.

The Review and Feedback activity provided an opportunity to assess the work completed during an iteration. The researchers reviewed developed features with the business owner and considered comments about their functionality and suitability for business use. Feedback could lead to changes in the existing implementation or influence the

priorities of later iterations. This recurring review helped keep development responsive to the business owner's input.

During Deployment, the completed system was prepared for use in its intended environment. This activity involved making the system available to authorized users and considering the preparation required for operational use. Deployment followed the development and evaluation work needed to establish that the system's key features were ready for use.

The Retrospective activity involved reflecting on the development process after an iteration or stage of work. The researchers considered what had gone well, what challenges had occurred, and what adjustments could improve subsequent work. This reflection supported ongoing refinement of the development approach as well as the system itself.

These activities were performed iteratively. Information gathered during requirements analysis could inform design; design decisions guided development; testing and review could identify issues or changes; and feedback could update the Product Backlog and influence future planning. This cycle allowed the researchers to refine the POS system throughout development while maintaining a structured process for completing its features.

Product Backlog

The researchers prepared the Product Backlog as an organized list of the features, requirements, and priorities identified for the Kape Tubong Point-of-Sale (POS) System. It provided a shared reference for what the system was expected to support and helped connect the business's operational needs with the development work. The backlog included functions related to order taking, inventory management, sales reporting, customer transactions, product management, and revenue analytics.

Each backlog item represented a function or requirement that needed to be considered during development. For example, order taking and customer transactions addressed the recording of sales activity, while product and inventory management supported the handling of the items available for sale. Sales reporting and revenue analytics focused on organizing transaction information so that business performance could be reviewed. Together, these items described the principal areas the POS system needed to address.

The researchers prioritized backlog items according to their importance to Kape Tubong's operations. This helped identify which functions should receive attention earlier in development and which could be scheduled for later iterations. Prioritization provided a practical basis for organizing the work around the system's most relevant needs while accounting for the available development time and resources.

The Product Backlog guided the planning of development activities. Before work began on an iteration, the researchers could refer to it to select features for development and determine the intended scope of that work. This helped clarify what the team needed to implement and provided a way to track progress as features moved from planned work into development, testing, and review.

The backlog also supported communication about the system's scope. By recording expected functions and their relative priorities, it gave the researchers and the business owner a reference for discussing what the POS system should provide. If a feature needed clarification, the backlog could help identify the requirement for further discussion before implementation.

As development continued, the Product Backlog was reviewed and updated when additional requirements, clarifications, or changes were identified. New items could be added for consideration, and existing priorities could be revised to reflect updated business needs or feedback. This allowed the development plan to remain responsive while preserving a structured list of work.

Planning

During the Planning phase, the researchers selected the features to be addressed in the current sprint and organized the work needed to develop them. They reviewed the prioritized items in the Product Backlog and identified which features could be handled within the sprint, taking into account the project objectives, available time, and development resources. This helped keep each sprint focused on a manageable set of tasks.

The selected features were broken down into smaller development activities. These could include clarifying requirements, preparing or revising designs, implementing the function, connecting it with related system components, testing its behavior, and making revisions based on the results. Breaking work into these activities helped the researchers understand what needed to be completed and made it easier to monitor progress during the sprint.

The researchers established a project schedule to guide the sequence and timing of development work. The schedule helped coordinate coding, testing, evaluation, and revision, while accounting for the dependencies among tasks. For example, a feature could require a design decision or a related data structure to be completed before it could be implemented and reviewed. Planning these dependencies helped the researchers organize work in a practical order.

Expected outputs were also defined for each sprint. These outputs described the work the researchers intended to complete, such as a developed feature or an improved system function ready for review. Clear expectations gave the development team a reference for assessing whether the sprint's objectives had been met and whether any remaining work should be carried forward.

Planning also helped the researchers allocate available time and resources. They considered the effort required to develop each selected feature and reserved time for testing and revision, rather than treating coding as the only activity. This allowed the team to identify issues, evaluate the completed work, and incorporate necessary changes before proceeding to the next development activity.

As sprint work progressed, the plan provided a basis for reviewing progress and adjusting task priorities when needed. If a requirement needed clarification or a task revealed additional work, the researchers could consider the

effect on the sprint's expected outputs and schedule. This supported an organized development process while allowing the team to respond to information discovered during implementation.

Requirements Analysis

During the Requirements Analysis phase, the researchers examined the features selected for development to determine what each one needed to do and how it would support Kape Tubong's operations. They analyzed the functional requirements, business rules, workflows, and system specifications for each feature. This involved considering how the intended users would interact with the system, what information they would provide or retrieve, and what results the system would need to produce.

The analysis was grounded in Kape Tubong's operational processes and the information needs of its intended users. The researchers considered how sales transactions were handled, how inventory records were maintained, how reports were prepared, and how user accounts were used. They also examined how these activities related to one another. For instance, a sales transaction could affect recorded sales information and inventory quantities, while the resulting transaction data could later support reports and analytics.

For each selected feature, the researchers clarified the expected workflow from the user's perspective. They identified the steps required to complete an activity, the information needed at each step, and the system's expected response. This helped define how users would carry out tasks such as recording sales, managing products, reviewing inventory, accessing reports, or working with account-related functions. Mapping these workflows also helped identify dependencies among features and information.

Business rules were analyzed to establish how the system should handle operational information. These rules could define how sales transactions were recorded, how product and inventory information was managed, how reports were generated, and how user access was controlled. Clarifying these rules before implementation helped provide consistent expectations for system behavior and reduced ambiguity during development.

The researchers also considered the information requirements for each feature. They examined what data would need to be entered, stored, retrieved, or calculated to support sales transactions, inventory records, reporting, user accounts, and analytics. This included considering how information used by one function might be needed by another, such as sales records supporting reports or inventory information being associated with products.

System specifications were developed to describe the expected behavior of the selected features in sufficient detail for design and development. These specifications helped define how the system should respond to user actions, what information should be displayed, and how related records should be processed. The researchers also considered relevant constraints and conditions that could affect how a feature operated.

This phase helped ensure that each feature was clearly defined before development began. When a requirement was unclear or incomplete, it could be reviewed and clarified so that the researchers had a shared understanding of the

intended outcome. Establishing this understanding helped reduce the risk of implementing a feature that did not match the operational process or user need it was intended to support.

The results of Requirements Analysis served as the basis for designing system functions and organizing their implementation. The identified requirements guided decisions about how sales transactions, inventory records, reports, user accounts, and analytics would operate. They also provided criteria that could later be used to review whether the developed features behaved as expected.

Design

During the Design phase, the researchers translated the requirements identified for the Kape Tubong Point-of-Sale (POS) System into plans for its interface, database, and workflows. The design described how users would interact with the application, how business information would be organized, and how the system would process and store that information. It provided the structure that would guide development and help ensure that the planned functions supported Kape Tubong's operational needs.

The user interface was designed to be clear and practical for the intended users. The researchers planned the application's page layouts, navigation, forms, tables, and other controls so users could locate and use the required functions. Transaction-related screens were designed to support the entry and review of sales information, while inventory screens were planned to present product and stock information in an organized way. The design also considered how users would move between related tasks, such as recording a transaction and reviewing relevant product or inventory details.

The researchers considered how information should be displayed so that users can distinguish between different types of records and understand the available actions. Consistent layouts and controls were planned to make similar functions behave in familiar ways across the application. The interface design aimed to reduce confusion during routine activities while keeping transaction records, inventory information, and sales data accessible for review.

The database structure was designed to organize the information required by the system and establish relationships among users, sales, products, recipes, and ingredients. The researchers considered what information each type of record would need to contain and how records would connect. For example, sales records needed to relate to the products included in transactions, while product information could connect to recipes and ingredients where required by the system. User records also needed to support the management of accounts and access to system functions.

Planning these relationships helped establish how information would be stored and retrieved consistently. The database design provided a basis for organizing records so the application could use them for transaction processing, inventory management, and sales review. It also supported the system's ability to connect relevant information across functions rather than treating each set of records as unrelated.

The overall system workflow was planned to describe how information would move among the user interface, application processes, and database. When a user performed an action, such as entering a transaction or reviewing

inventory, the application would process the request, access or update the relevant records, and return the appropriate information to the interface. Mapping these steps helped clarify how user activities related to the system's underlying processing and data storage.

The researchers also considered how transaction, inventory, and sales information would be handled across related workflows. The design established a reference for how records would be entered, processed, stored, retrieved, and presented. This helped identify connections and dependencies among system functions before implementation began.

Development

During the Development phase, the researchers implemented the approved system modules according to the requirements and designs established in the preceding phases. This work converted planned workflows, interface layouts, and data structures into working features of the Kape Tubong Point-of-Sale (POS) System. Development followed the Agile sprint plan, with selected functions built and reviewed in manageable increments.

The coding work covered order processing, inventory management, sales recording, user account management, reporting, and revenue analytics. For each module, the researchers used the approved design and requirements to determine how users would interact with the feature, what information it needed, and how the system should respond. The modules were developed with attention to their connections to the database and to other parts of the application.

Order processing was developed to support the system's transaction workflow. This included implementing the relevant steps for handling order information and connecting those steps with sales recording. Inventory management was developed to support the system's handling of product and stock information. Sales recording provided a way for transaction data to be stored and made available for review and reporting.

User account management was implemented to support the creation and management of system accounts and access associated with them. Reporting and revenue analytics were developed to use stored sales information to provide summaries and results for business review. These modules depended on the availability and organization of transaction, product, inventory, and account information, so the researchers considered their interactions during implementation.

Each module was developed incrementally in alignment with the sprint plan. The researchers worked on selected features, integrated them with relevant system components, and reviewed their behavior as development progressed. This allowed completed functions to become part of the evolving application while related modules were still being developed. It also provided opportunities to identify dependencies, clarify requirements, and address problems before they affected additional areas.

After implementation, completed modules were integrated into the system and prepared for testing. Integration involved connecting user-facing functions with the application processes and database operations they required. The

researchers considered whether information could move between related modules and whether the combined functions supported the intended workflow. For example, order processing and sales recording needed to work together, while reports and revenue analytics depended on stored transaction information.

Incremental development helped the researchers identify issues within individual modules while their scope was still manageable. If a feature did not behave as expected, its implementation could be reviewed and corrected before further integration. This approach also made it easier to examine whether a change to one module affected connected functions.

The Development phase produced the working modules of the POS system for evaluation. Through implementation in Agile iterations and integration of completed features, the researchers established the system functions needed to support order processing, inventory management, sales recording, account management, reporting, and revenue analysis. These modules were then ready for testing and review.

Testing

Testing was conducted after the development of each module to determine whether its features behaved according to the specified requirements. The researchers used software testing activities to identify defects, examine system behavior, and assess whether the completed functions produced the expected results [19]. Testing took place throughout development so issues could be addressed as modules were built and integrated.

The researchers examined the system's main functions, including transaction processing, sales calculations, inventory updates, report generation, and user authentication. They performed relevant actions and compared the system's responses with the expected behavior. For transaction processing, they reviewed whether the system could handle the information needed to complete a sales transaction. They checked whether sales calculations reflected the transaction details and whether recorded sales information was available for later review.

Inventory updates were tested to determine whether changes associated with transactions were reflected in the relevant inventory records. The researchers also examined report generation to check whether reports used the available system information and presented the expected results. Authentication testing focused on whether users could access the system with valid credentials and whether access to protected features was appropriately controlled.

Testing considered different inputs and outcomes relevant to a feature. This helped the researchers examine how the system responded during ordinary use and whether it handled invalid or incomplete information appropriately. The interface and system responses were also reviewed to identify unexpected behavior that could prevent users from completing their tasks.

Errors or defects identified during testing were documented so the researchers could review and correct them. After a correction was made, the affected feature was checked again to determine whether the problem had been resolved. This process helped ensure that functions were reviewed after changes rather than assuming that a code correction had addressed the issue.

Testing was also performed after modifications to the system. When a feature was changed or improved, the researchers checked whether previously working functions continued to operate properly. This regression testing helped identify unintended effects on connected modules. For example, a change to transaction processing could affect sales records or inventory updates, while a change to stored information could affect reports that relied on those records.

The researchers also reviewed interactions among modules as the system was integrated. This helped determine whether connected functions exchanged and used information as intended. Testing these interactions was important because individual modules could work correctly on their own but still encounter problems when used together.

Review and Feedback

At the end of each sprint, the researchers presented the modules developed during that iteration to the business owner for review. These reviews gave the business owner an opportunity to examine the features, consider how they related to Kape Tubong's operational needs, and provide comments before the researchers moved on to subsequent work.

The feedback focused on the functionality, usability, and operation of the developed features. The business owner could comment on whether a module supported the intended task, whether its information and controls were understandable, and whether its workflow was suitable for the business. Comments, suggestions, and recommendations were recorded for consideration by the researchers.

The researchers reviewed this feedback to identify aspects of a feature that needed clarification, correction, or improvement. They considered how suggested changes related to the system's requirements and the current development priorities. Relevant revisions were incorporated into subsequent development activities, either by refining the existing module or by adding work to a later sprint. This process helped ensure that development remained responsive to the business owner's input and continued to reflect Kape Tubong's operational requirements.

Deployment

After the developed modules had undergone the required testing and evaluation, the researchers prepared the system for use in its intended environment at Kape Tubong. Deployment activities included installing the application, configuring the database, and preparing the components needed for implementation. These steps moved the system from development into an environment where its functions could be accessed for operational use.

Once the application and database were configured, the researchers verified the major system functions in the intended environment. They checked user authentication, transaction processing, inventory updates, reports, and

other important features to confirm that these functions operated after deployment. This verification helped identify environment-related issues that might not have appeared during development or earlier testing.

The researchers also checked that the application could communicate with the configured database and that relevant records could be processed by the system. Confirming these connections was important because transaction handling, inventory management, and reporting depended on the application's ability to retrieve and store information. Any issues observed during verification could be reviewed and addressed as part of the deployment process.

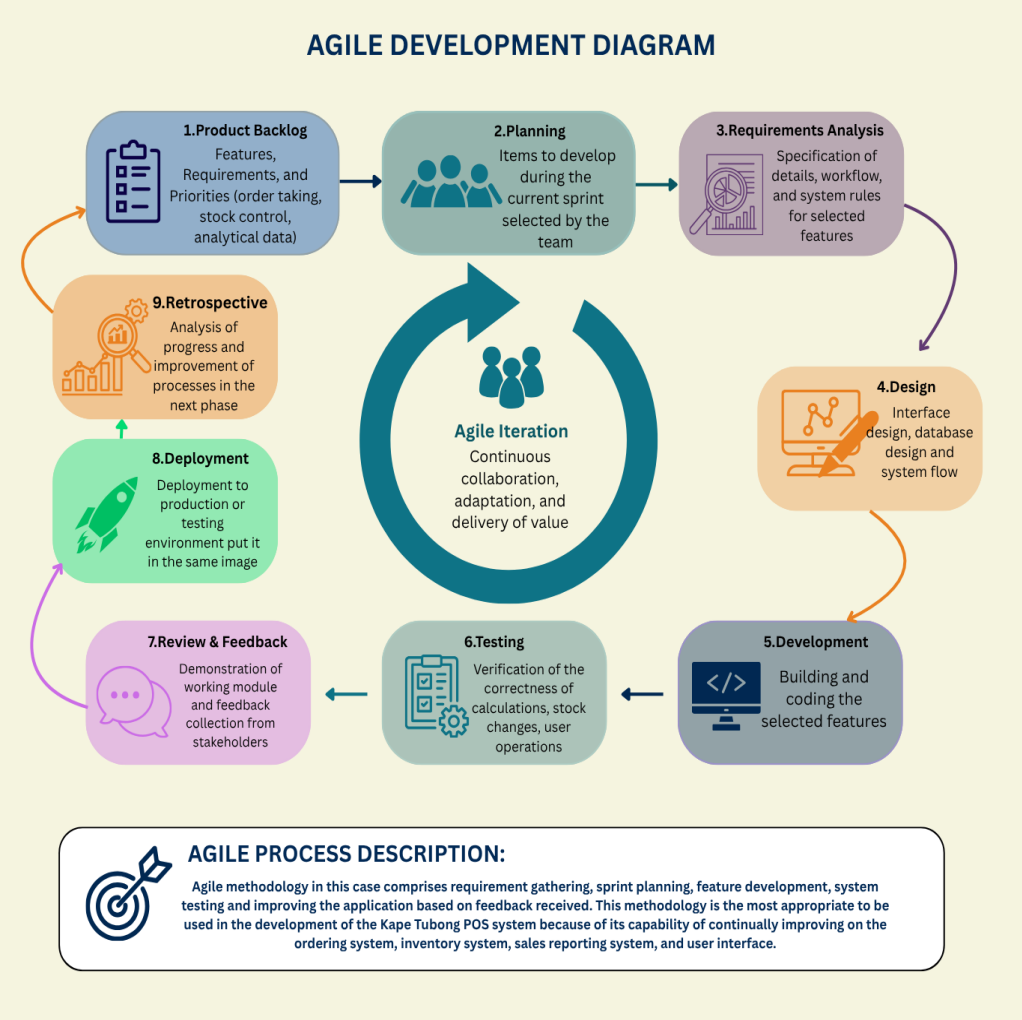
Retrospective

After deployment activities and at the completion of each sprint, the researchers conducted a retrospective to review how the development work had progressed. The retrospective provided an opportunity to discuss challenges encountered during the sprint, identify practices that had worked well, and determine where the process could be improved. This review considered both development activities and the coordination needed to complete and evaluate the planned work.

The researchers used the retrospective to reflect on issues such as difficulties in implementing a feature, unclear requirements, delays, integration concerns, or problems identified during testing and review. They also recognized effective practices that helped the team complete work or respond to feedback. Recording these observations provided a basis for deciding how to approach similar tasks in future iterations.

Lessons learned during each retrospective were applied to succeeding development activities. The researchers could adjust task organization, communication, testing, or review practices based on the issues identified. This continuous reflection helped the team improve its process over the course of the project and supported consistent attention to system quality.

Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.



Agile Process Description

The Agile process was a software development approach that focused on developing the system through small and manageable iterations called sprints. Instead of completing the entire application in a single development cycle, the researchers implemented system features incrementally while continuously gathering feedback from stakeholders. This approach allowed the researchers to identify issues at an early stage, implement necessary improvements, and verify that the developed system met the operational requirements of Kape Tubong [18].

For the development of the Kape Tubong POS System, the Agile methodology was selected because it provided flexibility in accommodating changes in business requirements while promoting continuous collaboration and system improvement. Throughout the development process, the researchers prepared the Product Backlog, planned each sprint, analyzed the system requirements, designed the application, developed the system modules, conducted testing, gathered stakeholder feedback, deployed completed features, and evaluated each sprint through retrospectives.The iterative process allowed the researchers to improve the system based on the results of testing and

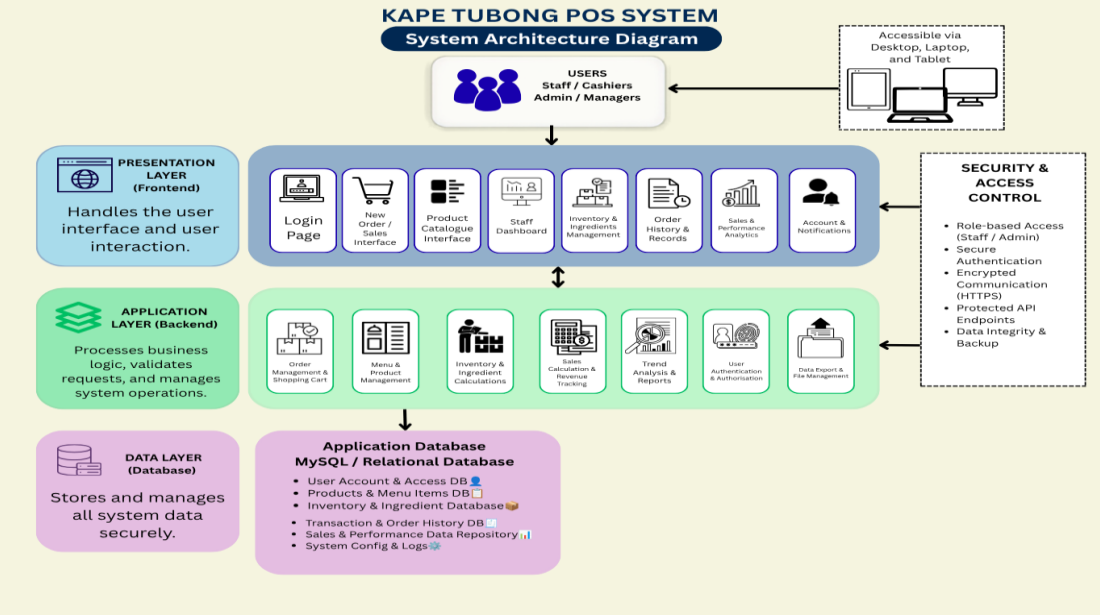
feedback gathered during development. By following this methodology, the researchers developed a system that supported order processing, inventory management, sales reporting, and revenue analytics for Kape Tubong.

4.4 System Architecture

The developed system followed a three-layer architecture that separated the Presentation Layer, Application Layer, and Data Layer into distinct components. This architectural design allowed each layer to perform its assigned responsibilities independently while communicating with the other layers when necessary. The separation of functions improved system organization, simplified maintenance, and provided a structured framework for processing, storing, and presenting business information.

The three-layer structure also supported the organization of the system's different functions. The Presentation Layer handled user interaction, the Application Layer processed system operations and business rules, and the Data Layer managed the storage and retrieval of information. This structure allowed the system to process sales, inventory, and business information in an organized manner.

Figure 4.3 illustrates the architecture of the developed system.



The architecture consisted of the following components:

Presentation Layer

The Presentation Layer served as the part of the system that users directly interacted with. It consisted of the visible interfaces that allowed staff, cashiers, and administrators to access the system and perform their assigned tasks. This layer displayed information retrieved from the lower layers while collecting user inputs such as sales transactions, inventory updates, and report requests.

The collected information was forwarded to the Application Layer for processing. The interface was developed using responsive web technologies to allow the system to function on tablets and smartphones used during daily business operations. It could also be accessed through desktop and laptop computers when needed.

The major interface components included:

- Login Page
- New Order / Sales Interface
- Product Catalogue Interface
- Staff Dashboard
- Inventory and Ingredients Management Interface
- Order History and Records Interface
- Sales and Performance Analytics Interface
- Account and Notifications Interface

Application Layer

The Application Layer served as the core processing component of the system. It received requests from the Presentation Layer, validated user inputs, applied the necessary business rules, and communicated with the Data Layer to retrieve or store information.

This layer processed customer orders, calculated transaction totals, updated inventory records, generated sales reports, analyzed revenue patterns, and managed user authentication and authorization. It also ensured that information exchanged between the user interface and database remained accurate and consistent throughout system operation.

The major functional components included:

- Order Management and Shopping Cart
- Menu and Product Management
- Inventory and Ingredient Usage Calculations
- Sales Calculation and Revenue Tracking
- Trend Analysis and Report Generation
- User Authentication and Authorization
- Data Export and File Management

Data Layer

The Data Layer managed the storage, organization, and retrieval of all system information using a MySQL relational database. It securely stored user accounts, product information, inventory records, sales transactions, recipes, ingredients, and other business data required by the system.

The Data Layer maintained relationships among the different database entities using primary keys and foreign keys. Foreign key constraints in MySQL help establish relationships between related tables and maintain referential integrity [20]. These relationships allowed the system to retrieve and process related information when required by different system functions.

The key data storage components included:

- User Accounts and Access Database
- Products and Menu Items Database
- Inventory and Ingredients Database
- Transaction and Order History Database
- Sales and Performance Data Repository
- System Configuration and Logs

4.5 Hardware and Software Requirements

This section presented the hardware and software requirements used in the development and implementation of the BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong. These requirements ensured that the system operated properly during development, testing, and implementation.

The hardware specifications provided the resources required for system operation, data processing, report generation, and dashboard visualization. The software requirements identified the programming languages, frameworks, database management system, development tools, browser, and other technologies used in developing and testing the system.

4.5.1 Hardware Requirements

Table 4.1 presented the minimum and recommended hardware specifications used in the development and implementation of the BrewInsight BI system. These specifications were identified to provide sufficient resources for system access, data processing, report generation, and dashboard visualization.

Table 4.1 Hardware Requirements

Components	Minimum Specification	Recommended Specification
Processor	Quad-core 1.8 GHz	Octa-core 2.0 GHz or Higher
Memory (RAM)	3 GB	4 GB or higher
Storage	32 GB	64 GB or higher
Display	8-inch HD (1280x800)	10-inch Full HD (1920x1080)
Internet Connection	10 Mbps	25 Mbps or higher
Mobile Device	Android 10	Android 13 or higher

The minimum hardware configuration provided the basic resources required to access and operate the system. It allowed users to perform regular activities such as recording sales transactions, managing products, monitoring inventory, and viewing reports.

The recommended hardware configuration provided additional processing and memory resources for smoother system operation. It supported the display of analytical dashboards, processing of business information, generation of reports, and other system activities with improved responsiveness.

4.5.2 Software Requirements

Table 4.2 presented the software requirements used in the development and implementation of the BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong. These software tools were utilized for system development, database management, user interface design, testing, data visualization, and version control.

Table 4.2 Software Requirements

Software	Purpose
Window 11	Operating System
Visual Studio Code	Sources Code Editor
Python 3.11	Backend Programming Language
Flash	Web Application Framework
MySQL	Database Management System
HTML,CSS,and JavaScript	Frontend Development
Bootstrap 5	Responsive User Interface Development
Chart.js	Data Visualization and Analytics Charts
Google Chrome	Web Browser and System Testing
Git	Version Control
XAMPP	Local Development Environment (Apache and MySQL)

The software tools were selected based on their suitability for developing the web-based system. Python and Flask were used for backend development, while HTML, CSS, JavaScript, and Bootstrap were used for the frontend interface. MySQL was used for database management, while Chart.js was used to present sales and analytical information through graphical visualizations.

Visual Studio Code was used as the source code editor, while Google Chrome was used to access and test the web application. XAMPP provided the local development environment, and Git was used to manage source code changes throughout the development process.

4.6 Database Design

The database for **BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong** was designed using a relational structure composed of six tables: **Users, Sales, Sales Items, Products, Recipes, and Ingredients**. Each table represented a specific business entity and stored data according to its intended function. The Users table managed account information, the Sales and Sales Items tables handled transaction records, the Products table maintained menu details, the Recipes table defined product ingredients, and the Ingredients table recorded raw-material inventory. This separation provided a clear organization of the different types of data required for sales and inventory operations.

The structure used primary keys and foreign keys to establish associations between related records. Primary keys uniquely identified entries within each table, while foreign keys referenced corresponding records in related tables. MySQL supports foreign key constraints for maintaining referential relationships between tables [20]. The use of these keys helped maintain data consistency and allowed related records to be accessed when required. For example, transaction records could be associated with the responsible user and purchased products, while product records could be traced to their required ingredients.

The **Users** table stored account information for staff and administrators. It contained the records needed to identify authorized personnel and associate their accounts with activities performed within the application. The `user_id` field served as the identifier used to associate a user with the sales transactions they processed, providing a basis for tracking transaction responsibility.

The **Sales** table stored the main details of completed transactions, including the total amount, payment method, sale date, and responsible user. The `sales_id` field uniquely identified each transaction and was used to associate it with the individual products recorded under the corresponding sale. This arrangement separated overall transaction information from the specific items purchased.

The **Sales Items** table recorded the individual products included in each transaction. It contained the quantity, selling price, and subtotal for each item. The `sales_id` field associated each entry with its corresponding transaction, while `product_id` identified the menu item involved. This provided detailed item-level records that supported the examination of product sales and transaction composition.

The **Products** table contained information about Kape Tubong's menu items, including product names, prices, stock levels, and categories. Product records served as references for items included in sales transactions and were also associated with their respective recipes. This allowed each menu item to be linked to the raw materials required for its preparation.

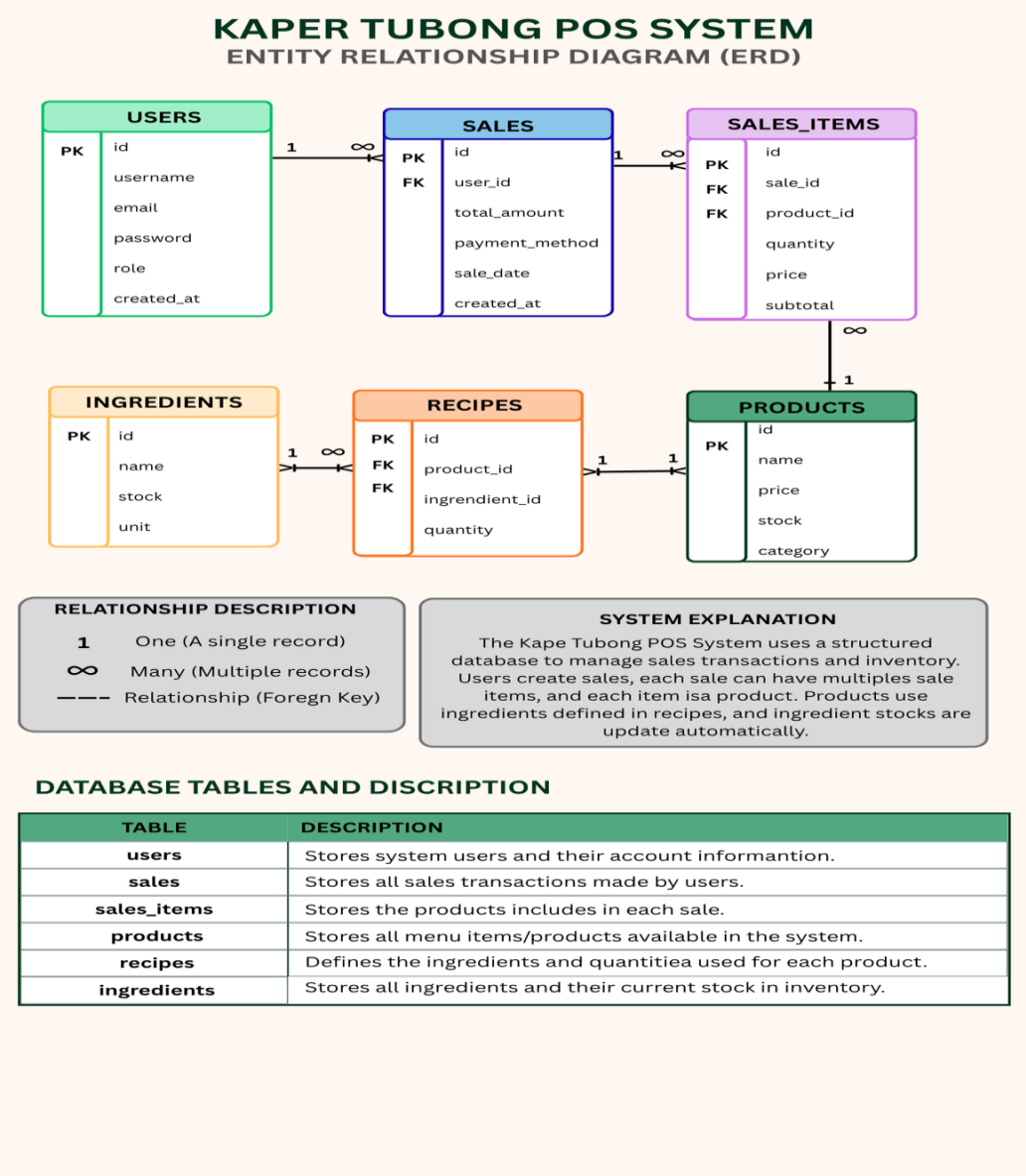
The **Recipes** table defined the ingredients and quantities required for preparing each product. It provided the information needed to determine the raw materials consumed by a particular menu item. Through the associated

product and ingredient identifiers, the required components of each recipe could be determined when processing product sales and updating inventory.

The **Ingredients** table maintained records of raw materials used in preparing Kape Tubong's products. It stored information such as ingredient names, available quantities, and units of measurement. The recorded quantities served as the basis for monitoring inventory levels and determining changes in available stock resulting from product consumption.

Overall, the six tables provided the data structure required to support the main operations of BREWINSIGHT BI. The arrangement followed the flow of business activities from user-processed transactions and purchased products to recipes and ingredient consumption. This structure supported sales processing, inventory monitoring, reporting, and revenue analysis while keeping each type of business data within its appropriate table.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.



DATABASE TABLES AND DISCRIPTION

TABLE	DESCRIPTION
users	Stores system users and their account informantion.
sales	Stores all sales transactions made by users.
sales_items	Stores the products includes in each sale.
products	Stores all menu items/products available in the system.
recipes	Defines the ingredients and quantitiea used for each product.
ingredients	Stores all ingredients and their current stock in inventory.

The primary entities of the database include:

- Users
- Sales
- Sale Items
- Products
- Recipes
- Ingredients

Users Table

This table stores account credentials and user information of all authorized users of the system.

Field	Description
id	Primary Key
Username	Username of the system user
Email	User email address
password	User password
Role	User role (Admin or Staff)
Create_at	Date the account was create

Sales Table

This table stored all sales transactions processed by the system.

Field	Description
id	Primary Key
user_id	Foreign Key to User
total_amount	Total amount of the transaction
payment_method	Payment method used
sale_date	Date of the sale
create_at	Date of the transaction was recorded

Sales Items Table

This table stored the individual products included in each sales transaction.

Field	Description
id	Primary Key
sales_id	Foreign Key to Sales
product_id	Foreign Key to Products
quantity	Quantity of the product sold
price	Selling price of the product
subtotal	Total amount of the selected product

Products Table

This table stored all menu items available in the system.

Field	Description
id	Primary Key
name	Product name
price	Selling price of the product
stock	Available stock quantity
Category	Product category

Recipes Table

This table stored the ingredients required for preparing each product.

Field	Description
id	Primary Key
Product_id	Foreign key to Products
Ingredients_id	Foreign key to ingredients
quantity	Quantity of the ingredient required for the product

Ingredients Table

This table stored all ingredients used in preparing menu items.

Field	Description
id	Primary Key
name	Ingredients name
stock	Available stock quantity
unit	Unit of measurement (e.g., Grams, milliliters, pieces)

Database Summary

The system database was designed for around six interconnected entities: Users, Sales, Sales Items, Products, Recipes, and Ingredients. These entities were connected through primary keys and foreign keys to maintain data consistency and establish clear relationships among system records [20].

The database structure supported user management, sales transaction processing, product management, recipe management, and inventory monitoring. The relationships among the tables also allowed the system to retrieve related information for reports and sales analysis.

The database also supported the automatic updating of ingredient inventory whenever a sales transaction was completed. The system used the recipe information associated with each product to determine the ingredients and quantities that needed to be deducted from inventory.

Overall, the relational database provided an organized and reliable foundation for the core functions of the BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong. It supported centralized record management, transaction processing, inventory monitoring, reporting, and revenue analytics.

References

[17] ISO/IEC 25010:2023, *Systems and software engineering—Systems and software Quality Requirements and Evaluation (SQuaRE)—Product quality model*, 2nd ed., 2023. [Online]. Available: ISO/IEC 25010:2023

[18] K. Beck, M. Beedle, A. van Bennekum, A. Cockburn, W. Cunningham, M. Fowler, J. Grenning, J. Marick, R. C. Martin, S. Mellor, K. Schwaber, K. Sutherland, and D. Thomas, "Manifesto for Agile Software Development," Agile Manifesto, 2001. [Online]. Available: Agile Manifesto

[19] ISO/IEC/IEEE 29119-1:2022, *Software and systems engineering—Software testing—Part 1: General concepts*, 2nd ed., 2022. [Online]. Available: ISO/IEC/IEEE 29119-1:2022

[20] Oracle Corporation, *MySQL 8.4 Reference Manual: FOREIGN KEY Constraints*. [Online]. Available: MySQL 8.4 Reference Manual – FOREIGN KEY Constraints

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presented the conclusion and recommendations of the study based on the development, implementation, testing, and evaluation of **BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong**. The discussion was grounded in the results obtained from the development process, testing procedures, evaluation activities, and feedback from the business owner. These results provided the basis for determining whether the objectives of the study were achieved and whether the identified concerns in Kape Tubong's sales and inventory management were addressed.

The developed solution addressed the identified challenges associated with Kape Tubong's sales and inventory management. The previous manual processes required additional time and effort for recording sales transactions, maintaining records, consolidating information, and preparing reports. These activities could also result in inaccurate records, duplicate entries, incomplete information, and delays in obtaining updated business data. Such limitations made the retrieval and review of records more difficult, particularly when information was needed for monitoring sales activities, inventory conditions, and business performance.

BREWINSIGHT BI provided a centralized database for storing and managing sales, product, and inventory information. It also incorporated interactive dashboards and reports that presented revenue patterns, sales performance, product information, and inventory summaries. These features allowed relevant business data to be organized, retrieved, and reviewed more efficiently. The visual presentation of sales data also provided a clearer view of business performance, while the reporting functions supported the preparation and examination of summarized records for operational monitoring and analysis.

The chapter also presented recommendations based on the findings and conclusions of the study. These recommendations included possible improvements to the developed solution, areas that could be explored in future research, and implementation strategies that could support its continued use. The proposed improvements addressed areas such as additional functionality, security, reporting, analytical capabilities, and accessibility, while the implementation strategies focused on user training, maintenance, data backup, and proper handling of business records. Together, these recommendations provided direction for maintaining the usefulness of BREWINSIGHT BI and supporting its continued development in accordance with Kape Tubong's future needs.

5.2 Summary of Findings

The findings of the study showed that the developed system achieved the objectives established for BREWINSIGHT BI. The system integrated sales management, product management, inventory monitoring, reporting, and Business Intelligence features into a centralized web-based platform.

The following findings were identified:

- The system provided a secure login and user authentication feature that controlled access to the system and its available functions.
- The Sales Transaction Module provided a centralized process for recording sales transactions and storing sales information in the database.
- The Product Management Module allowed users to manage and maintain product information in an organized manner.
- The Inventory Management Module used recipe information to monitor the ingredients associated with products and update inventory based on recorded sales transactions.
- The Reports Module provided organized sales and inventory information that could be retrieved and reviewed for business monitoring.
- The Business Intelligence Dashboard presented sales information through interactive dashboards and visualizations. It provided information related to revenue patterns, product performance, peak sales periods, and other sales-related trends.
- The database design centralized the records of users, sales, sale of items, products, recipes, and ingredients. The relationships between these records supported the proper flow of information within the system. MySQL foreign key relationships were used to connect related records and maintain data consistency [20].
- The system followed the Agile Software Development Methodology, which supported the development through planning, analysis, design, implementation, testing, deployment, and maintenance activities [18].
- Testing procedures were conducted to determine whether the developed features operated according to the intended requirements. The testing process included unit testing, system testing, and user acceptance testing [19].
- The evaluation results showed that the developed system was functional, easy to use, reliable, and appropriate for its intended purpose. The system supported the management of sales, inventory, reporting, and revenue analysis.

Overall, the findings indicated that BREWINSIGHT BI provided a centralized environment that helped organize Kape Tubong's sales information and supported the analysis of business data.

5.3 Conclusion

Based on the findings of the study, BREWINSIGHT BI successfully addressed the identified problems associated with the manual management and analysis of Kape Tubong’s sales information. The developed platform centralized sales transactions, product management, inventory monitoring, and report generation within a single environment. This integration organized essential business records and provided users with a more systematic means of accessing and managing sales-related information.

The integration of the Business Intelligence dashboard enabled sales information to be presented through organized visualizations and summarized data. It displayed revenue patterns, product performance, sales periods, and inventory conditions, providing management with a clearer view of business operations. These analytical features supported the

assessment of business performance and supplied relevant information for decision-making based on available sales and inventory data.

The database design and architecture provided a structured foundation for managing the various types of information processed throughout the platform. The use of the Agile methodology supported development through iterative planning, implementation, testing, review, and improvement [18]. Testing and evaluation activities demonstrated that the developed features performed their intended functions and met the requirements of the intended users. These results indicated that the development approach effectively supported the implementation and evaluation of the project.

Therefore, the study concluded that BREWINSIGHT BI achieved its objectives of providing Kape Tubong with a centralized sales and Business Intelligence platform. The project improved the organization of sales records, inventory monitoring, reporting, and revenue analysis while providing relevant information for management and decision-making. Overall, the developed solution addressed the identified need for a more organized and accessible approach to managing and analyzing business information.

5.4 Recommendations

Based on the findings and conclusions of the study, the following recommendations were proposed to improve the functionality, security, usability, and long-term application of BREWINSIGHT BI:

1. System Improvements.

The study recommended the integration of automated notifications for low ingredient stock, unusual sales activities, and slow-moving products. These features were intended to assist users in identifying operational concerns more efficiently and reducing the need for manual monitoring of individual records. The study also recommended the implementation of additional security measures, including stronger password requirements, account recovery mechanisms, limits on consecutive failed login attempts, and automatic logout after a period of inactivity. The reporting module was recommended for enhancement through the inclusion of export options in PDF, Excel, and CSV formats. Additional dashboard visualizations covering monthly revenue, seasonal sales patterns, product categories, and inventory movement were also recommended to provide more comprehensive business information.

2. Future Research.

The study recommended that future researchers investigate the integration of predictive analytics and machine learning techniques into the system. These technologies were identified as potential approaches for forecasting future sales, identifying recurring purchasing patterns, and analyzing historical sales data for additional business insights. The study also recommended further research on mobile accessibility and the integration of additional business modules, including supplier management, expense monitoring, payroll management, customer loyalty programs, online ordering, and customer feedback. A longer evaluation period was further recommended to provide a more comprehensive assessment of the system's performance, usability, reliability, and usefulness under different operating conditions.

3. Implementation Strategies.

The study recommended the provision of proper user training to ensure that personnel understood the system's functions, workflows, and appropriate data-entry procedures. The study also emphasized the importance of recording transactions accurately and promptly so that the information presented in reports and dashboards remained current and reliable. Regular database backups, system maintenance, and security updates were recommended to reduce the risk of data loss and maintain system reliability. Periodic reviews of system records were also recommended to identify and correct inconsistencies or errors in the stored information. In addition, the study recommended the provision of a user manual containing instructions for common operations, system procedures, and basic troubleshooting to assist users in performing routine tasks and addressing basic system concerns.

5.5 Impact of the Study

The developed system provided Kape Tubong with a centralized platform for handling sales, product, inventory, and reporting activities. By consolidating these functions into one system, business information could be recorded and accessed in a more structured manner, reducing the need to manage multiple records or sources of information. This organization helped streamline routine tasks and made relevant sales and inventory data more readily available to authorized users. As a result, the system contributed to a more systematic approach to managing day-to-day business operations and maintaining essential business records.

The Business Intelligence dashboard enhanced the way management could examine and interpret available business data. Through visual representations of revenue patterns, product performance, sales trends, and other relevant indicators, users could gain a clearer view of the business's performance over specific periods. Rather than relying solely on individual transaction records, management could use summarized information to identify patterns and areas that may require attention. These capabilities could support planning, performance monitoring, and other management activities by providing relevant information in a more understandable and accessible format.

Beyond its application at Kape Tubong, the study demonstrated the potential of Business Intelligence technologies in supporting the operations of small food and beverage businesses. The development of BREWINSIGHT BI showed how sales data, inventory information, reporting functions, and analytical dashboards can be integrated into a single system to address practical business needs. The study may also provide a useful reference for future students, researchers, and system developers working on related projects involving sales analytics, centralized databases, automated reporting, dashboards, and decision-support systems. Its development process and identified areas for improvement may serve as a basis for exploring similar solutions in other small-business settings.

5.6 Summary

The study developed **BREWINSIGHT BI**, a web-based Business Intelligence system designed to centralize Kape Tubong's sales and business information. The system combined key business functions, including sales transaction management, product management, inventory monitoring, report generation, and Business Intelligence dashboards, within a single platform. This integration provided a structured approach to handling business records and reduced the need to manage information across separate systems or records. By consolidating these functions, the system

established a more organized environment for managing and accessing data relevant to the daily operations of Kape Tubong.

The findings indicated that BREWINSIGHT BI fulfilled its intended objectives by providing accessible and organized sales and inventory information. Its dashboards and reports presented relevant business data in summarized and visual formats, allowing users to examine revenue patterns, product performance, sales periods, and other operational indicators. These features supported the monitoring of business performance and helped transform stored transaction data into information that could be used for analysis and management activities. Thus, the system served not only as a platform for recording transactions but also as a tool for interpreting business data and supporting data-informed decision-making.

The study also identified several areas that may be considered for future development. These include automated notifications, stronger security measures, report exporting, customer management, predictive analytics, mobile accessibility, and additional business modules. Such enhancements could expand the system's functionality as the operational requirements of Kape Tubong evolve. To maintain its reliability and usefulness, regular maintenance, data backups, security monitoring, and appropriate user training should also be carried out. Continuous evaluation and improvement of the system would further allow BREWINSIGHT BI to adapt to changing business needs and provide sustained support for sales management, inventory monitoring, reporting, and business analysis.

References

[18] K. Beck, M. Beedle, A. van Bennekum, A. Cockburn, W. Cunningham, M. Fowler, J. Grenning, B. Marick, R. C. Martin, [Online]. Available: Agile Manifesto

[19] ISO/IEC/IEEE 29119-1:2022, *Software and systems engineering—Software testing—Part 1: General concepts*, 2nd ed., 2022. [Online]. Available: ISO/IEC/IEEE 29119-1:2022

[20] Oracle Corporation, *MySQL 8.4 Reference Manual: FOREIGN KEY Constraints*. [Online]. Available: MySQL 8.4 Reference Manual: FOREIGN KEY Constraints